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Dipartimento di Informatica e Automazione
Computer Networks Research Group

netkit lab

bgp: transit as

Version	1.7
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Description	possible architectures for a transit provider, bad interactions between igp and bgp routing protocols, configuration of tunnels

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disclaimer

- this is a rather complex lab
- some of the proposed experiences rely on particular event timings and may not be observable in all situations
- other experiences describe ways to force timings and should therefore be guaranteed observable
- last, observed phenomena may be due to zebra-specific implementation choices and may not apply to more recent releases of zebra
- please read carefully through this documentation before concluding that the lab is not working properly

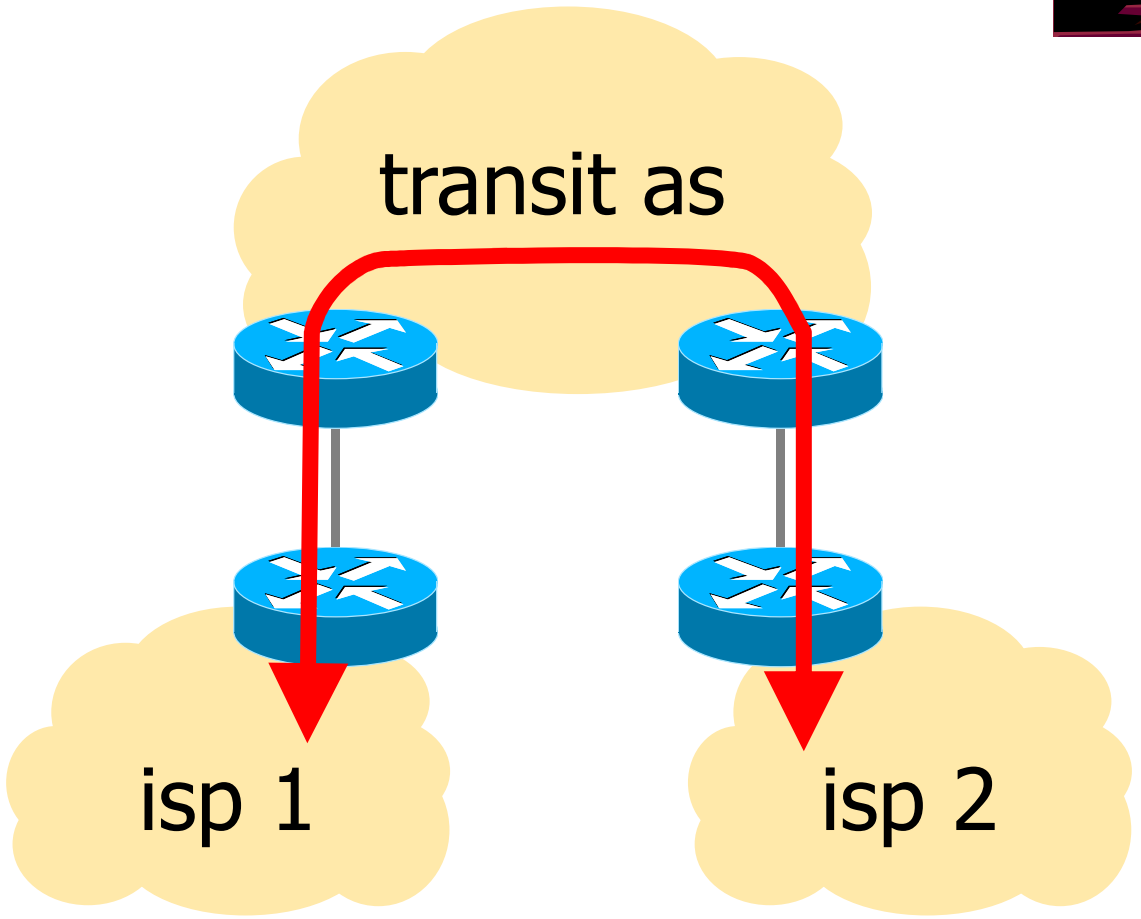
scenario



- a transit as
  - receives and propagates the full bgp routing table from/to its neighbors (customers, peers, providers)
  - receives and forwards traffic across its neighbors

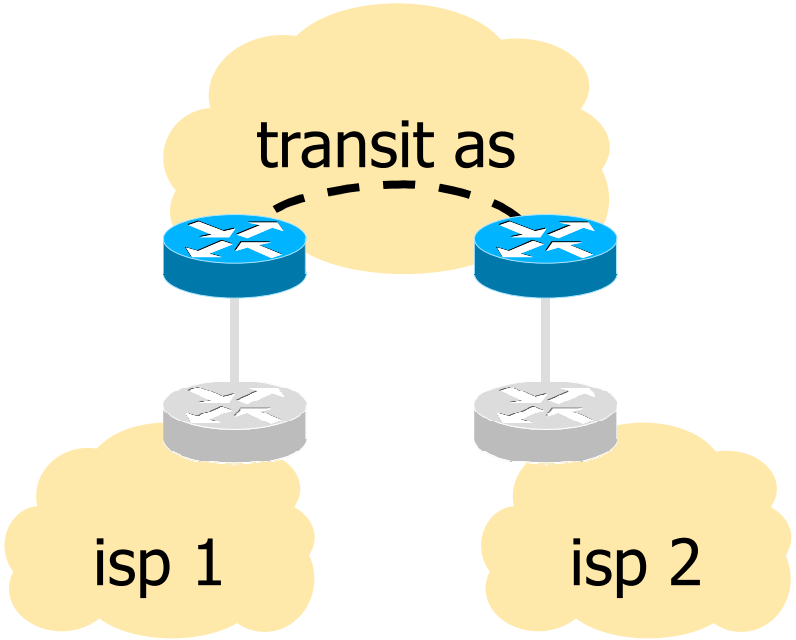
scenario





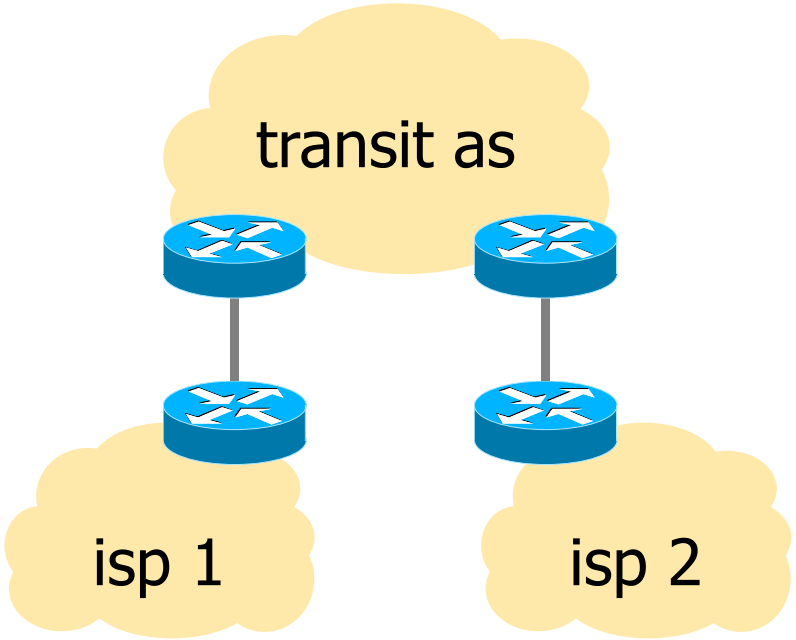
transit as: requirements

- problem: border routers must know each other's routes
- solution: ibgp peerings (possibly full mesh)



transit as: requirements

- problem: someone must tell bgp how to reach external next hops (recursive lookup)
- solution: igp

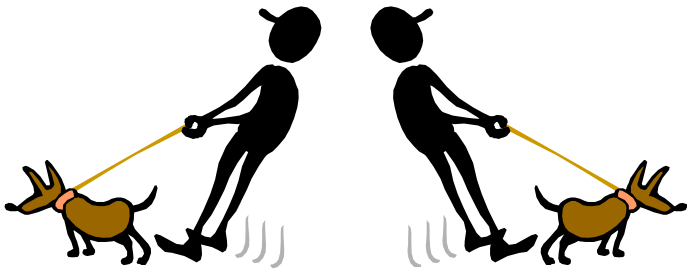


transit as: requirements

- problem:
  - ibgp carries announcements between border routers
  - igp carries traffic between border routers

so...

  - consistent routing between ibgp and igp must be guaranteed
    - even in the presence of bgp routing policies
  - ibgp and igp should never disagree on the route to a destination
- solution: ???



transit as: degrees of freedom

- internal routers must support traffic flows from/to neighboring ases
  - choice 1: redistribute bgp routes into the igp
    - overgrowth of igp routing tables 
    - update churn from bgp affects the igp 
  - choice 2: route traffic flowing through via an ad-hoc overlay
    - internal routers know about border routers only
    - bgp updates leak into the igp only if this changes the choice of the egress routers

how to read

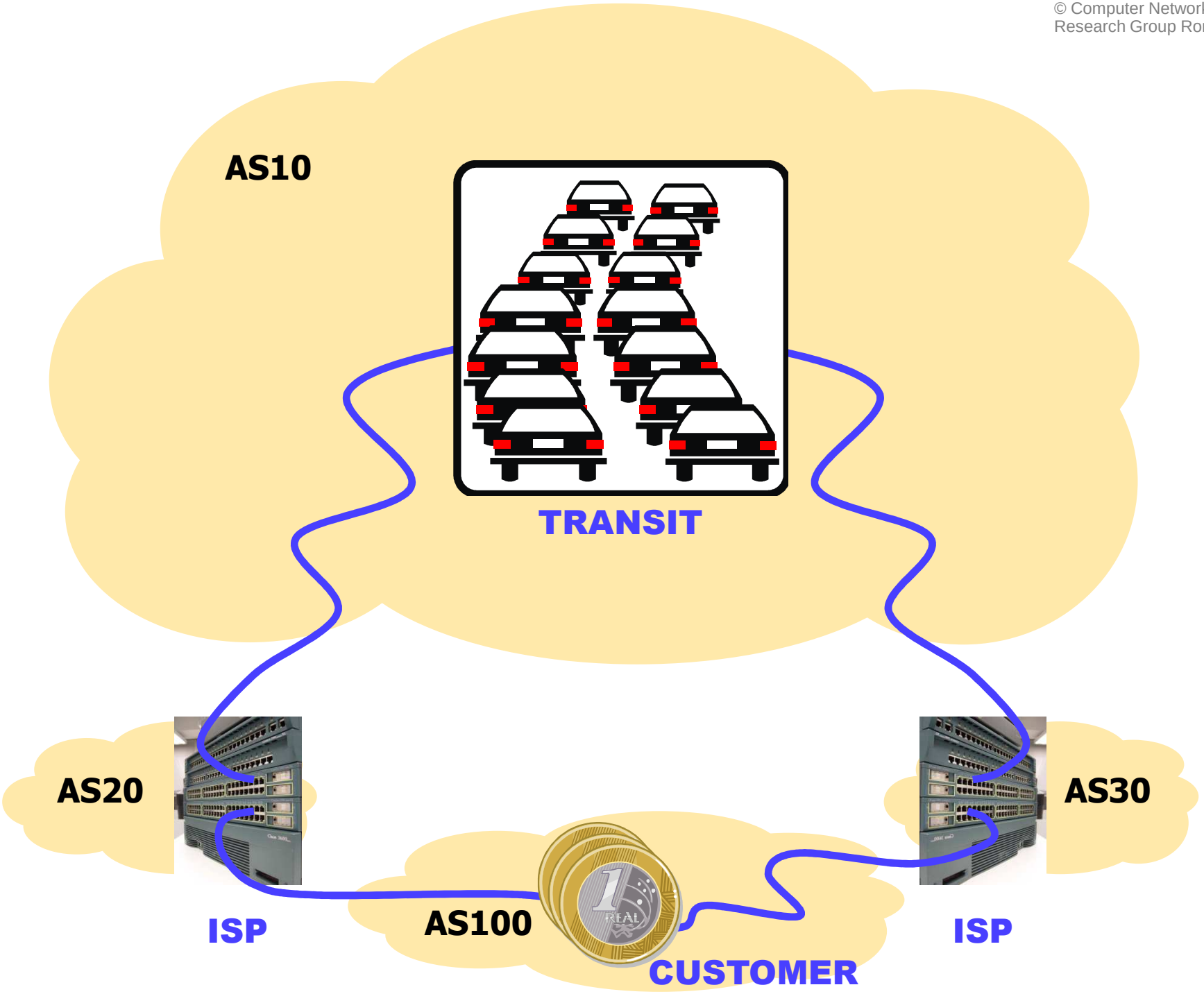


- browse through all the slides
- concentrate on specific topics
  - ✓ redistribution of bgp into the igp
  - ✓ peering setup using loopback interfaces
  - ✓ forwarding loops
  - ✓ bad cross-protocol event timings
  - ✓ administrative distance
  - ✓ setup of an overlay network using tunnels
  - ✓ asymmetric routing

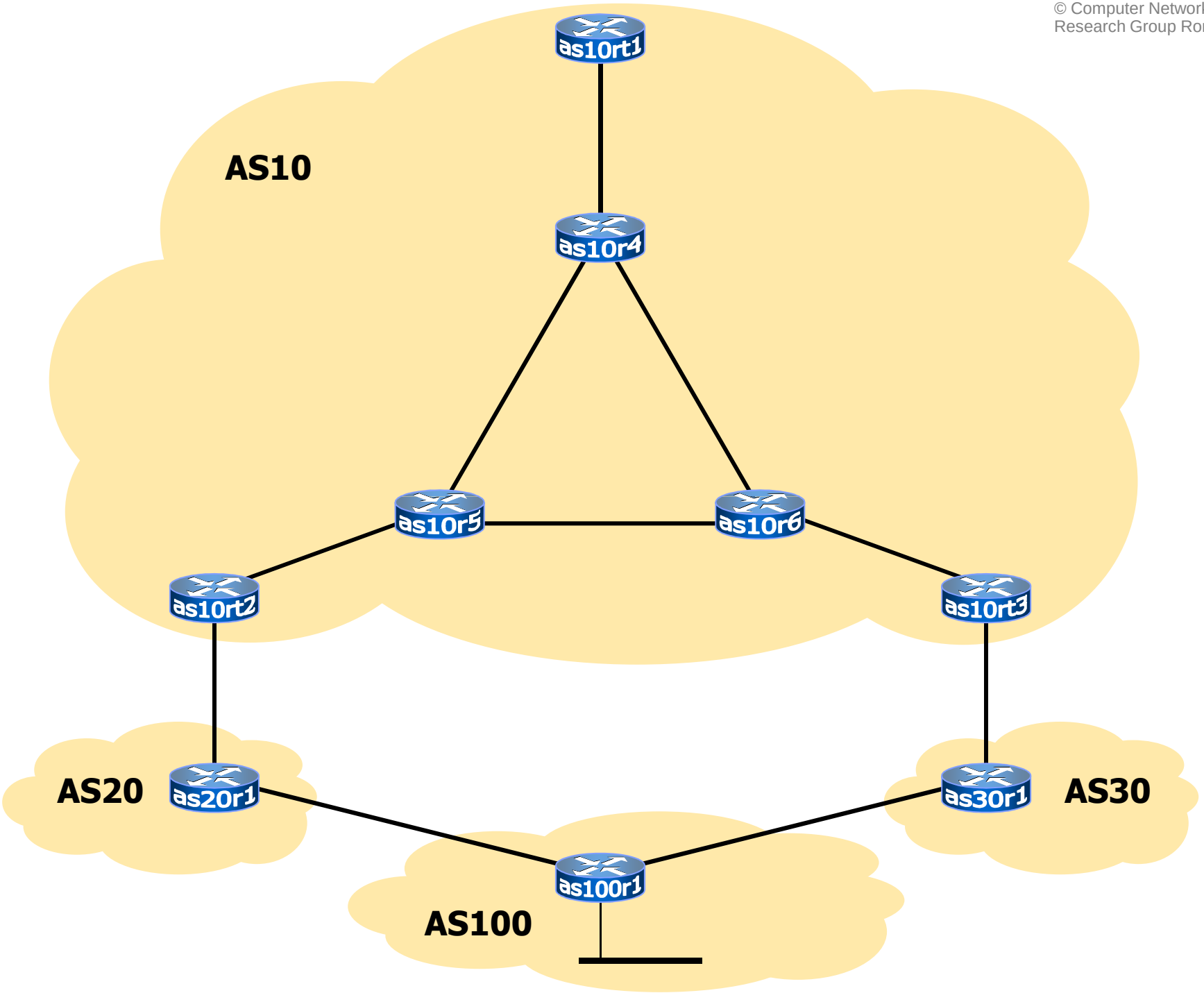
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network topology

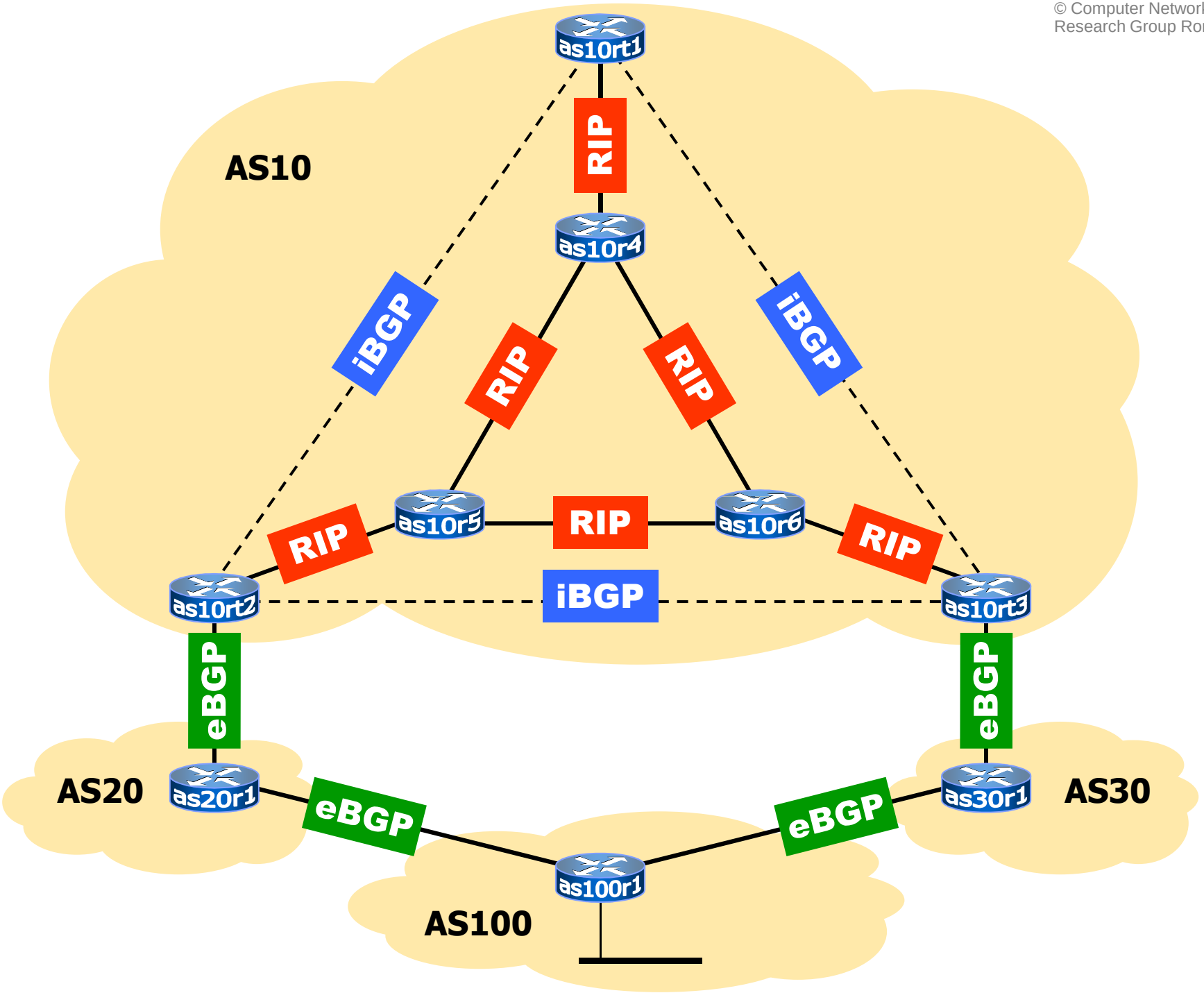
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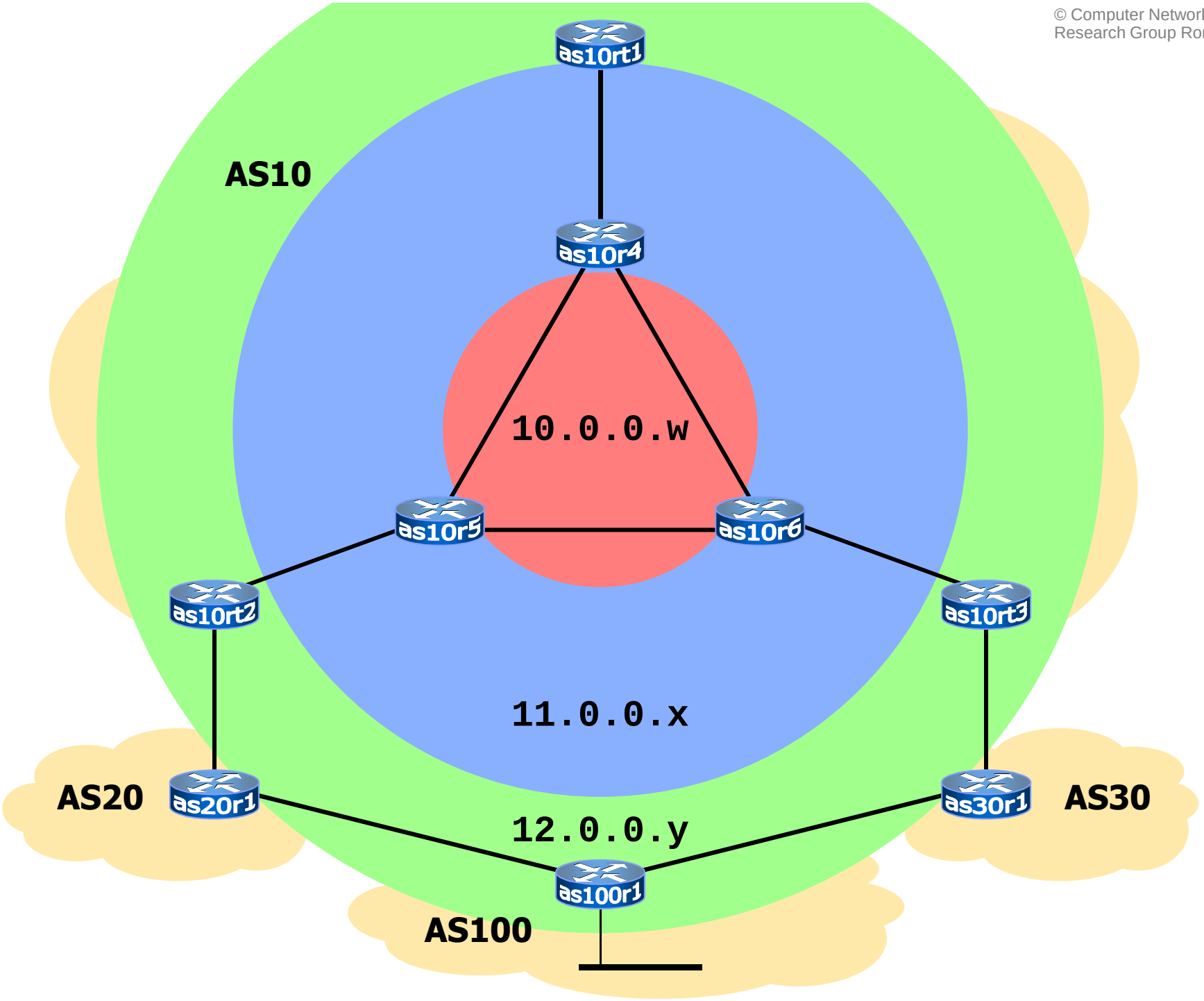
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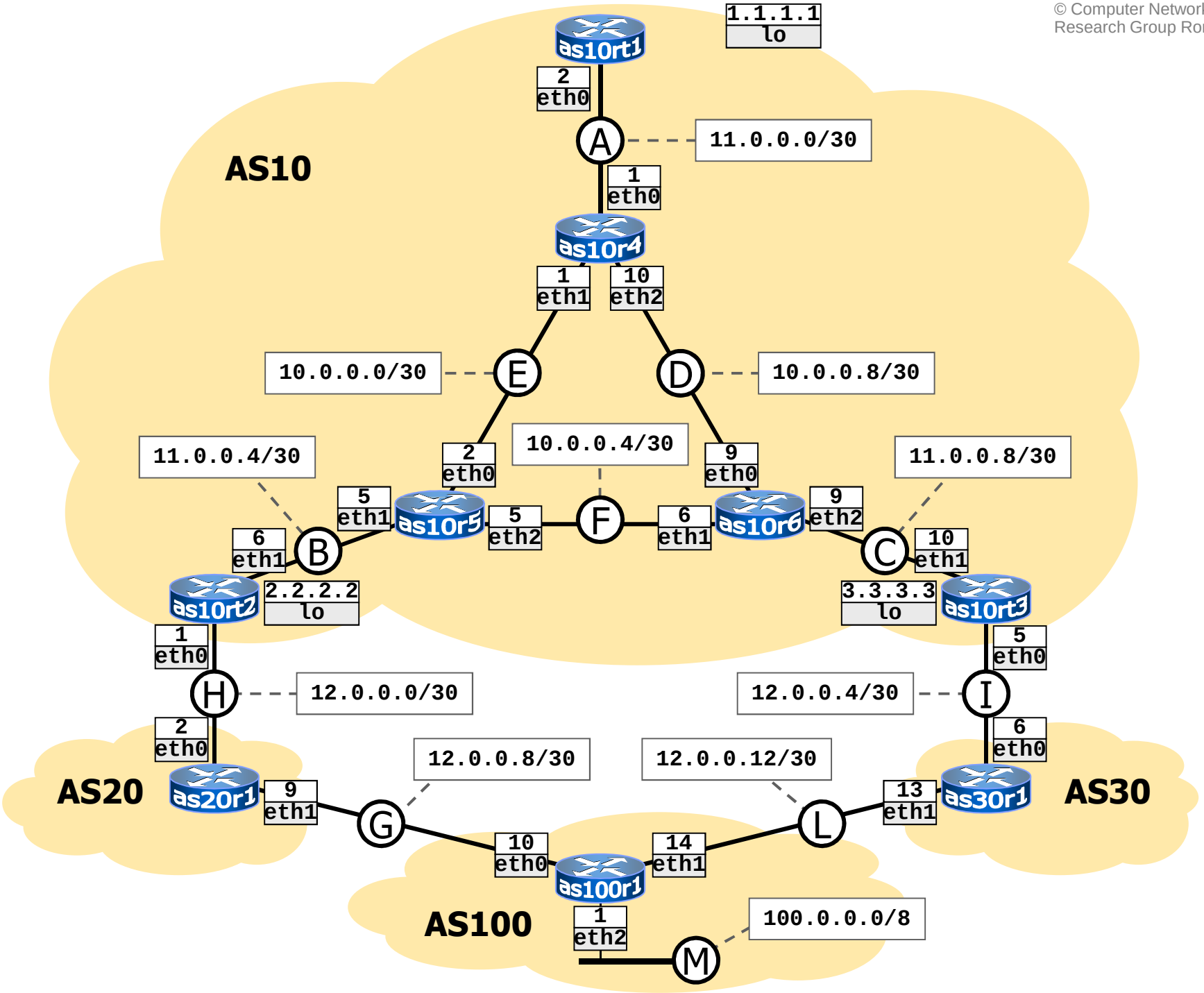
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choice 1

redistribution

transit as: interesting configurations

zebra bgp configuration file

```
router bgp 10
   network 10.0.0.0/8
   network 12.0.0.0/30
   neighbor 1.1.1.1 remote-as 10
   neighbor 1.1.1.1 update-source lo
   neighbor 1.1.1.1 description as10rt1(iBGP)
   neighbor 3.3.3.3 remote-as 10
   neighbor 3.3.3.3 update-source lo
   neighbor 3.3.3.3 description as10rt3(iBGP)
```

- the transit as announces its own network (**10.0.0.0/8**) but no default

transit as: interesting configurations

zebra rip configuration file

```
router rip
    network eth1
    redistribute connected
    redistribute bgp
```

- bgp routing information is injected into rip

transit as: interesting configurations

zebra rip configuration file

```
router rip
    network eth1
    redistribute connected
    redistribute bgp
```

rip speaking interfaces can be specified by their name or network address

- bgp routing information is injected into rip

transit as: bgp peerings

- bgp peerings are established on loopback interfaces
  - improved resiliency
    - the peering stays up even if all the router's physical interfaces are down
- two loopbacks for each border router of as10
  - **`ifconfig lo:1 2.2.2.2 netmask 255.255.255.255 up`**
    - **`lo:1`** is an ip alias used for the peerings
    - the usual loopback address, **`lo`**, is still available



beware: using **`ifconfig lo:1 2.2.2.2/32`** sets up a **`/0`** netmask instead(!)

- a default route would unexpectedly be announced when loopback interfaces are redistributed in an igp

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transit as: bgp peerings

- be careful when configuring peerings on the loopbacks
  - bgp complains if the source address of OPEN messages from a neighbor does not match the neighbor's address configured in the peering (in this case, the loopback address)
  - bgp messages come out of a physical interface, whose address is different from the loopback's
  - need to force the source address of bgp messages
    - **`update-source`**
- cisco says:

  You only have to use the ***`update-source`*** *command when someone is peering to your loopback address*

transit as: bgp peerings

zebra bgp configuration file

```
router bgp 10
   network 10.0.0.0/8
   network 12.0.0.0/30
   neighbor 1.1.1.1 remote-as 10
   neighbor 1.1.1.1 update-source 2.2.2.2
   neighbor 1.1.1.1 description as10rt1(iBGP)
   neighbor 3.3.3.3 remote-as 10
   neighbor 3.3.3.3 update-source 2.2.2.2
   neighbor 3.3.3.3 description as10rt3(iBGP)
```

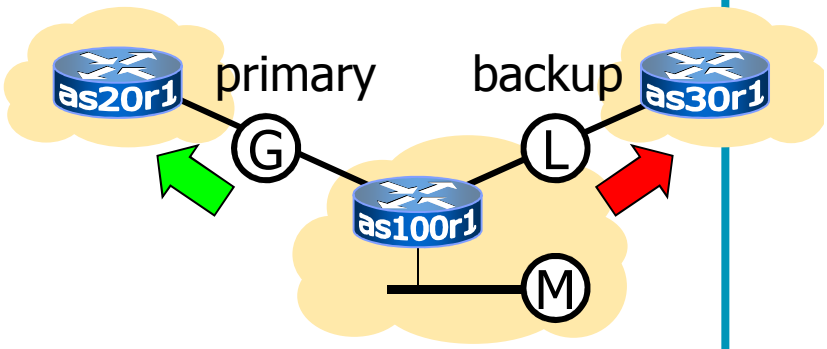
- note
  - **`update-source`** accepts an ip address or an interface name
  - zebra does not allow to set the **`update-source`** to an alias interface (e.g., **`lo:1`**)

transit as: some other flavouring

as100r1

```
as100r1:~# less /etc/zebra/bgpd.conf
hostname as100r1-bgpd
password zebra
!
ip prefix-list mineOut permit 100.0.0.0/8
!
route-map lowerPreference permit 10
   set local-preference 10
!
router bgp 100
   network 100.0.0.0/8
   neighbor 12.0.0.9 remote-as 20
   neighbor 12.0.0.9 description as20r1
   neighbor 12.0.0.9 prefix-list mineOut out
   neighbor 12.0.0.13 remote-as 30
   neighbor 12.0.0.13 description as30r1
   neighbor 12.0.0.13 prefix-list mineOut out
   neighbor 12.0.0.13 route-map lowerPreference in
```

/etc/zebra/bgpd.conf



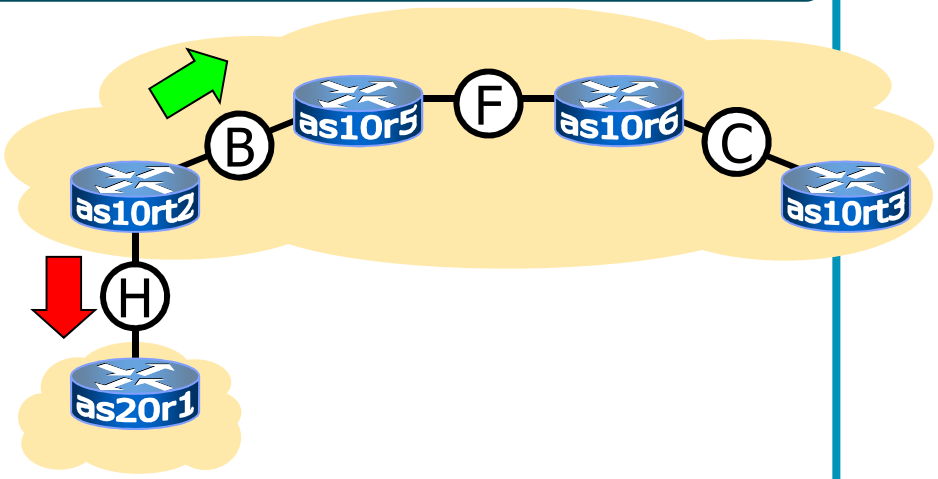
the customer prefers using link **G**

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transit as: some other flavouring

as10rt2

```
as10rt2:~# less /etc/zebra/bgpd.conf
hostname as10rt2-bgpd
password zebra
...
!
route-map dePref permit 10
   set local-preference 10
!
router bgp 10
   network 10.0.0.0/8
   network 12.0.0.0/30
   neighbor 1.1.1.1 remote-as 10
   neighbor 1.1.1.1 update-source lo
   neighbor 1.1.1.1 description as10rt1(iBGP)
   neighbor 3.3.3.3 remote-as 10
   neighbor 3.3.3.3 update-source lo
   neighbor 3.3.3.3 description as10rt3(iBGP)
   neighbor 12.0.0.2 remote-as 20
   neighbor 12.0.0.2 description as20r1(eBGP)
   neighbor 12.0.0.2 route-map dePref in
   neighbor 12.0.0.2 prefix-list noDefault in
/etc/zebra/bgpd.conf
```



as10rt2 prefers using the egress router **as10rt3**

transit as

- start the lab

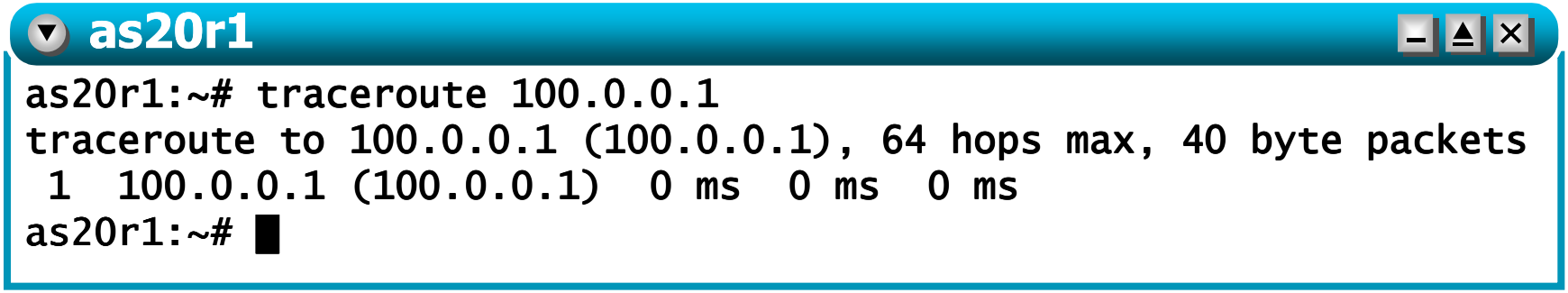
host machine

```
user@localhost:~$ cd netkit-lab_bgp-transit-as-forwarding-loop
user@localhost:~/netkit-lab_bgp-transit-as-forwarding-loop$ lstart
```

- wait a few minutes for the routing system(s) to converge

transit as

- check connectivity



  - **as20r1** correctly chooses to reach **100.0.0.1** via the direct link **G**

transit as

- check connectivity

as10r6

```
as10r6:~# traceroute 100.0.0.1
traceroute to 100.0.0.1 (100.0.0.1), 64 hops max, 40 byte packets
 1  11.0.0.10 (11.0.0.10)  0 ms  0 ms  0 ms
 2  * * *
 3  * * *
 4  * * *
 5
```

- **as10r6** fails to reach **100.0.0.1** even if bgp routes are being redistributed in rip!

transit as

- let's have a closer look at **as10r6**...

as10r6

```
as10r6-ripd> show ip rip
Codes: R - RIP, C - connected, O - OSPF, B - BGP
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

     Network            Next Hop         Metric From            Time
R(n) 1.1.1.1/32         10.0.0.10             3 10.0.0.10      02:54
R(n) 2.2.2.2/32         10.0.0.5              3 10.0.0.5       02:28
R(n) 3.3.3.3/32         11.0.0.1              2 11.0.0.10      02:52
R(n) 10.0.0.0/30        10.0.0.1              2 10.0.0.10      02:54
C(i) 10.0.0.4/30        0.0.0.0               1 self
C(i) 10.0.0.8/30        0.0.0.0               1 self
R(n) 11.0.0.0/30        10.0.0.1              2 10.0.0.10      02:54
R(n) 11.0.0.4/30        10.0.0.5              2 10.0.0.10      02:28
C(i) 11.0.0.8/30        0.0.0.0               1 self
R(n) 12.0.0.0/30        10.0                  2 11.0.0.10      02:28
R(n) 12.0.0.4/30        11                    2 11.0.0.10      02:52
R(n) 12.0.0.8/30        1.0.0.10              2 11.0.0.10      02:52
R(n) 12.0.0.12/30       11.0.0.10             2 11.0.0.10      02:52
R(n) 20.0.0.0/8         10.0.0.5              3 10.0.0.5       02:28
R(n) 30.0.0.0/8         11.0.0.10             2 11.0.0.10      02:52
R(n) 100.0.0.0/8        11.0.0.10             2 11.0.0.10      02:52
as10r6-ripd>
```

a route to **100.0.0.0/8** exists and correctly uses the border router **as10rt3**!

transit as

- let's have a closer look at **as30r1** then...

as10r6

```
as10r6:~# traceroute -I icmp 100.0.0.1
traceroute to 100.0.0.1 (100.0.0.1), 64 hops max
 1  11.0.0.10 (11.0.0.10)  2 ms  3 ms
 2  * * *
 3
```

force **traceroute** to use icmp packets instead of the default UDP packets

as30r1

```
as30r1:~# tcpdump -i eth0 icmp
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 96 bytes
15:04:22.738639 IP 11.0.0.9 > 100.0.0.1: icmp 8: echo request seq 1024
15:04:25.694192 IP 11.0.0.9 > 100.0.0.1: icmp 8: echo request seq 1280
15:04:28.701455 IP 11.0.0.9 > 100.0.0.1: icmp 8: echo request seq 1536
15:04:31.701105 IP 11.0.0.9 > 100.0.0.1: icmp 8: echo request seq 1792

4 packets captured
4 packets received by filter
0 packets dropped by kernel
```

echo requests come from **11.0.0.9**...!!

transit as

- the mystery unrevealed:

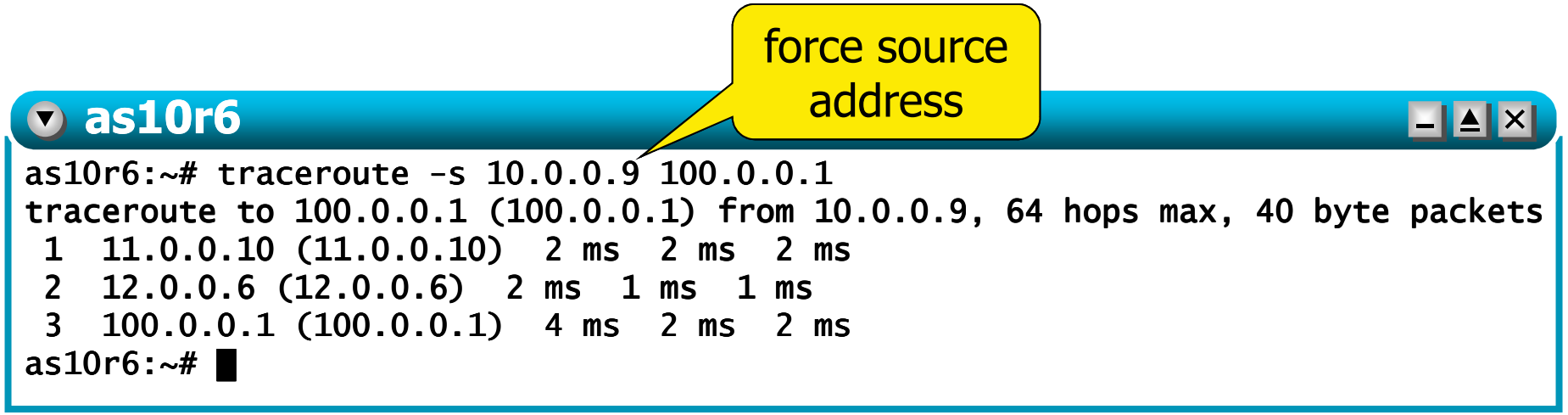
as30r1

```
as30r1:~# route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
12.0.0.4        0.0.0.0         255.255.255.252 U     0      0        0 eth0
12.0.0.0        12.0.0.5        255.255.255.252 UG    0      0        0 eth0
12.0.0.12       0.0.0.0         255.255.255.252 U     0      0        0 eth1
12.0.0.8        12.0.0.5        255.255.255.252 UG    0      0        0 eth0
20.0.0.0        12.0.0.5        255.0.0.0       UG    0      0        0 eth0
100.0.0.0       12.0.0.14       255.0.0.0       UG    0      0        0 eth1
10.0.0.0        12.0.0.5        255.0.0.0       UG    0      0        0 eth0
as30r1:~#
```

- **as30r1** is not offered the default route, therefore is unable to send icmp replies back to **11.0.0.9**

transit as

- things start working if we set a source address falling in the transit as' network



  - now **`as10r6`** correctly reaches the customer using the nearest egress point **`as10rt3`**

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transit as: routing tables

as10r6

```
as10r6-ripd> show ip rip
Codes: R - RIP, C - connected, O - OSPF, B - BGP
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

     Network            Next Hop         Metric From            Time
R(n) 1.1.1.1/32         10.0.0.10             3 10.0.0.10      02:58
R(n) 2.2.2.2/32         10.0.0.5              3 10.0.0.5       02:43
R(n) 3.3.3.3/32         11.0.0.10             2 11.0.0.10      02:55
R(n) 10.0.0.0/30        10.0.0.10             2 10.0.0.10      02:58
C(i) 10.0.0.4/30        0.0.0.0               1 self
C(i) 10.0.0.8/30        0.0.0.0               1 self
R(n) 11.0.0.0/30        10.0.0.10             2 10.0.0.10      02:58
R(n) 11.0.0.4/30        10.0.0.5              2 10.0.0.5       02:43
C(i) 11.0.0.8/30        0.0.0.0               1 self
R(n) 12.0.0.0/30        10.0.0.5              3 10.0.0.5       02:43
R(n) 12.0.0.4/30        11.0.0.10             2 11.0.0.10      02:55
R(n) 12.0.0.8/30        11.0.0.10             2 11.0.0.10      02:55
R(n) 12.0.0.12/30       11.0.0.10             2 11.0.0.10      02:55
R(n) 20.0.0.0/8         10.0.0.5              3 10.0.0.5       02:43
R(n) 30.0.0.0/8         11.0.0.10             2 11.0.0.10      02:55
R(n) 100.0.0.0/8        11.0.0.10             2 11.0.0.10      02:55
as10r6-ripd>
```

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transit as: routing tables

as10r6

```
as10r6-ripd> show ip rip
Codes: R - RIP, C - connected, O - OSPF, B - BGP
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
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     Network            Next Hop         Metric From            Time
R(n) 1.1.1.1/32         10.0.0.10             3 10.0.0.10      02:58
R(n) 2.2.2.2/32         10.0.0.5              3 10.0.0.5       02:43
R(n) 3.3.3.3/32         11.0.0.10             2 11.0.0.10      02:55
R(n) 10.0.0.0/30        10.0.0.10
C(i) 10.0.0.4/30        0.0.0.0
C(i) 10.0.0.8/30        0.0.0.0
R(n) 11.0.0.0/30        10.0.0.10
R(n) 11.0.0.4/30        10.0.0.5
C(i) 11.0.0.8/30        0.0.0.0
R(n) 12.0.0.0/30        10.0.0.5
R(n) 12.0.0.4/30        11.0.0.10
R(n) 12.0.0.8/30        11.0.0.10
R(n) 12.0.0.12/30       11.0.0.10
R(n) 20.0.0.0/8         10.0.0.5              3 10.0.0.5       02:43
R(n) 30.0.0.0/8         11.0.0.10             2 11.0.0.10      02:55
R(n) 100.0.0.0/8        11.0.0.10             2 11.0.0.10      02:55
as10r6-ripd>
```

routing tables of the internal routers of **as10** are unnecessarily large, due to redistribution of bgp routes

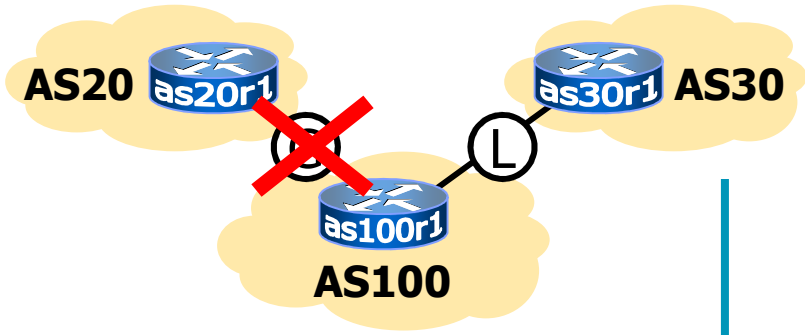
transit as: playing with the backup

- let's bring **as100**'s primary link down
  - expected result: traffic from **as20r1** to **as100r1** should traverse the transit as

as20r1

```
as20r1:~# telnet localhost bgpd
Trying 127.0.0.1...
Connected to as20r1.
Escape character is '^]'.

Hello, this is zebra (version 0.94).
...
as20r1-bgpd> enable
as20r1-bgpd# configure terminal
as20r1-bgpd(config)# router bgp 20
as20r1-bgpd(config-router)# neighbor 12.0.0.10 shutdown
```



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transit as: playing with the backup

- wait for the routing to converge, then check the bgp routing table of **as20r1**

as20r1

```
as20r1-bgpd> show ip bgp
BGP table version is 0, local router ID is 12.0.0.9
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 0.0.0.0          0.0.0.0                  0         32768 i
*> 10.0.0.0         12.0.0.1                 0             0 10 i
*> 12.0.0.0/30      12.0.0.1                 0             0 10 i
*> 12.0.0.4/30      12.0.0.1                               0 10 i
*> 12.0.0.8/30      0.0.0.0                  0         32768 i
*> 12.0.0.12/30     12.0.0.1                               0 10 30 i
*> 20.0.0.0         0.0.0.0                  0         32768 i
*> 30.0.0.0         12.0.0.1                               0 10 30 i
*> 100.0.0.0        12.0.0.1                               0 10 30 100 i

Total number of prefixes 9
as20r1-bgpd>
```

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transit as: playing with the backup

- wait for the routing to converge, then check the bgp routing table of **as20r1**

as20r1

```
as20r1-bgpd> show ip bgp
BGP table version is 0, local router ID is 12.0.0.9
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            M         LocPrf Weight Path
*> 0.0.0.0          0.0.0.0                                32768 i
*> 10.0.0.0         12.0.0.1                                   0 10 30 i
*> 12.0.0.0/30      12.0.0.1                                   0 10 30 i
*> 12.0.0.4/30      12.0.0.1                                   0 10 30 i
*> 12.0.0.8/30      0.0.0.0                                32768 i
*> 12.0.0.12/30     12.0.0.1                                   0 10 30 i
*> 20.0.0.0         0.0.0.0                  0             32768 i
*> 30.0.0.0         12.0.0.1                                   0 10 30 i
*> 100.0.0.0        12.0.0.1                                   0 10 30 100 i

Total number of prefixes 9
as20r1-bgpd>
```

bgp is now correctly using the backup link **L**

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transit as: playing with the backup

- let's check whether things work...

as20r1

```
as20r1:~# ping 100.0.0.1
PING 100.0.0.1 (100.0.0.1) 56(84) bytes of data.
From 11.0.0.5 icmp_seq=1 Time to live exceeded
From 11.0.0.5 icmp_seq=2 Time to live exceeded

--- 100.0.0.1 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1010ms

as20r1:~# traceroute 100.0.0.1
traceroute to 100.0.0.1 (100.0.0.1), 64 hops max, 40 byte packets
 1  12.0.0.1 (12.0.0.1)  0 ms  0 ms  0 ms
 2  11.0.0.5 (11.0.0.5)  0 ms  0 ms  0 ms
 3  12.0.0.1 (12.0.0.1)  0 ms  1 ms  8 ms
 4  11.0.0.5 (11.0.0.5)  1 ms  1 ms  0 ms
 5  * 12.0.0.1 (12.0.0.1)  1 ms  1 ms
 6  11.0.0.5 (11.0.0.5)  1 ms  1 ms *
 7  12.0.0.1 (12.0.0.1)  1 ms  1 ms  1 ms
 8  11.0.0.5 (11.0.0.5)  1 ms  1 ms  1 ms
```

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transit as: playing with the backup

- let's check whether things work...

as20r1

```
as20r1:~# ping 100.0.0.1
PING 100.0.0.1 (100.0.0.1) 56(84) bytes of data.
From 11.0.0.5 icmp_seq=1 Time to live exceeded
From 11.0.0.5 icmp_seq=2 Time to live exceeded

--- 100.0.0.1 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1010ms

as20r1:~# traceroute 100.0.0.1
traceroute to 100.0.0.1 (100.0.0.1), 64 hops max, 40 byte packets
 1  12.0.0.1 (12.0.0.1)  0 ms  0 ms  0 ms
 2  11.0.0.5 (11.0.0.5)  0 ms  0 ms  0 ms
 3  12.0.0.1 (12.0.0.1)  0 ms  1 ms  8 ms
 4  11.0.0.5 (11.0.0.5)  1 ms  1 ms  0 ms
 5  * 12.0.0.1 (12.0.0.1)  1 ms  1 ms
 6  11.0.0.5 (11.0.0.5)  1 ms  1 ms *
 7  12.0.0.1 (12.0.0.1)  1 ms  1 ms  1 ms
 8  11.0.0.5 (11.0.0.5)  1 ms  1 ms  1 ms
```

these are symptoms of a forwarding loop!!

transit as: playing with the backup

- same test from **as10rt2**

as10rt2

```
as10rt2:~# ping 100.0.0.1
PING 100.0.0.1 (100.0.0.1) 56(84) bytes of data.
From 11.0.0.5: icmp_seq=2 Redirect Host(New nexthop: 11.0.0.6)
From 11.0.0.5: icmp_seq=3 Redirect Host(New nexthop: 11.0.0.6)
From 11.0.0.5: icmp_seq=5 Redirect Host(New nexthop: 11.0.0.6)

--- 100.0.0.1 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4016ms

as10rt2:~#
```

transit as: playing with the backup

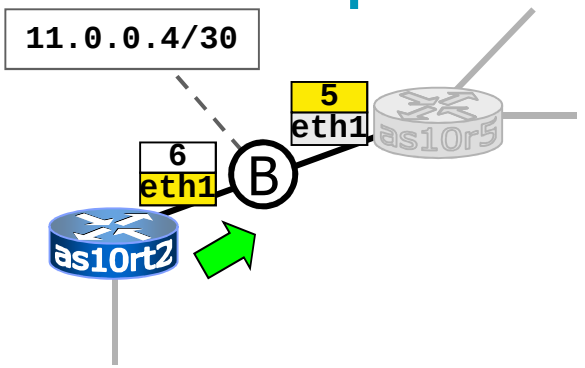
- investigating the loop

as10rt2

```
as10rt2:~# telnet localhost zebra
Trying 127.0.0.1...
Connected to as10rt2.
Escape character is '^]'.
...
Router> show ip route 100.0.0.0/8
Routing entry for 100.0.0.0/8
  Known via "bgp", distance 200, metric 0, best
  Last update 02:49:04 ago
  * 12.0.0.6, recursive via 11.0.0.5, eth1
```

ebgp next hop

actual next hop (obtained by recursive lookup)

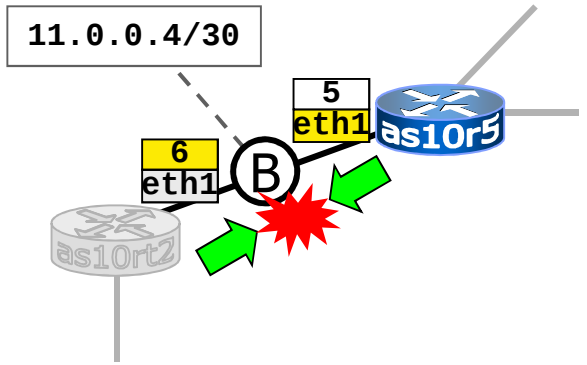


transit as: playing with the backup

- investigating the loop

as10r5

```
as10r5:~# telnet localhost zebra
Trying 127.0.0.1...
Connected to as10r5.
Escape character is '^]'.
...
Router> show ip route 100.0.0.0/8
Routing entry for 100.0.0.0/8
  Known via "rip", distance 120, metric 2, best
  Last update 02:50:02 ago
  * 11.0.0.6, via eth1
```



transit as: playing with the backup

- motivating the loop

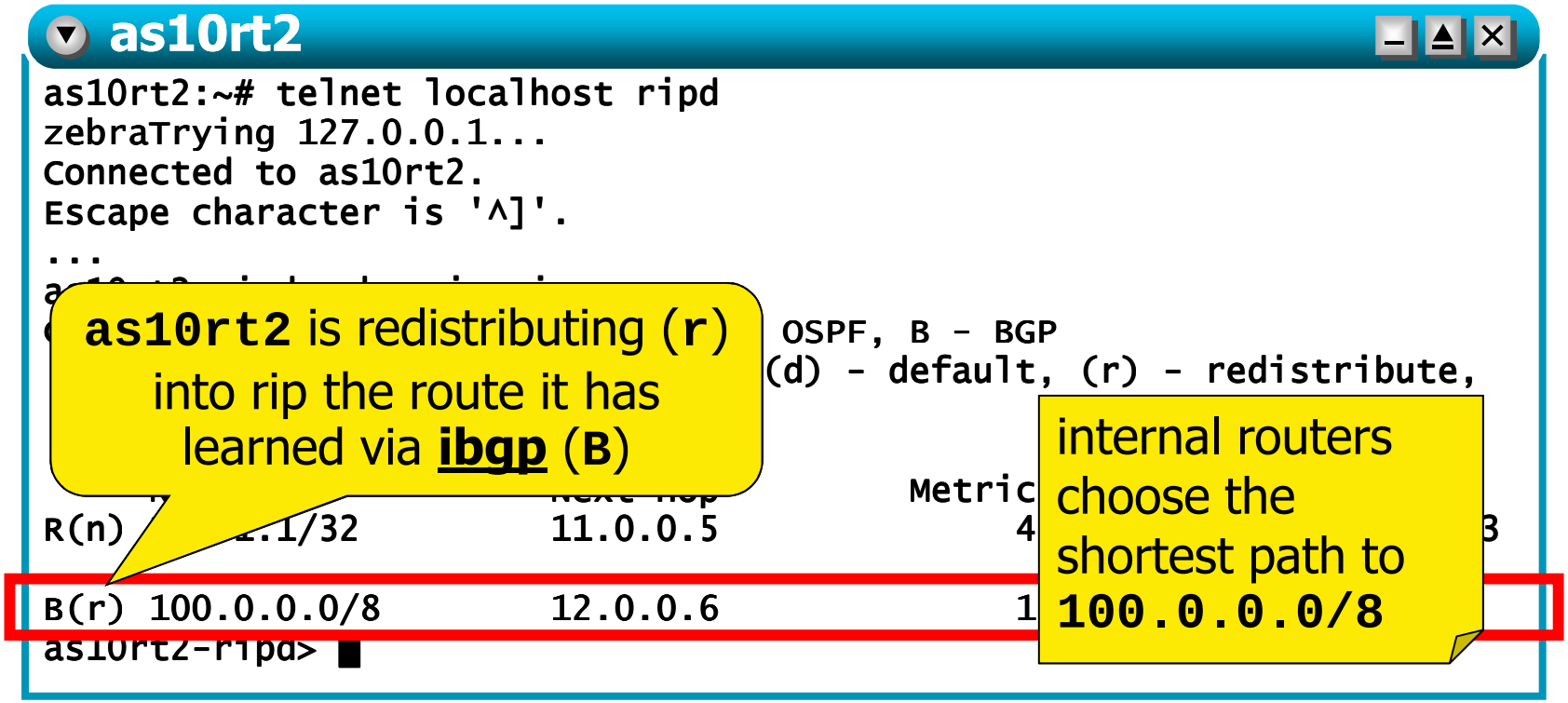
as10rt2

```
as10rt2:~# telnet localhost ripd
zebraTrying 127.0.0.1...
Connected to as10rt2.
Escape character is '^]'.
...
as10rt2-ripd> show ip rip
Codes: R - RIP, C - connected, O - OSPF, B - BGP
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

     Network            Next Hop         Metric From            Time
R(n) 1.1.1.1/32         11.0.0.5              4 11.0.0.5        02:43
...
B(r) 100.0.0.0/8        12.0.0.6              1 self
as10rt2-ripd>
```

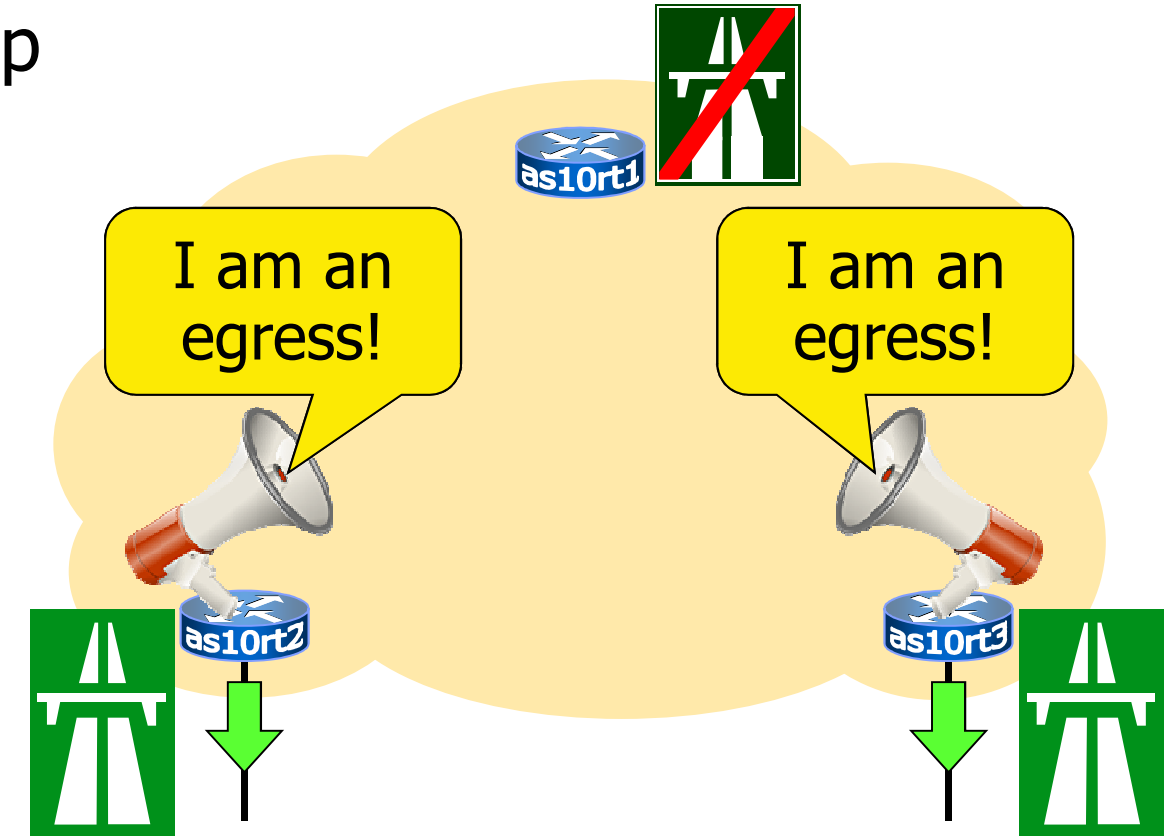
transit as: playing with the backup

- motivating the loop



transit as: playing with the backup

- how to fix?
  - tell rip not to inject information learned by ibgp



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transit as: playing with the backup

- how to fix?
  - tell rip not to inject information learned by ibgp
    - cisco (and juniper) say:



By default, iBGP redistribution into IGP is disabled. To enable redistribution of iBGP routes into IGP, issue the **`bgp redistribute-internal`** *command. Precautions should be taken to redistribute specific routes using route maps into IGP.*

Note: Redistributing internal Border Gateway Protocol (iBGP) routes into an Interior Gateway Protocol may cause routing loops within the Autonomous System (AS). This is not recommended. Route filters should be set to control the information which is imported into the IGP.

https://supportforums.cisco.com/document/8621/unable-redistribute-ibgp-learnt-routes-igp-such-eigrp-ospf-and

transit as: playing with the backup

- how to fix?
  - tell rip not to inject information learned by ibgp
- how to tell?
  - no way in zebra to say "redistribute ebgp" but...
  - ...**`route-map`**s can be applied on redistributed routes

transit as: playing with the backup

- fixing the loop

as10rt2

```
as10rt2:~# telnet localhost ripd
Trying 127.0.0.1...
Connected to as10rt2.
Escape character is '^]'.
...
as10rt2-ripd> enable
as10rt2-ripd# configure terminal
as10rt2-ripd(config)# ip prefix-list myNeighbors permit 12.0.0.0/30 le 32
as10rt2-ripd(config)# route-map eBGP permit 10
as10rt2-ripd(config-route-map)# match ip next-hop prefix-list myNeighbors
as10rt2-ripd(config-route-map)# exit
as10rt2-ripd(config)# router rip
as10rt2-ripd(config-router)# no redistribute bgp
as10rt2-ripd(config-router)# redistribute bgp route-map eBGP
as10rt2-ripd(config-router)# write file
```

  - restart zebra by typing **`/etc/init.d/zebra restart`**

transit as: playing with the backup

- fixing the loop

match all the more specifics of the **12.0.0.0/30** network (next-hops are single ip addresses)

as10rt2

```
as10rt2:~# telnet localhost ripd
Trying 127.0.0.1...
Connected to as10rt2.
Escape character is '^]'.
...
as10rt2-ripd> enable
as10rt2-ripd# configure terminal
as10rt2-ripd(config)# ip prefix-list myNeighbors permit 12.0.0.0/30 le 32
as10rt2-ripd(config)# route-map eBGP permit 10
as10rt2-ripd(config-route-map)# match ip next-hop prefix-list myNeighbors
as10rt2-ripd(config-route-map)# exit
as10rt2-ripd(config)# router rip
as10rt2-ripd(config-router)# no redistribute bgp
as10rt2-ripd(config-router)# redistribute bgp route-map eBGP
as10rt2-ripd(config-router)# write file
```

- restart zebra by typing **`/etc/init.d/zebra restart`**

transit as: playing with the backup

- fixing the loop

as10rt2

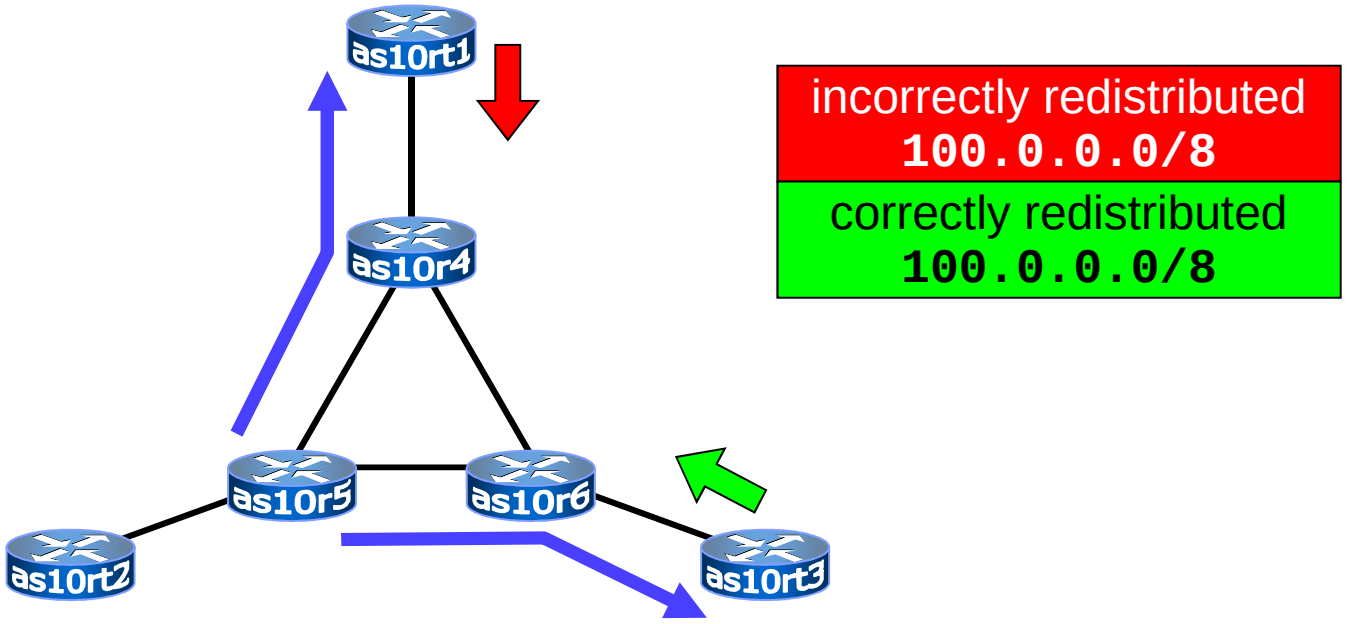
```
as10rt2:~# telnet localhost ripd
Trying 127.0.0.1...
Connected to as10rt2.
Escape character is '^]'.
...
as10rt2-ripd> enable
as10rt2-ripd# configure terminal
as10rt2-ripd(config)# ip prefix-list myNeighbors permit 12.0.0.0/30 le 32
as10rt2-ripd(config)# route-map eBGP permit 10
as10rt2-ripd(config-route-map)# match ip next-hop prefix-list myNeighbors
as10rt2-ripd(config-route-map)# exit
as10rt2-ripd(config)# router rip
as10rt2-ripd(config-router)# no redistribute bgp
as10rt2-ripd(config-router)# redistribute bgp route-map eBGP
as10rt2-ripd(config-router)# write file
```

this is the BGP next-hop (i.e., *before* the recursive lookup)

- restart zebra by typing **`/etc/init.d/zebra restart`**

transit as: playing with the backup

- fixing the loop
  - we need to fix **`as10rt1`** as well, because **`as10r5`** may still be offered two equally long paths among which to choose



transit as: playing with the backup

- fixing the loop

as10rt1

```
as10rt1:~# telnet localhost ripd
Trying 127.0.0.1...
Connected to as10rt1.
Escape character is '^]'.
...
as10rt1-ripd> enable
as10rt1-ripd# configure terminal
as10rt1-ripd(config)# route-map eBGP deny 10
as10rt1-ripd(config-route-map)# exit
as10rt1-ripd(config)# router rip
as10rt1-ripd(config-router)# no redistribute bgp
as10rt1-ripd(config-router)# redistribute bgp route-map eBGP
as10rt1-ripd(config-router)# write file
```

as10rt1 has no ebgp peering, so it should simply avoid redistributing anything into rip

- restart zebra by typing **/etc/init.d/zebra restart**

transit as: playing with the backup

- fixing the loop

as10rt1

```
as10rt1:~# telnet localhost ripd
Trying 127.0.0.1...
Connected to as10rt1.
Escape character is '^]'.
...
as10rt1-ripd> enable
as10rt1-ripd# configure terminal
as10rt1-ripd(config)# route-map eBGP deny 10
as10rt1-ripd(config-route-map)# exit
as10rt1-ripd(config)# router
as10rt1-ripd(config-router)#
as10rt1-ripd(config-router)#
as10rt1-ripd(config-router)#
```



notice: a similar configuration should be applied to **`as10rt3`**; however, this is enough for the purposes of this lab

- restart zebra by typing **`/etc/init.d/zebra restart`**

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transit as: playing with the backup

- after some time...

as10rt2

```
as10rt2:~# telnet localhost ripd
Trying 127.0.0.1...
Connected to as10rt2.
Escape character is '^]'.
...
as10rt2-ripd> show ip rip
Codes: R - RIP, C - connected, O - OSPF, B - BGP
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

     Network            Next Hop         Metric From            Time
R(n) 1.1.1.1/32         11.0.0.5              4 11.0.0.5        02:47
...
R(n) 30.0.0.0/8         11.0.0.5              4 11.0.0.5        02:46
B(r) 100.0.0.0/8        12.0.0.2              1 self
as10rt2-ripd>
```

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transit as: playing with the backup

- after some time...

as10rt2

```
as10rt2:~# telnet localhost ripd
Trying 127.0.0.1...
Connected
Escape ch
...
as10rt2-r
Codes: R                                          P
       (n)                                        , (r) - redistribute,
       (i)

     Network          Next Hop         Metric From              Time
R(n) 1.1.1.2          11.0.0.5              4 11.0.0.5          02:47
...
R(n) 30.0.0.0/8       11.0.0.5              4 11.0.0.5          02:46
B(r) 100.0.0.0/8      12.0.0.2              1 self
as10rt2-ripd>
```

the route may still marked as "redistributed" (**r**) because a **redistribute bgp** command still exists but...

transit as: playing with the backup

- after some time...

as10r5

```
as10r5:~# telnet localhost zebra
Trying 127.0.0.1...
Connected to as10r5.
Escape character is '^]'.
...
Router> show ip route 100.0.0.0/8
Routing entry for 100.0.0.0/8
  Known via "rip", distance 120, metric 3, best
  Last update 00:04:55 ago
  * 10.0.0.6, via eth2

Router>
```

...**as10r5** is no longer using **11.0.0.6** (**as10rt2**) as next hop

transit as: playing with the backup

- fault recovered!

as20r1

```
as20r1:~# traceroute 100.0.0.1
traceroute to 100.0.0.1 (100.0.0.1), 64 hops max, 40 byte packets
 1  12.0.0.1 (12.0.0.1)  1 ms  1 ms  1 ms
 2  11.0.0.5 (11.0.0.5)  1 ms  2 ms  1 ms
 3  10.0.0.6 (10.0.0.6)  2 ms  2 ms  2 ms
 4  11.0.0.10 (11.0.0.10)  2 ms  2 ms  3 ms
 5  12.0.0.6 (12.0.0.6)  2 ms  3 ms  2 ms
 6  100.0.0.1 (100.0.0.1)  3 ms  3 ms  3 ms
as20r1:~#
```

- observation: the same traceroute from **as10rt2** would fail because the source address **11.0.0.6** cannot be reached from outside the transit as

transit as: timings

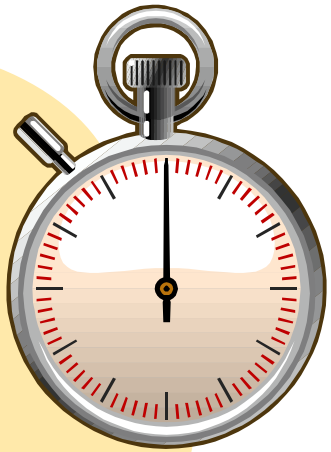
- why is zebra propagating ibgp routes?
  - a.k.a. "going deep into the cause of the forwarding loop"
- the cause is to be sought in the bgp redistribution into rip
  - once a bgp learned route has been installed in the rip routing table, no more rip alternatives for that route are accepted by zebra
- this has some consequences

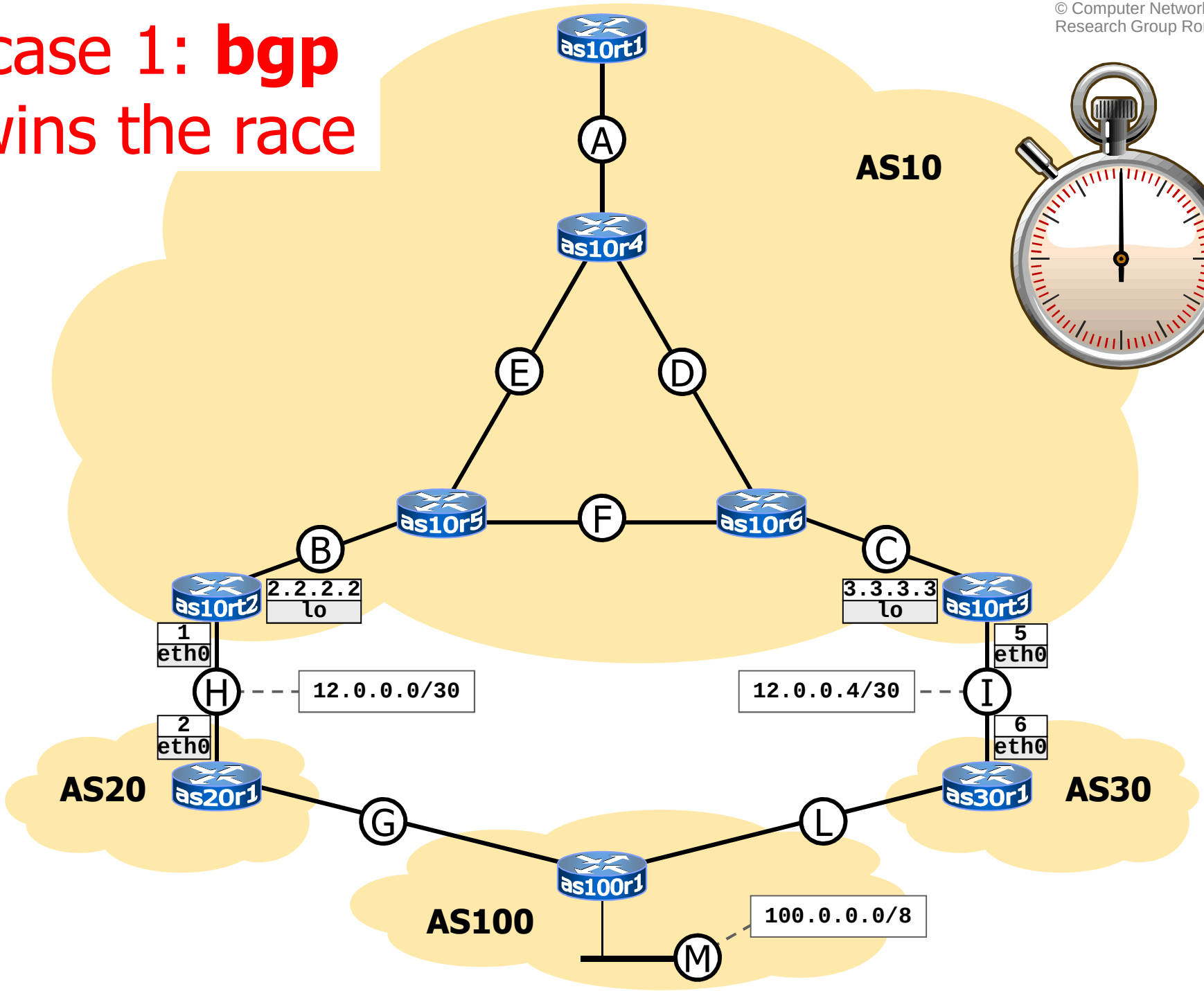
transit as: timings

- we now observe in detail the effect of bgp redistribution
- backtrack to the condition causing the forwarding loop
  - restart the lab
  - bring link **H** down
- timings play a crucial role in triggering the forwarding loop

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case 1: **bgp** wins the race

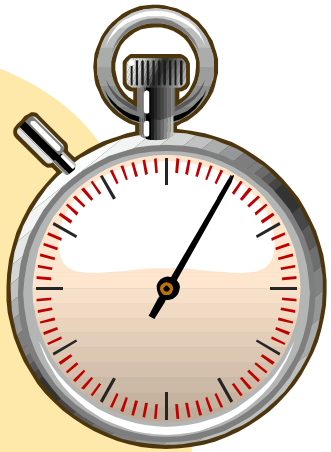


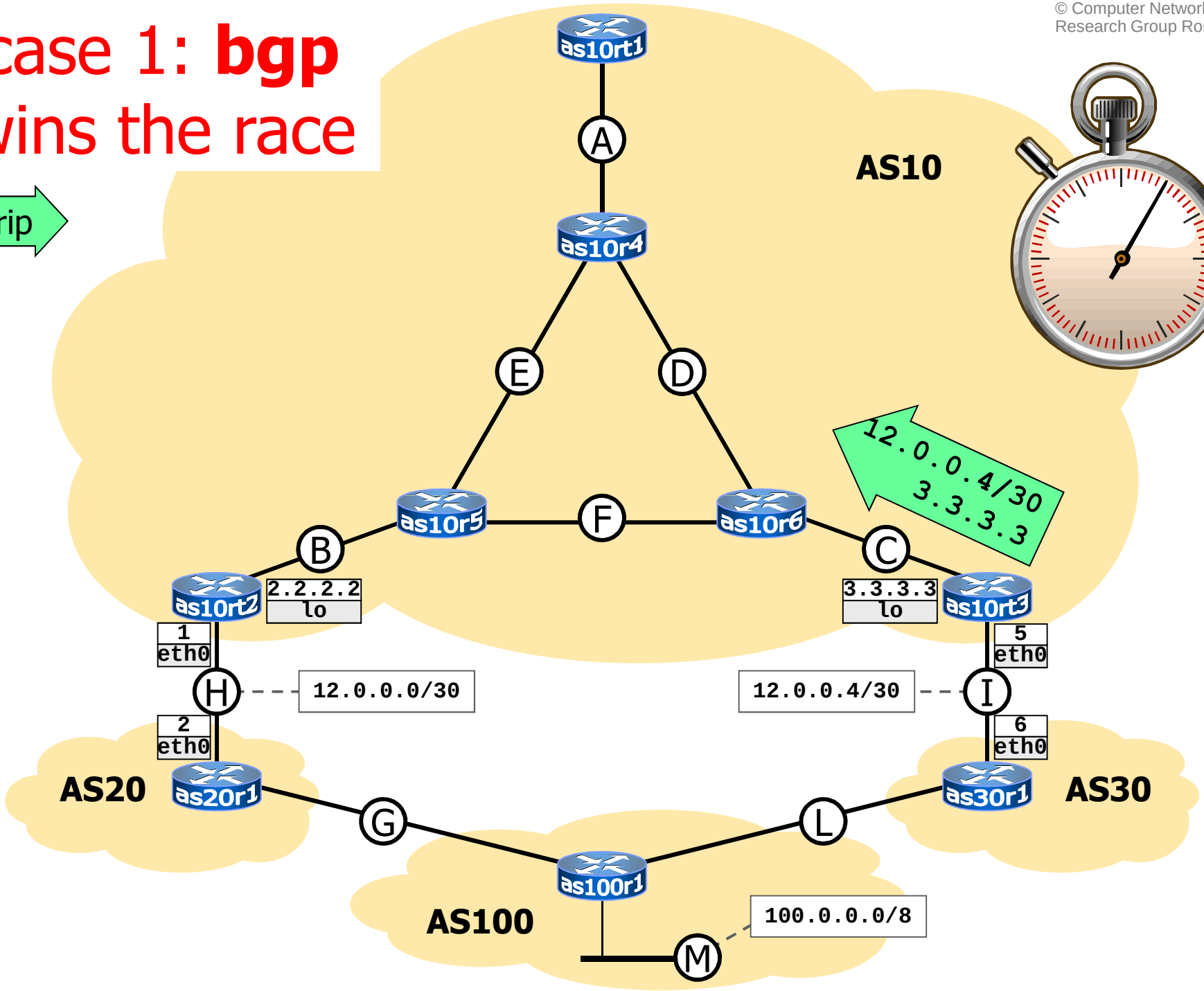


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case 1: **bgp** wins the race



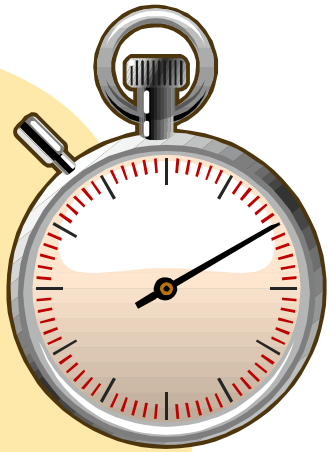


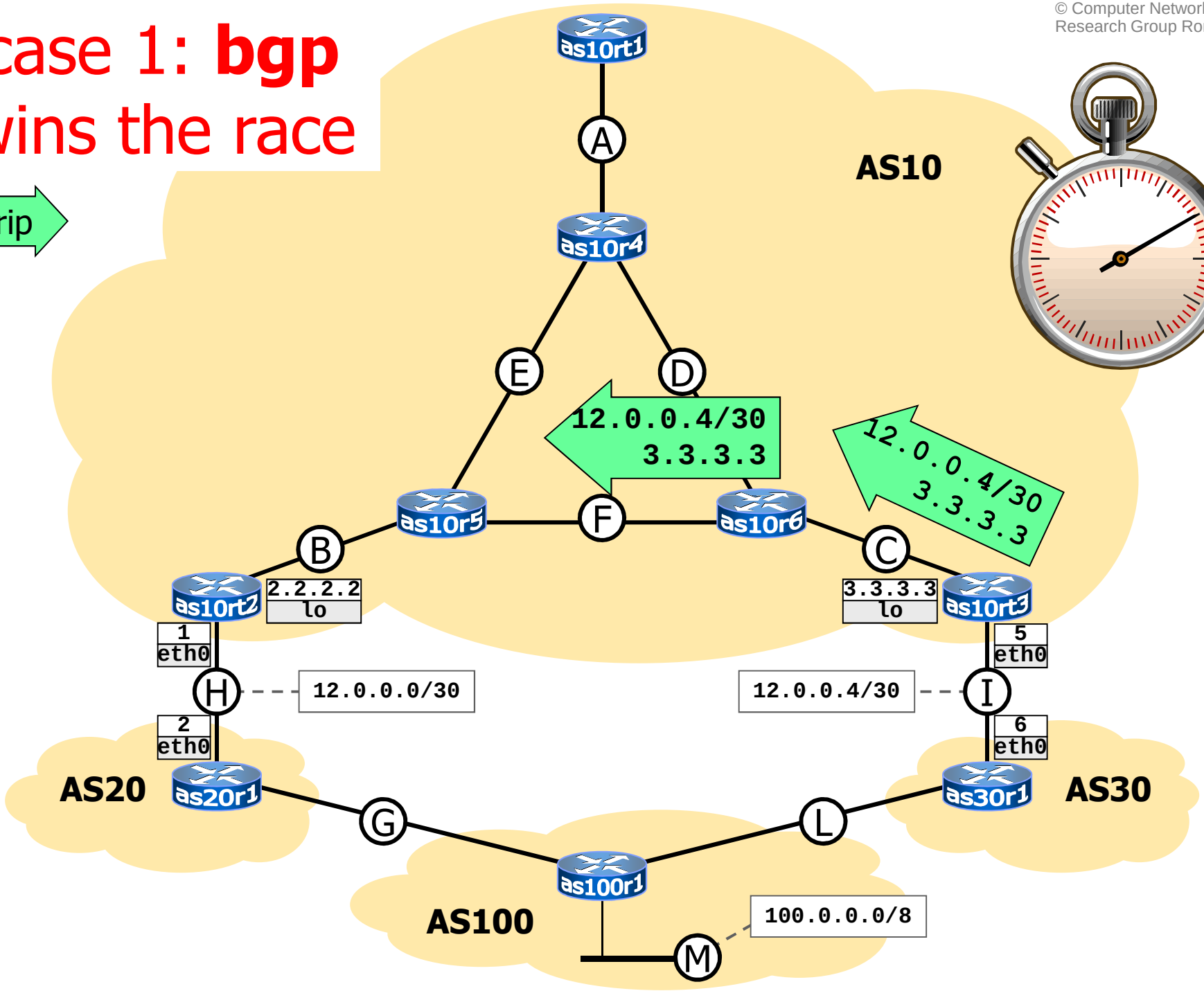


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case 1: **bgp** wins the race



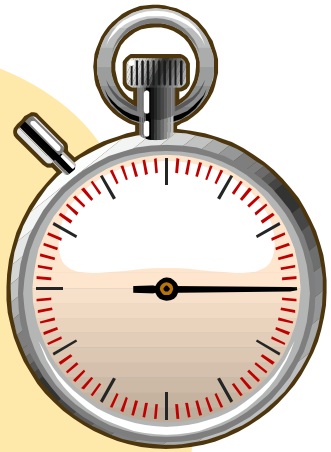


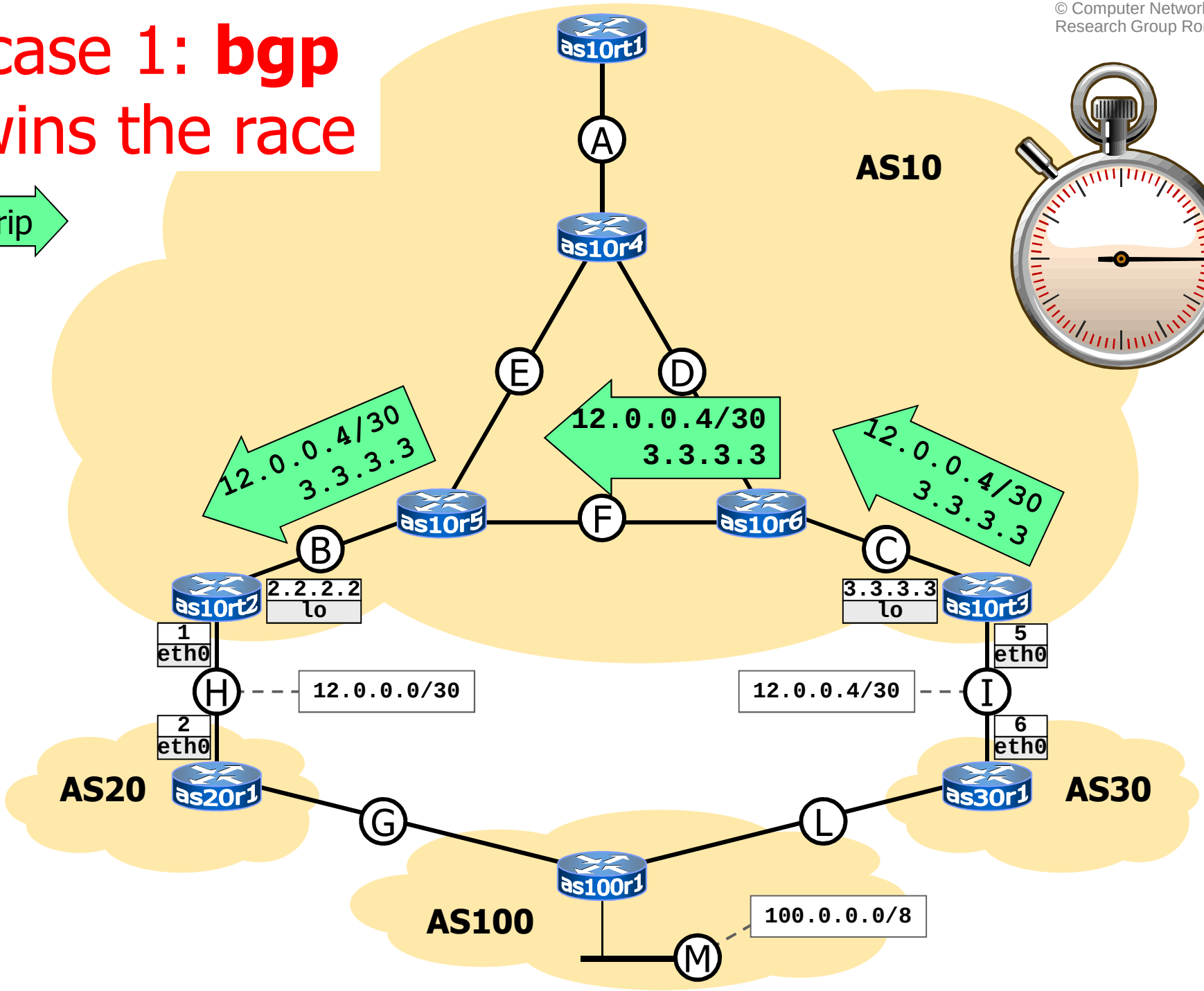


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case 1: **bgp** wins the race



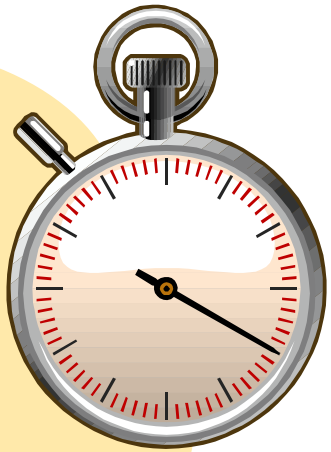


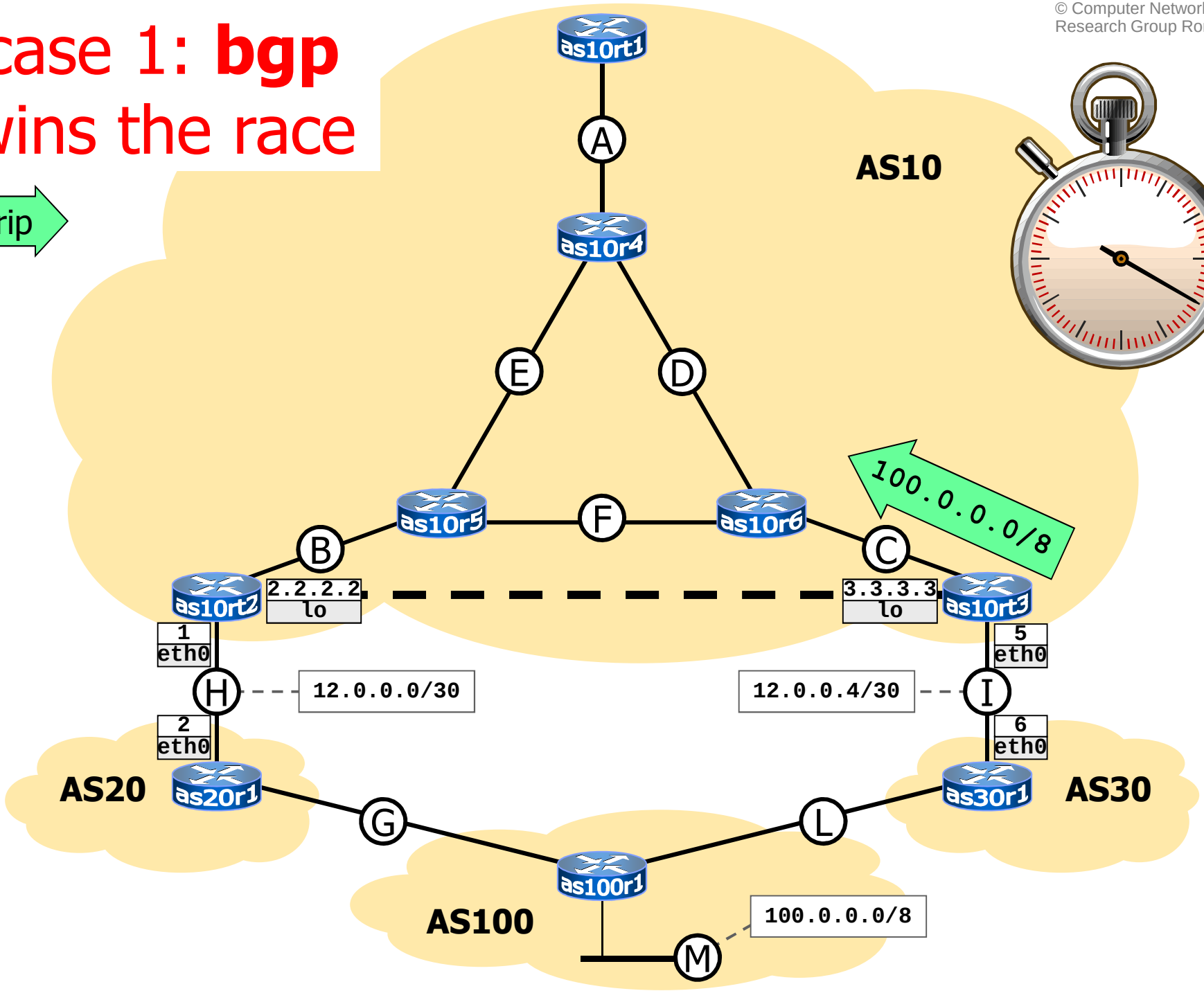


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case 1: **bgp** wins the race

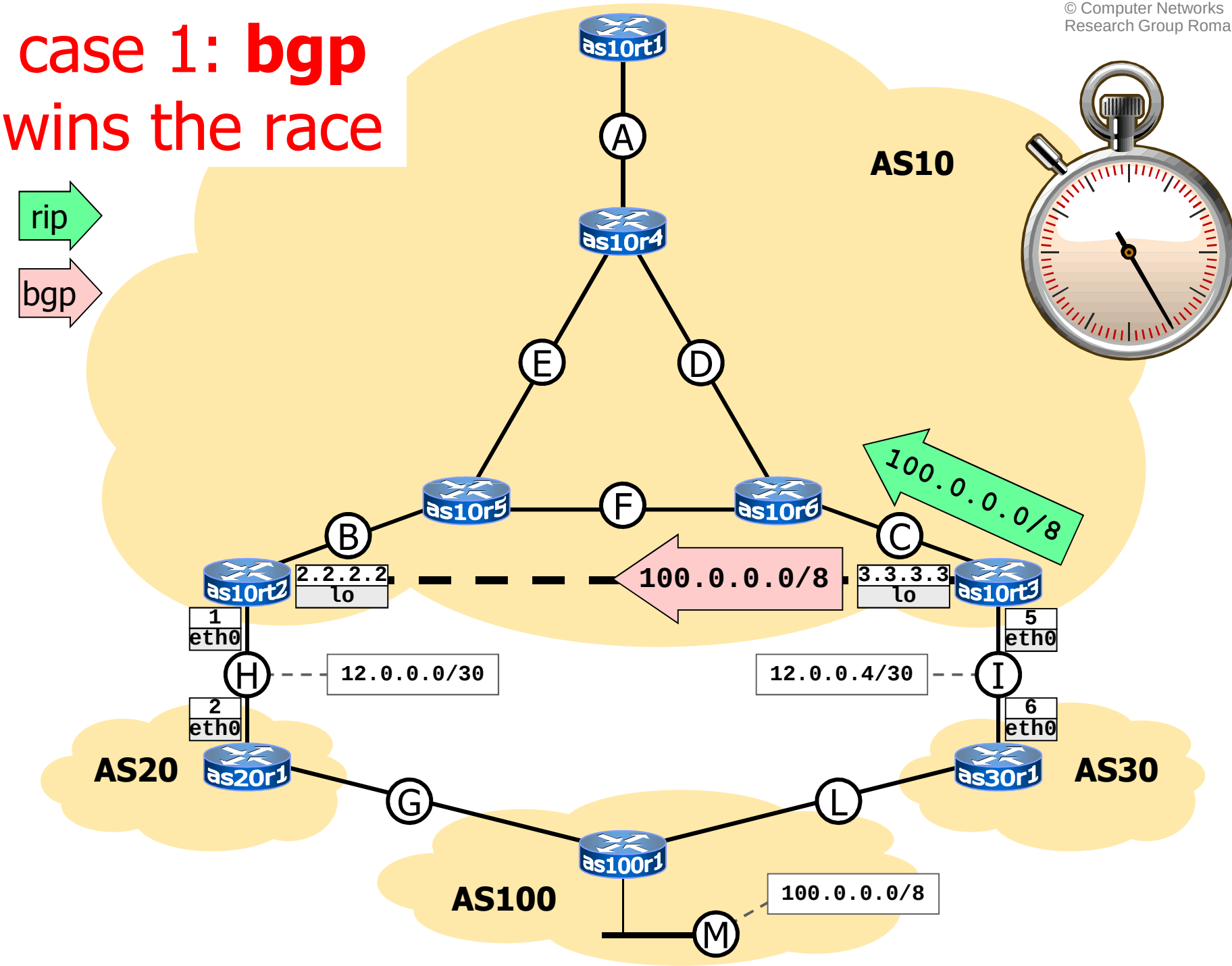






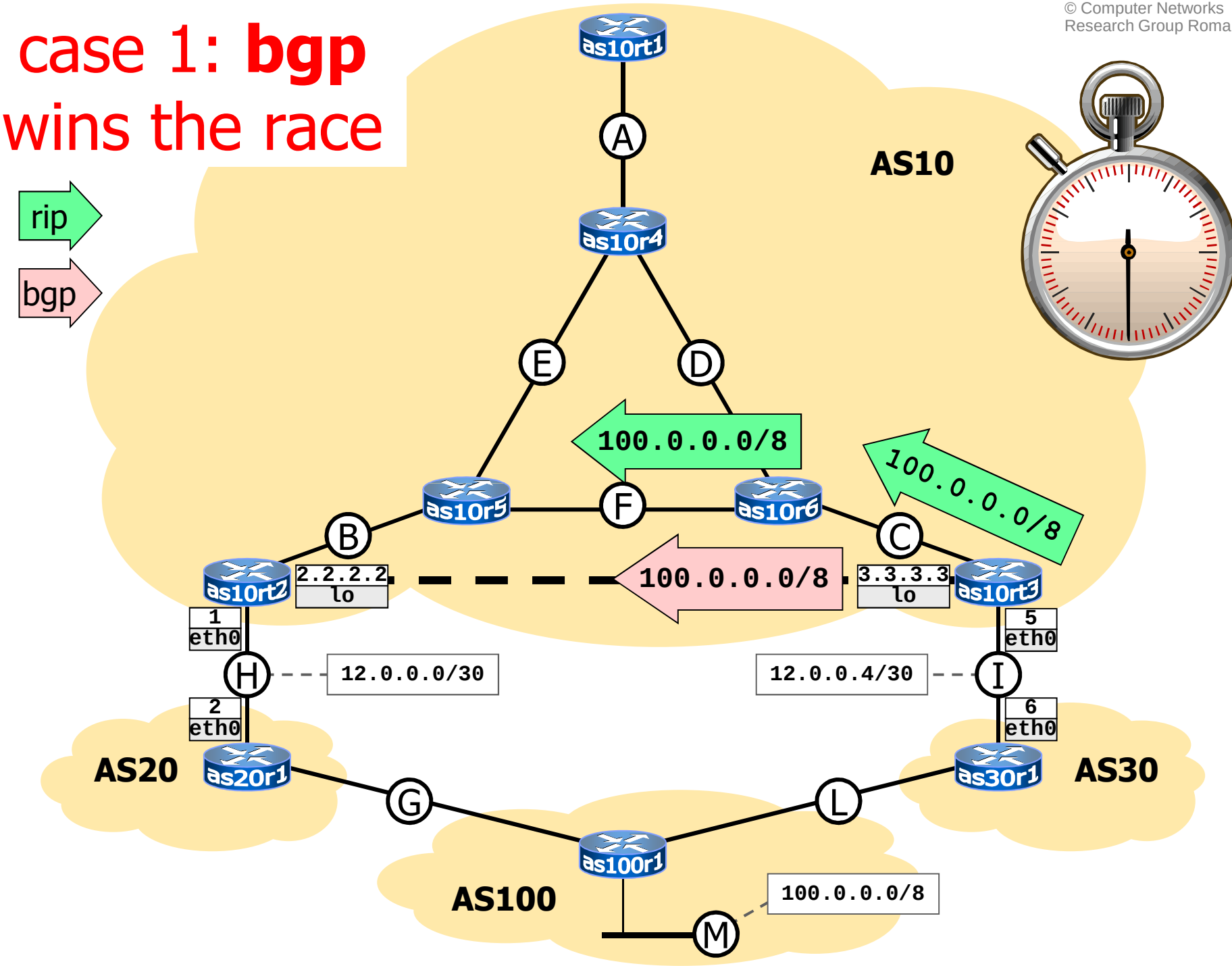
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case 1: **bgp** wins the race



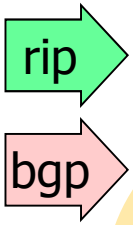
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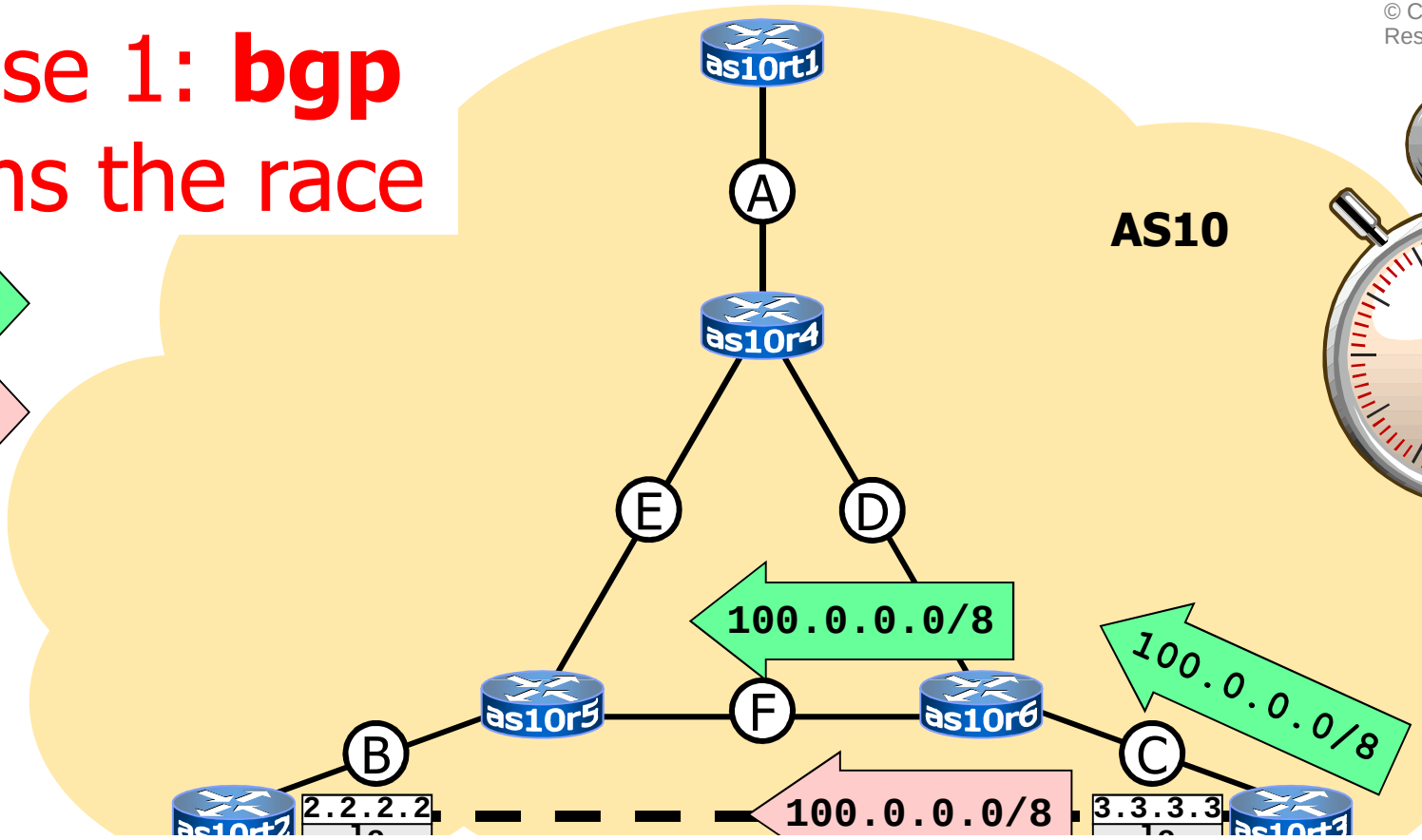
case 1: **bgp** wins the race

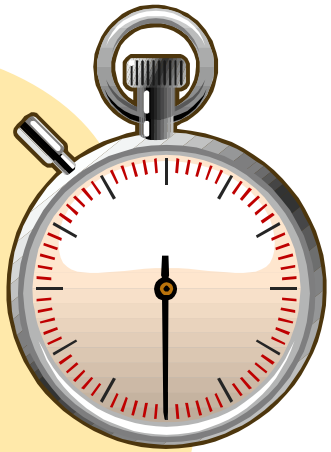


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case 1: **bgp** wins the race







AS

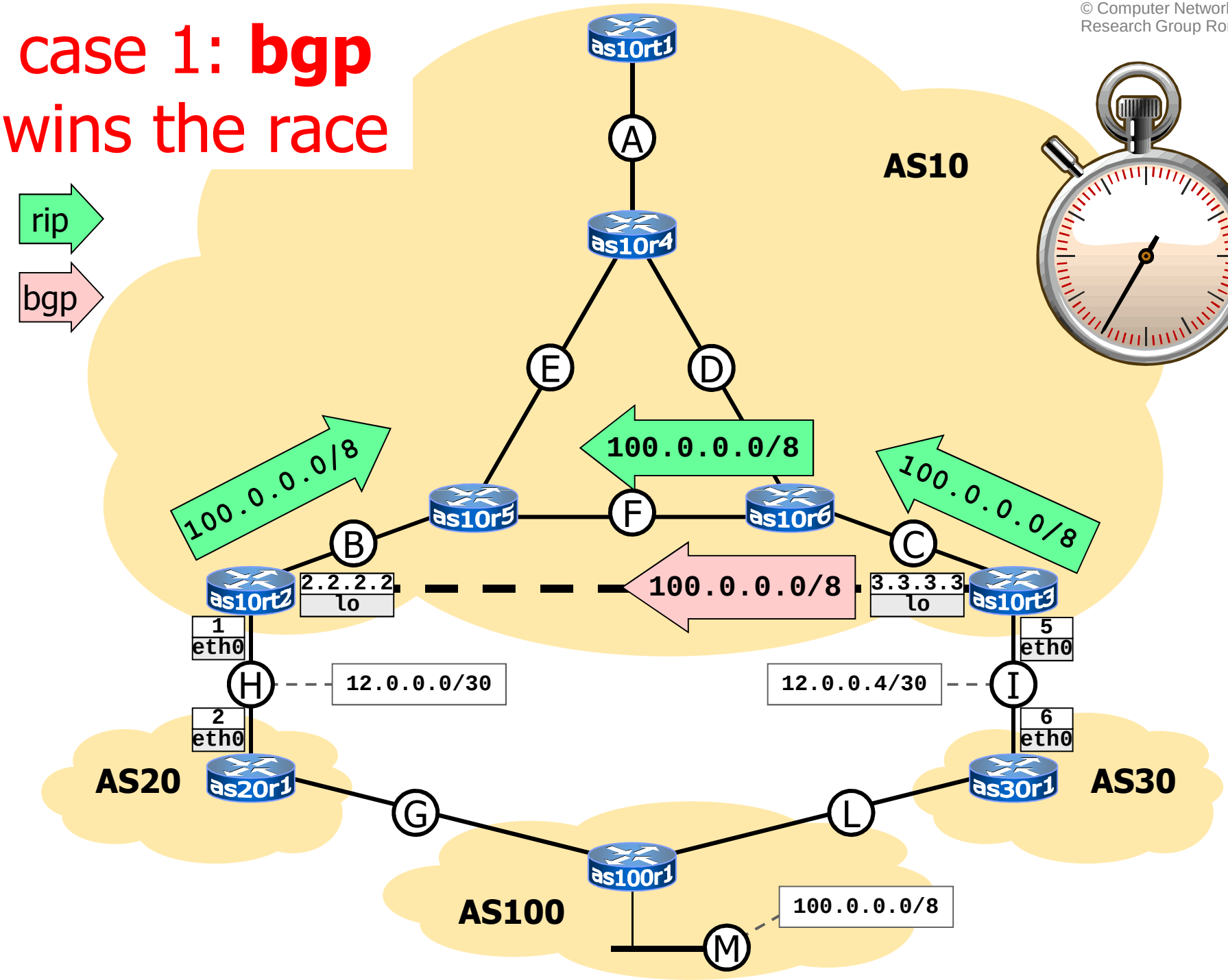
as10rt2

```
as10rt2:~# telnet localhost zebra
...
Router> show ip route 100.0.0.0/8
Routing entry for 100.0.0.0/8
  Known via "bgp", distance 200, metric 0, best
  Last update 02:39:10 ago
  * 12.0.0.6, recursive via 11.0.0.5, eth1

Router>
```

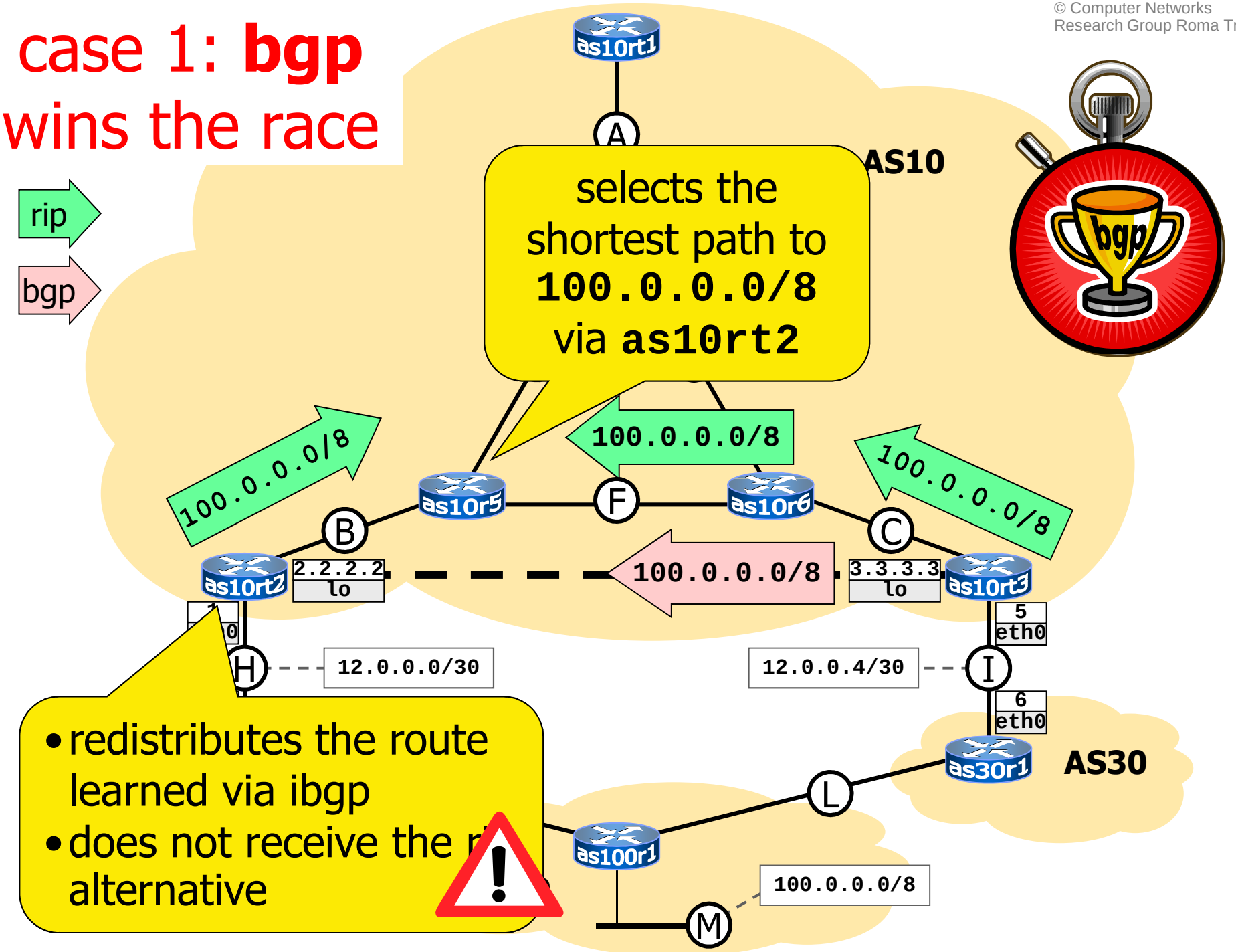
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case 1: **bgp** wins the race



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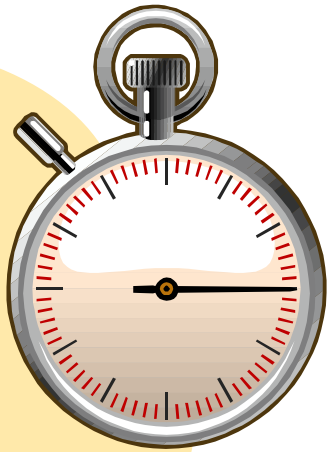
case 1: **bgp** wins the race

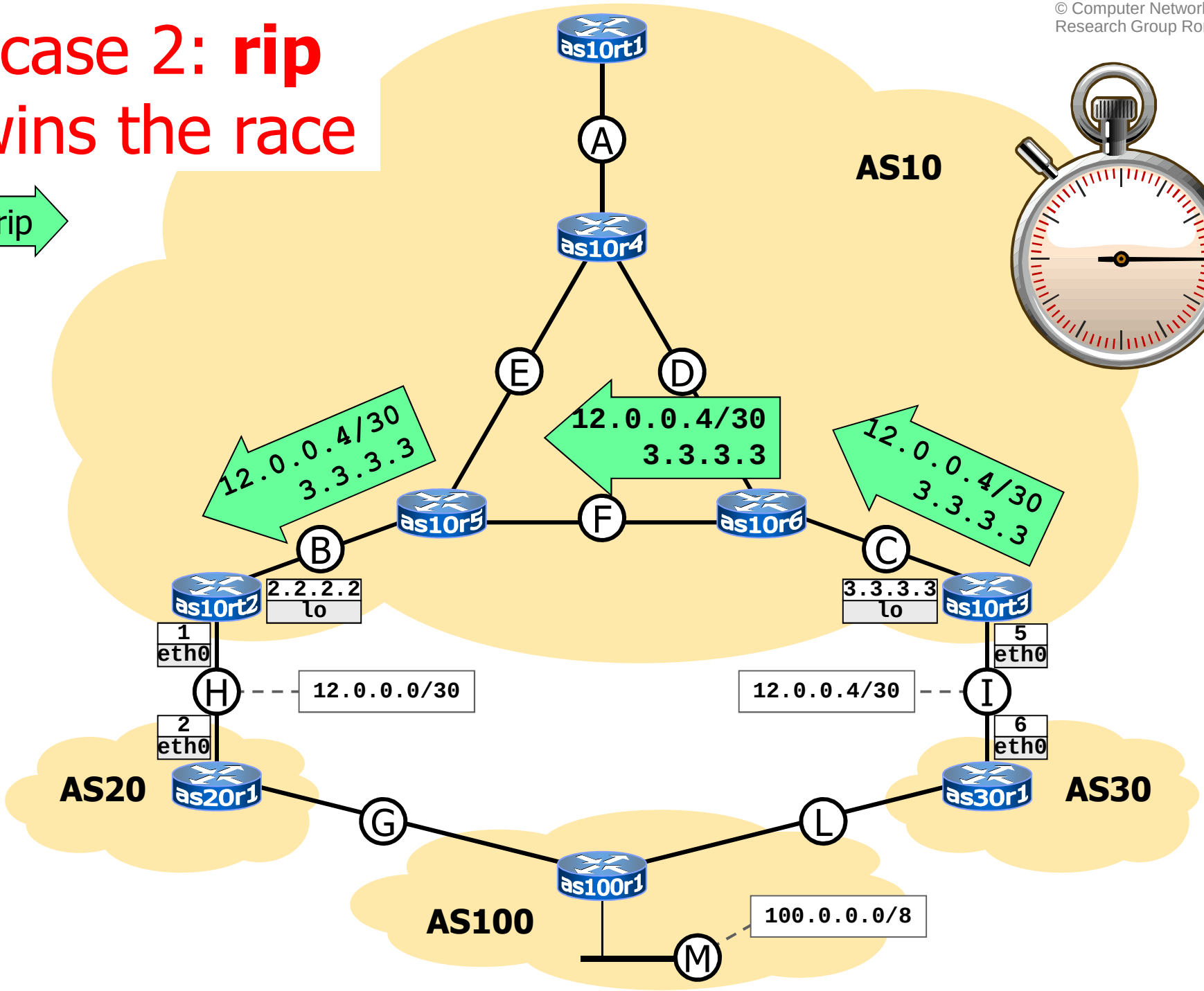


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case 2: **rip** wins the race



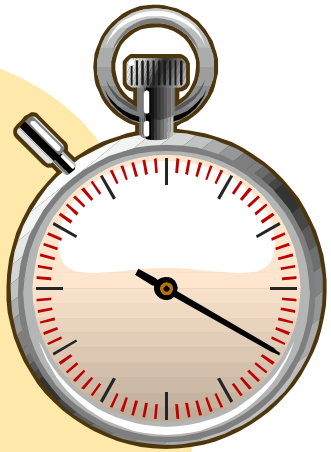


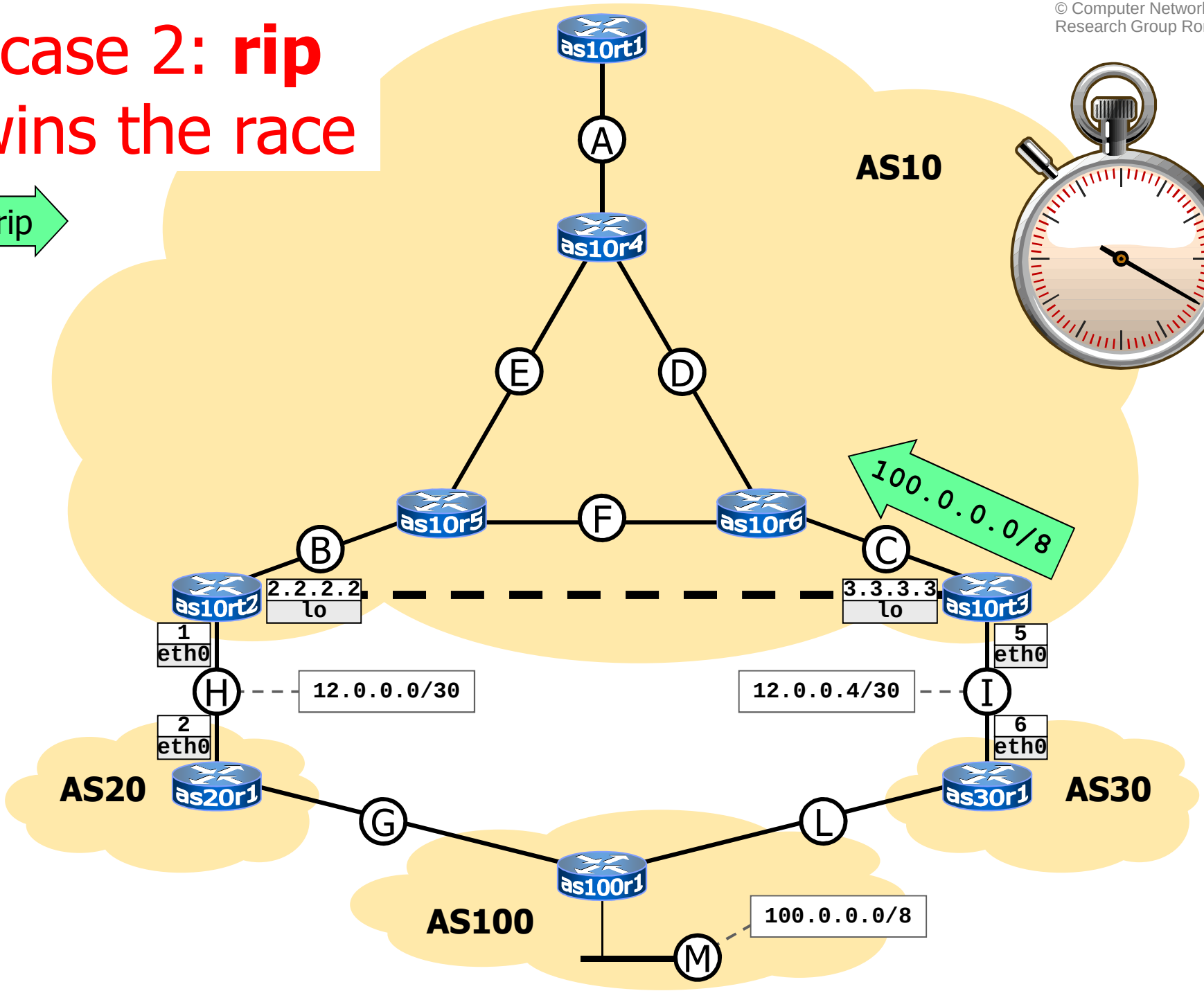


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case 2: **rip** wins the race



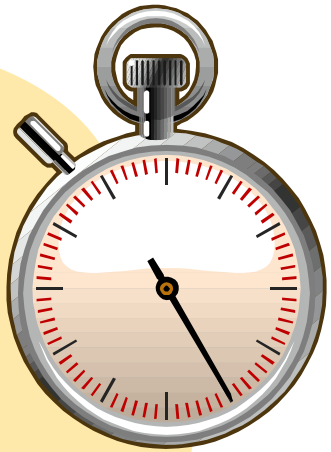


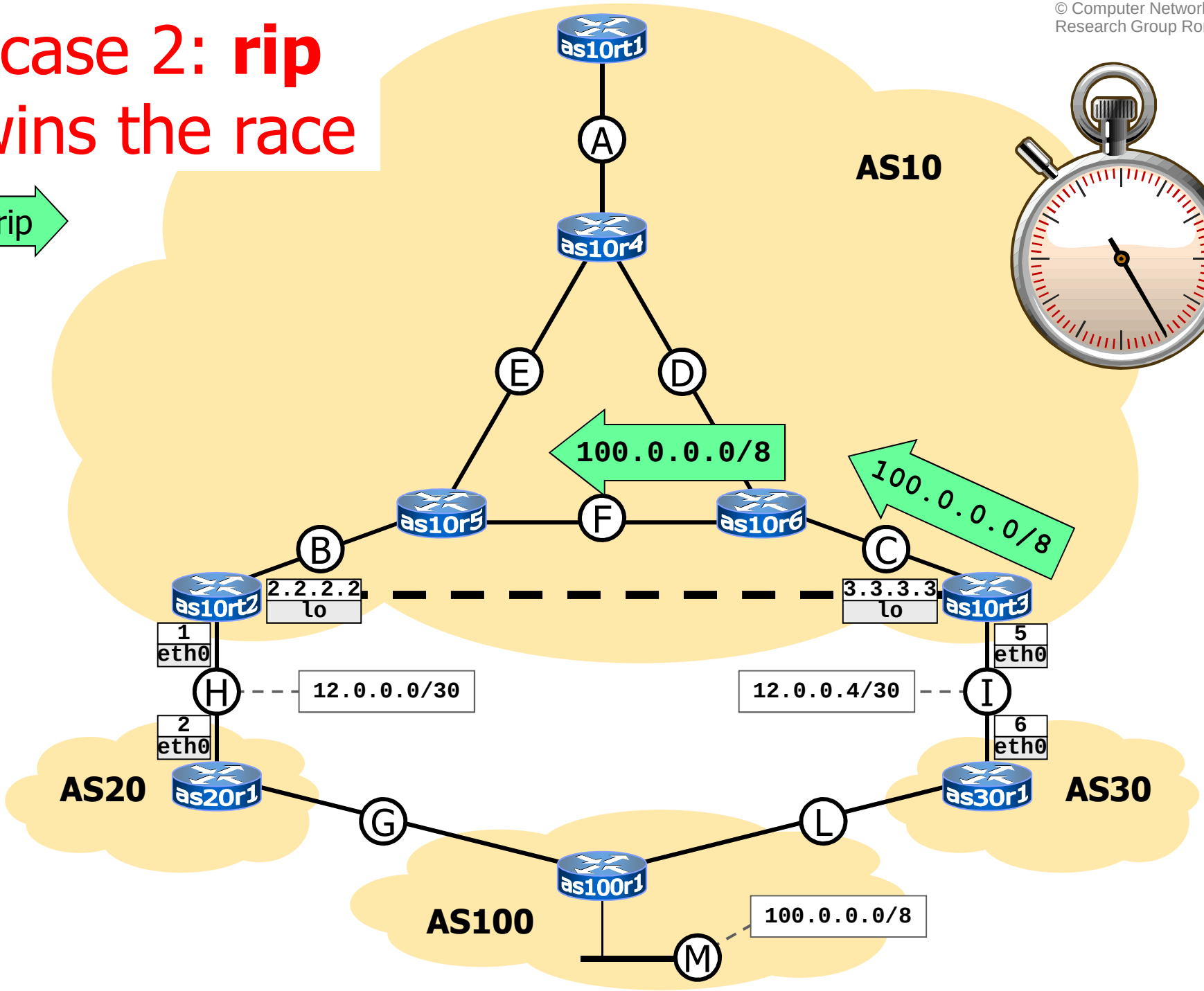


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case 2: **rip** wins the race



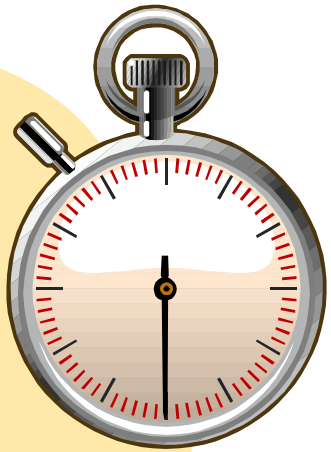


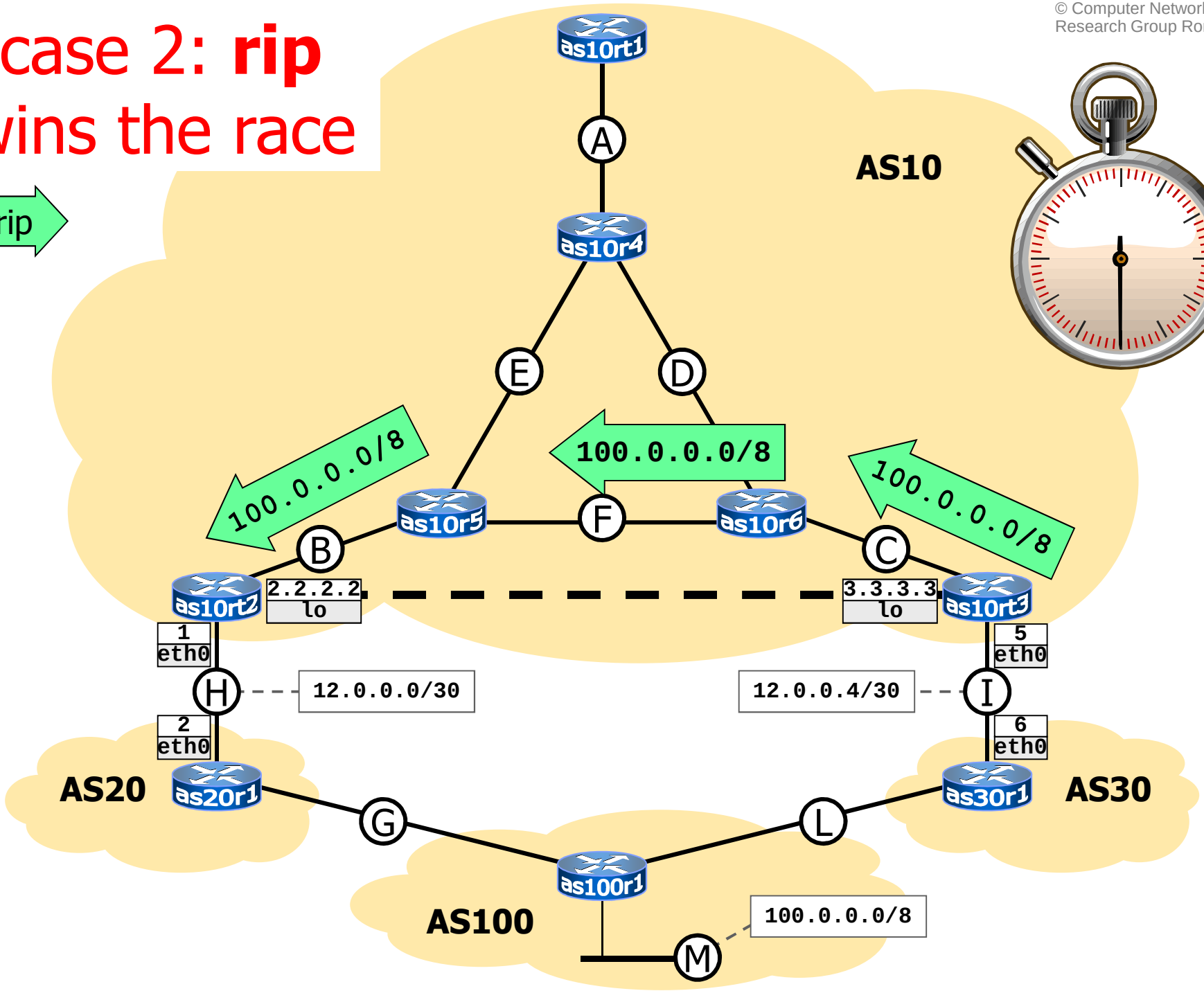


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case 2: **rip** wins the race

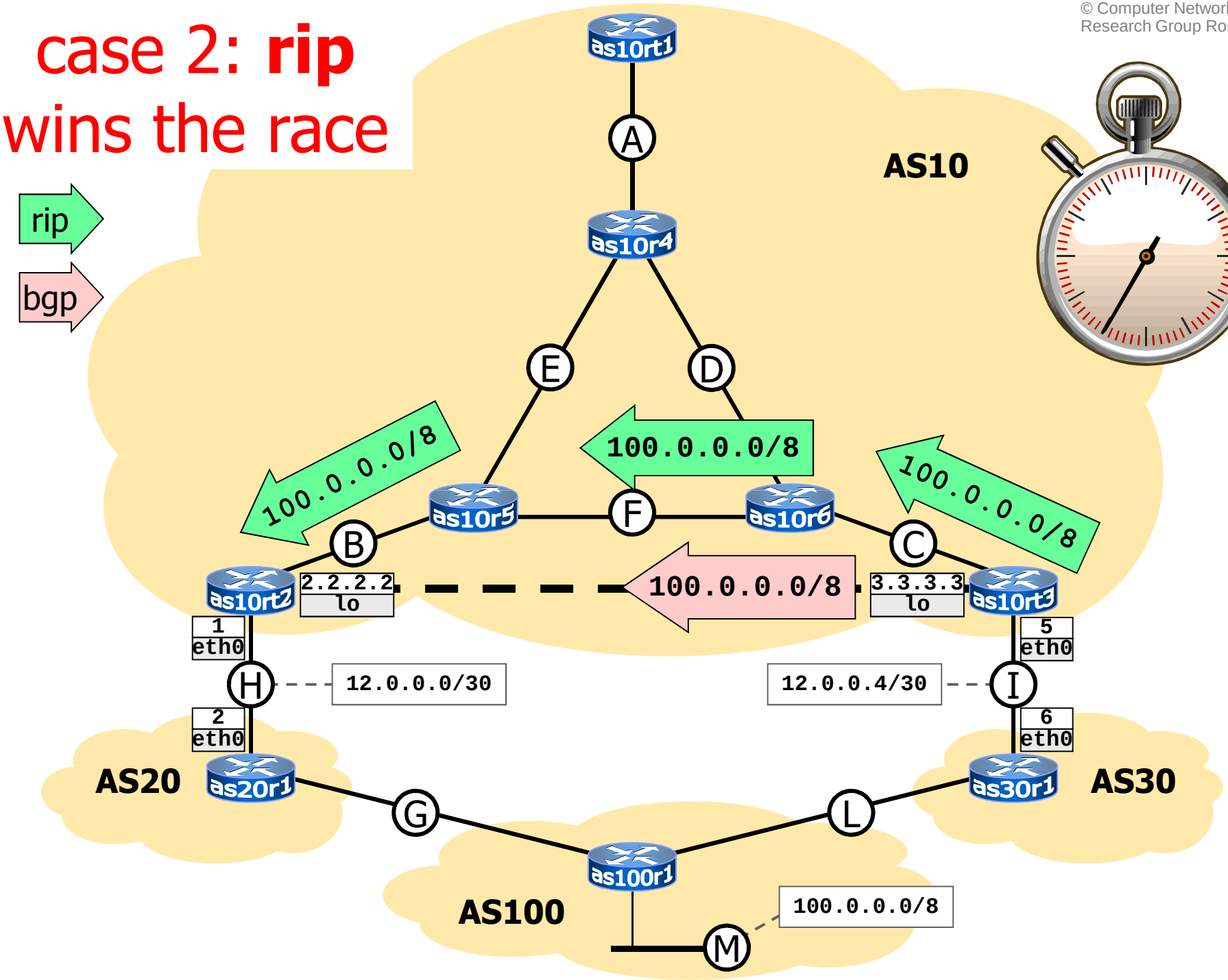






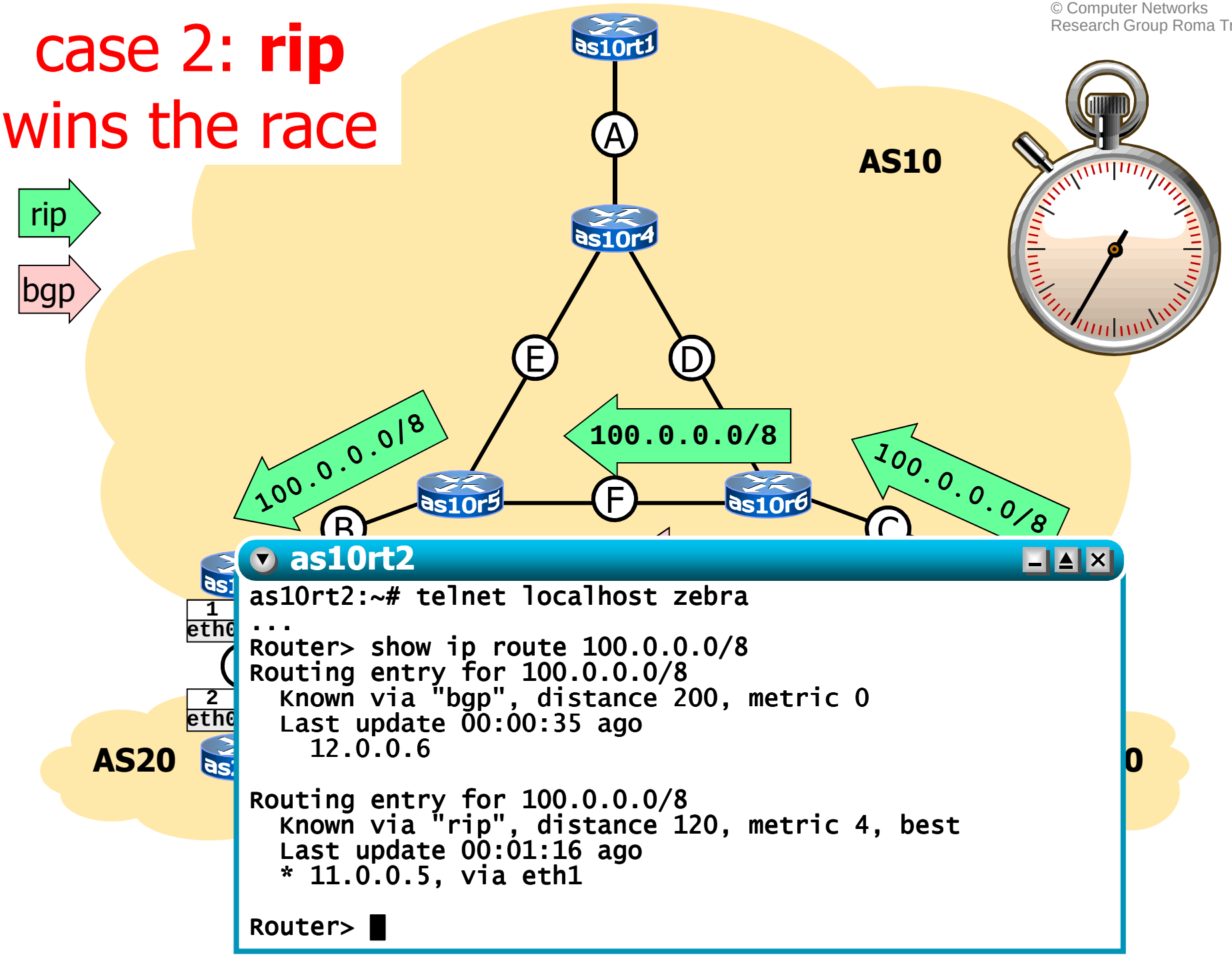
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case 2: **rip** wins the race



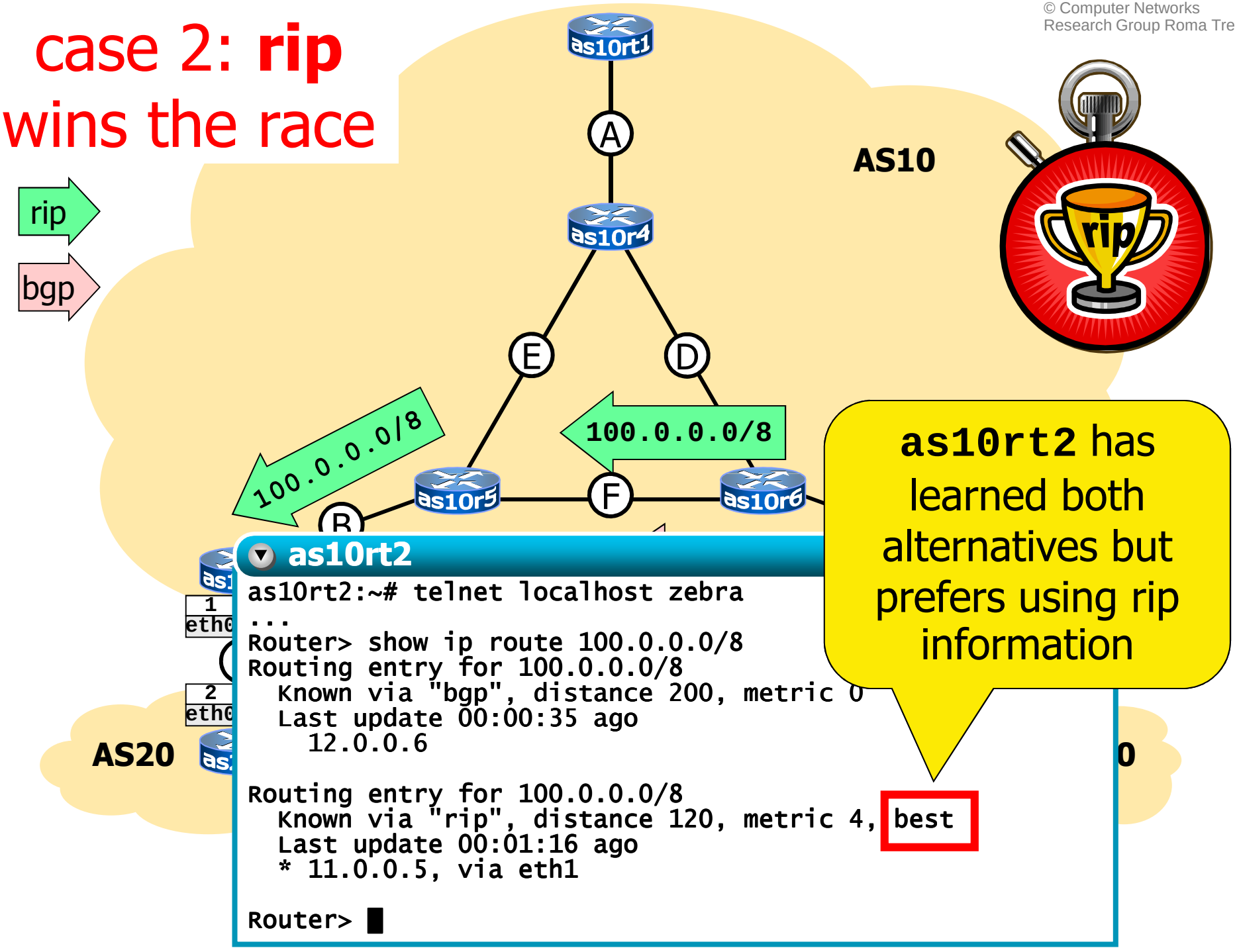
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case 2: **rip** wins the race



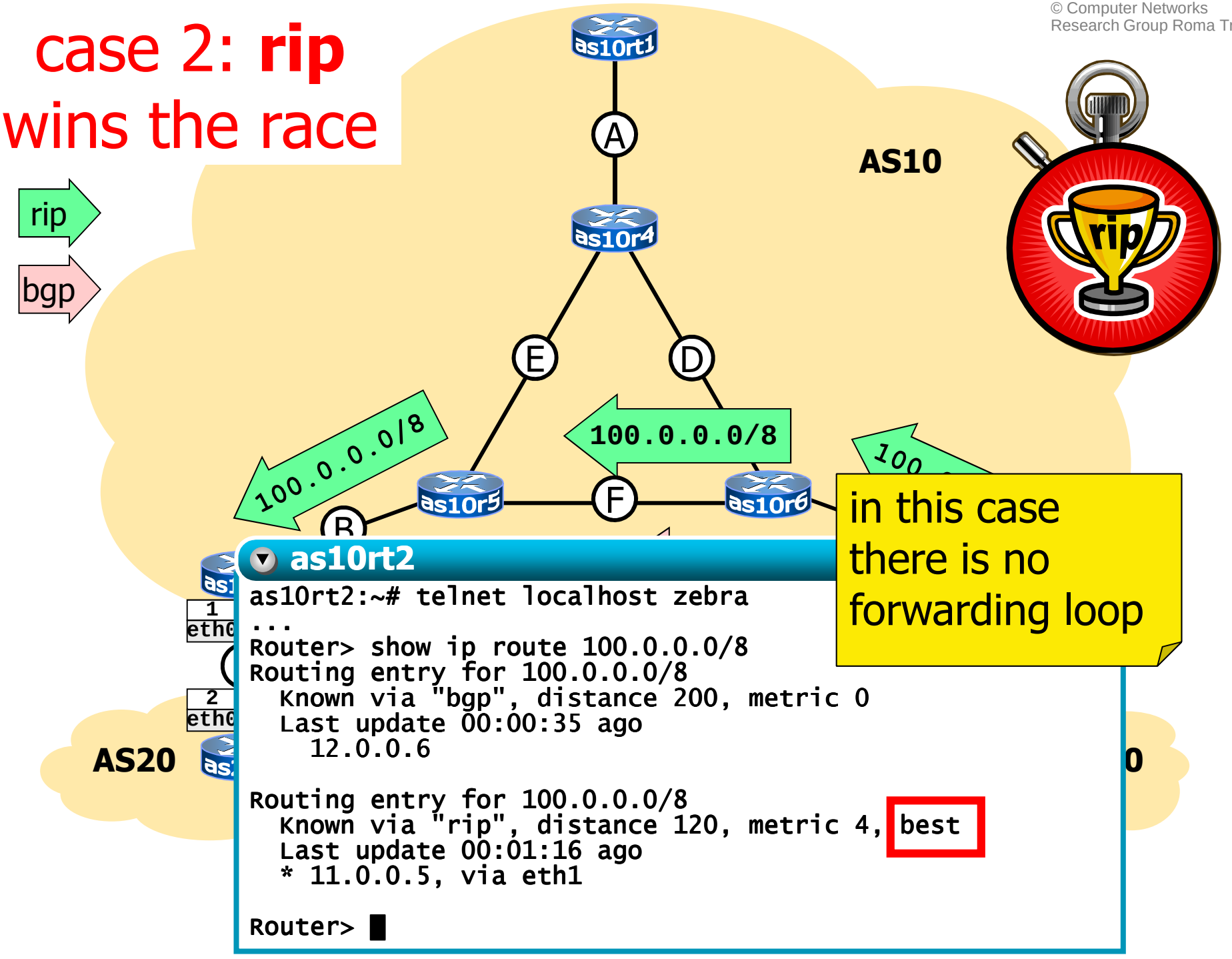
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case 2: **rip** wins the race



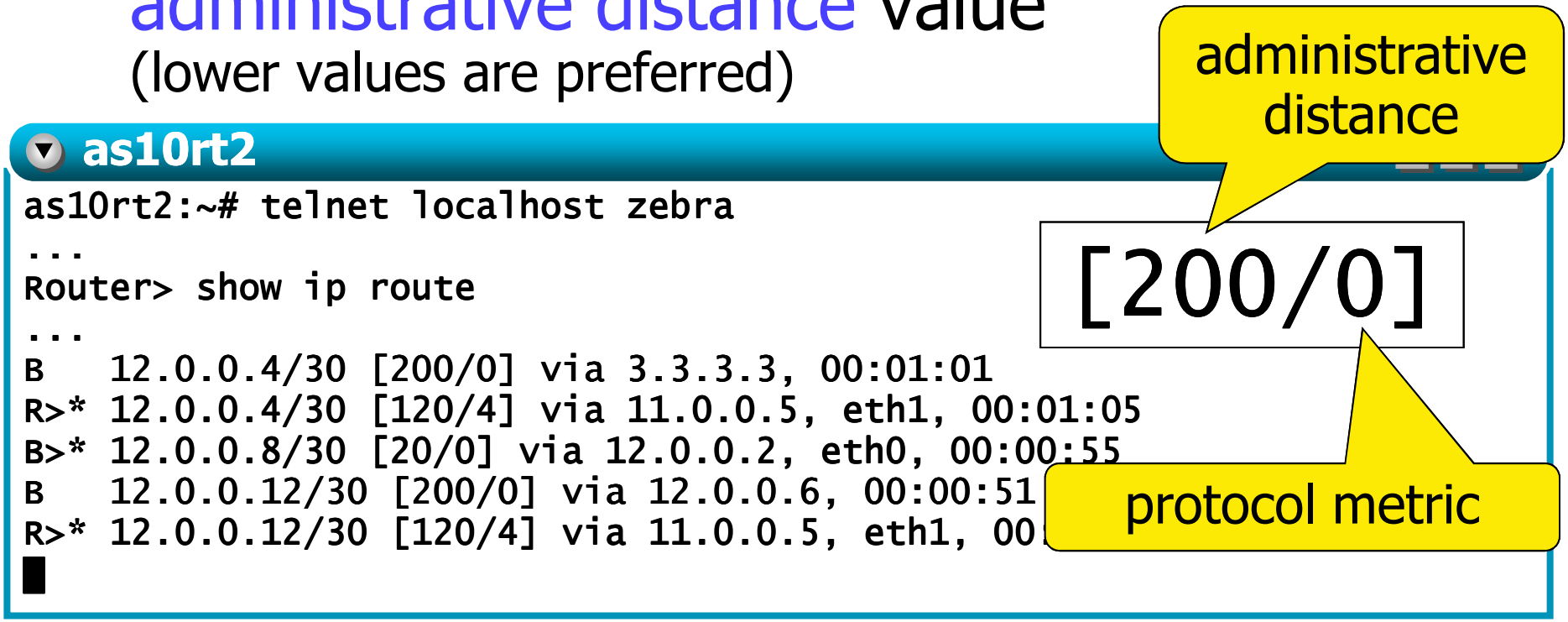
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case 2: **rip** wins the race

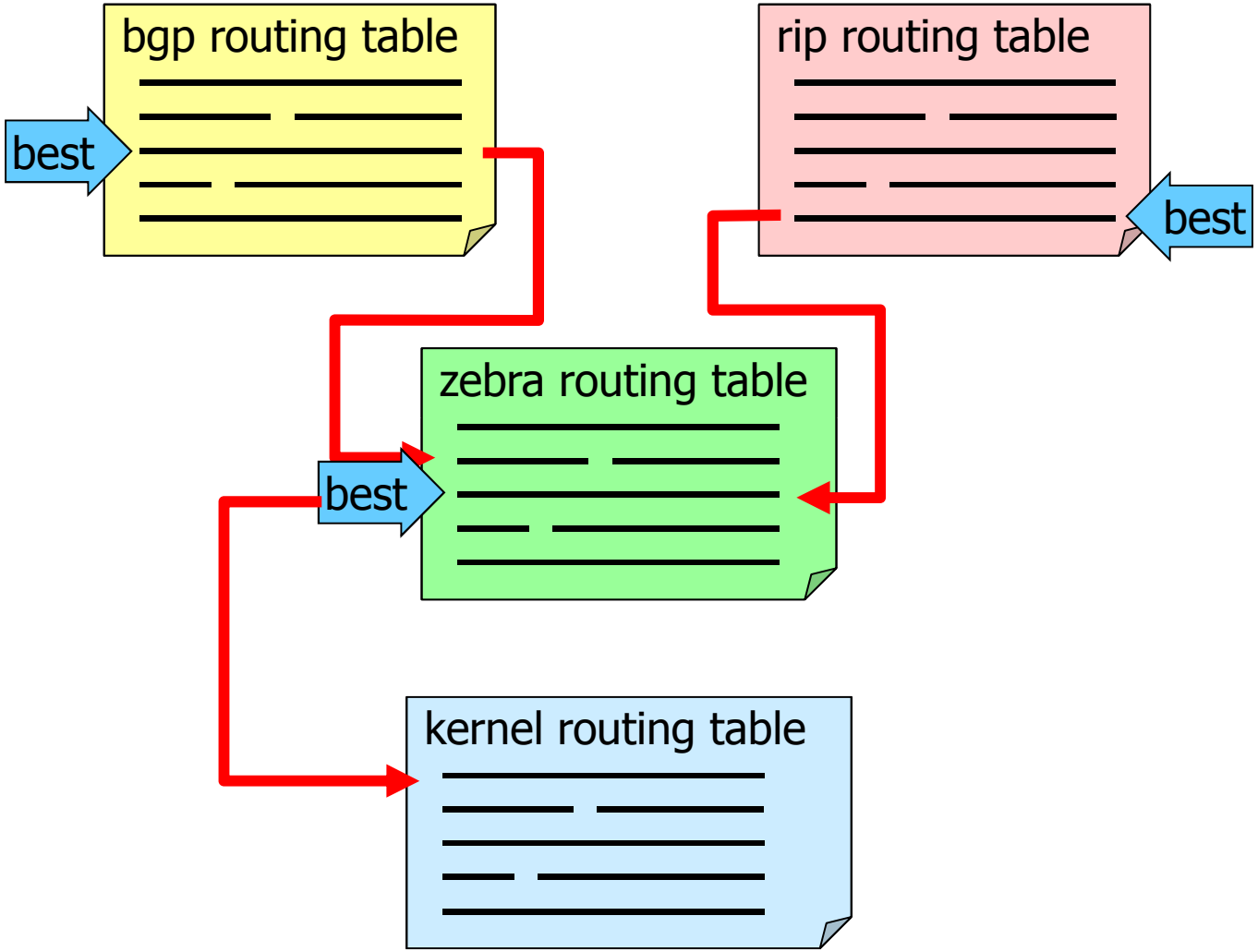


administrative distance

- if different routing protocols propose alternatives for the same route, zebra picks the best route based on an administrative distance value
  (lower values are preferred)



administrative distance

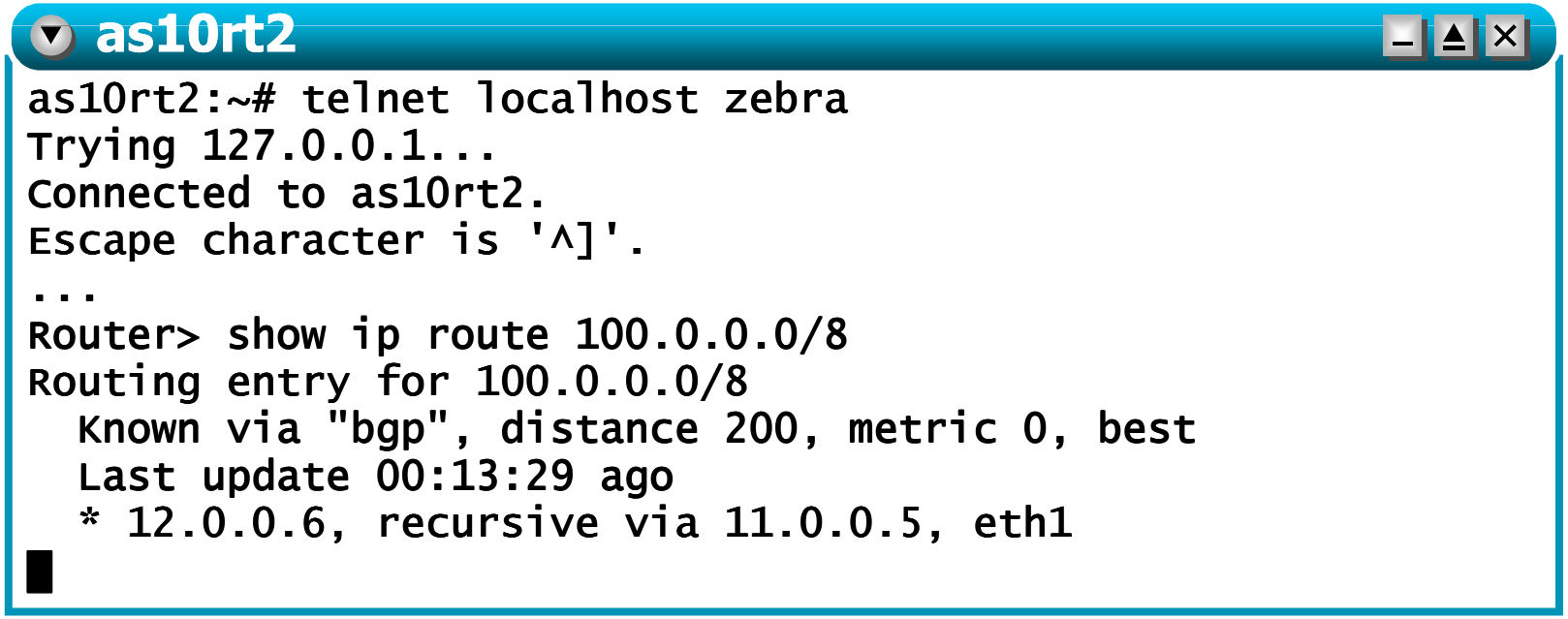


administrative distance

- some default values
  - ebgp: 20
  - rip: 120
  - ibgp: 200

timings

- let's try to force the timings
  - after bringing link **H** down, **as10rt2** is stuck on the bgp choice



timings

- let's try to force the timings
  - after bringing link **H** down, **as10rt2** is stuck on the bgp choice
  - stop **bgpd** and wait for the routing to converge



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timings

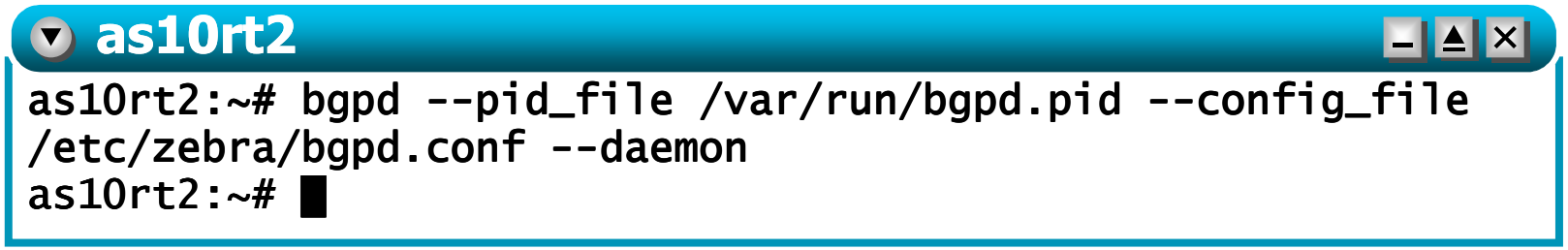
- let's try to force the timings
  - after bringing link **H** down, **as10rt2** is stuck on the bgp choice
  - stop **bgpd** and wait for the routing to converge
    - check if **as10rt2** has selected the rip alternative

as10rt2

```
as10rt2:~# telnet localhost zebra
Trying 127.0.0.1...
Connected to as10rt2.
Escape character is '^]'.
...
Router> show ip route 100.0.0.0/8
Routing entry for 100.0.0.0/8
  Known via "rip", distance 120, metric 4, best
  Last update 00:02:09 ago
  * 11.0.0.5, via eth1
```

timings

- let's try to force the timings
  - after bringing link **H** down, **as10rt2** is stuck on the bgp choice
  - stop **bgpd** and wait for the routing to converge
    - check if **as10rt2** has selected the rip alternative
  - restart **bgpd** and wait for the routing to converge



timings

- let's try to force the timings
  - after bringing link **`H`** down, **`as10rt2`** is stuck on the bgp choice
  - stop **`bgpd`** and wait for the routing to converge
    - check if **`as10rt2`** has selected the rip alternative
  - restart **`bgpd`** and wait for the routing to converge
    - check what **`as10rt2`** knows about **`100.0.0.0/8`**

timings

- once the rip alternative is known, zebra selects it as best and retains it

as10rt2

```
as10rt2:~# telnet localhost zebra
Trying 127.0.0.1...
Connected to as10rt2.
Escape character is '^]'.
...
Router> show ip route 100.0.0.0/8
Routing entry for 100.0.0.0/8
  Known via "bgp", distance 200, metric 0
  Last update 00:02:09 ago
    12.0.0.6

Routing entry for 100.0.0.0/8
  Known via "rip", distance 120, metric 4, best
  Last update 00:08:15 ago
  * 11.0.0.5, via eth1

Router>
```

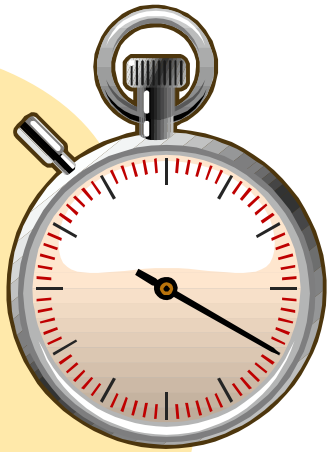
still about timings

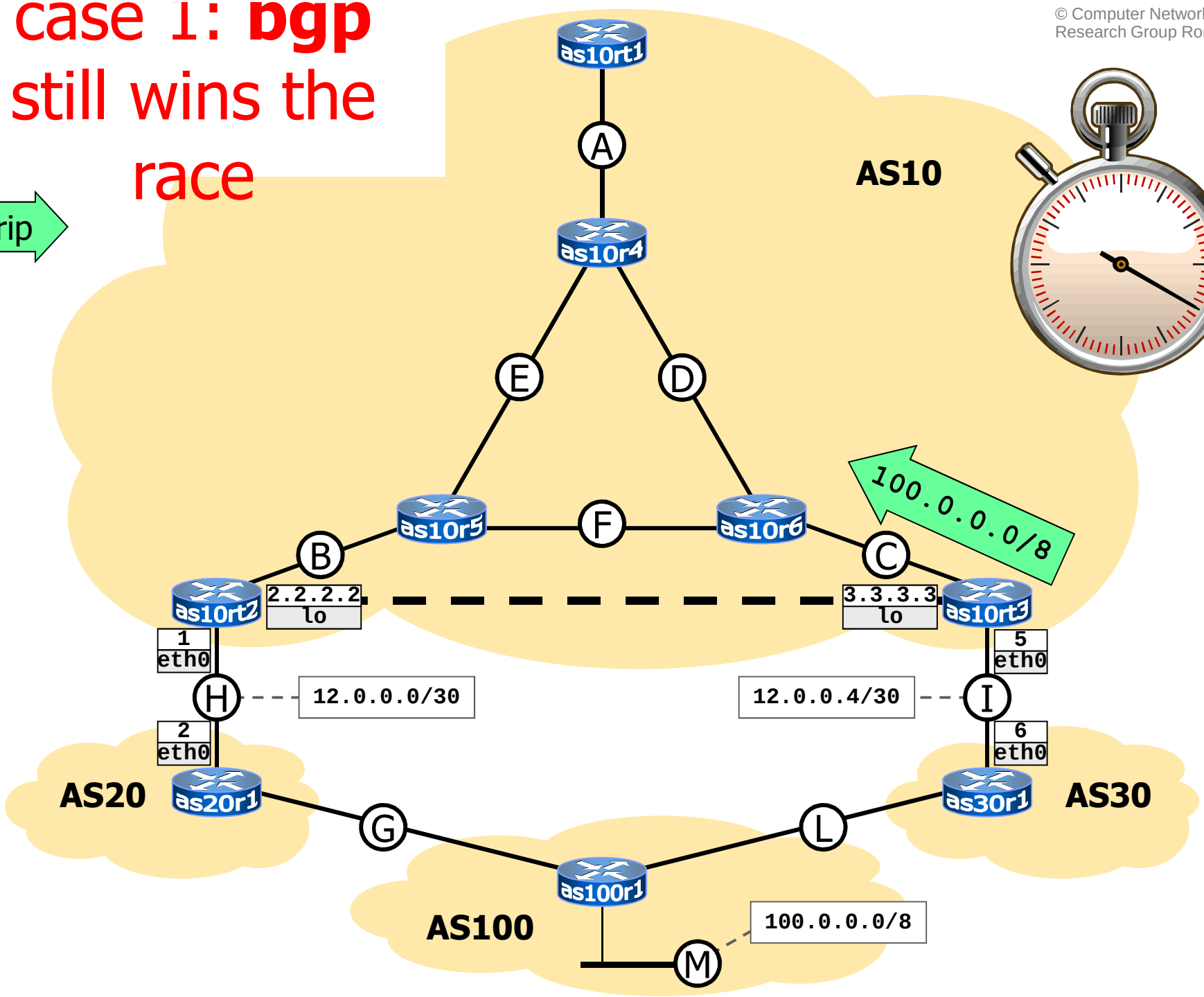
- note: even after setting seasonable **`route-map`**s to avoid redistribution of ibgp into rip, **`as10rt2`** *may* still select the bgp alternative depending on the timings
  - see next slides...
- the forwarding loop is however prevented

case 1: bgp still wins the race

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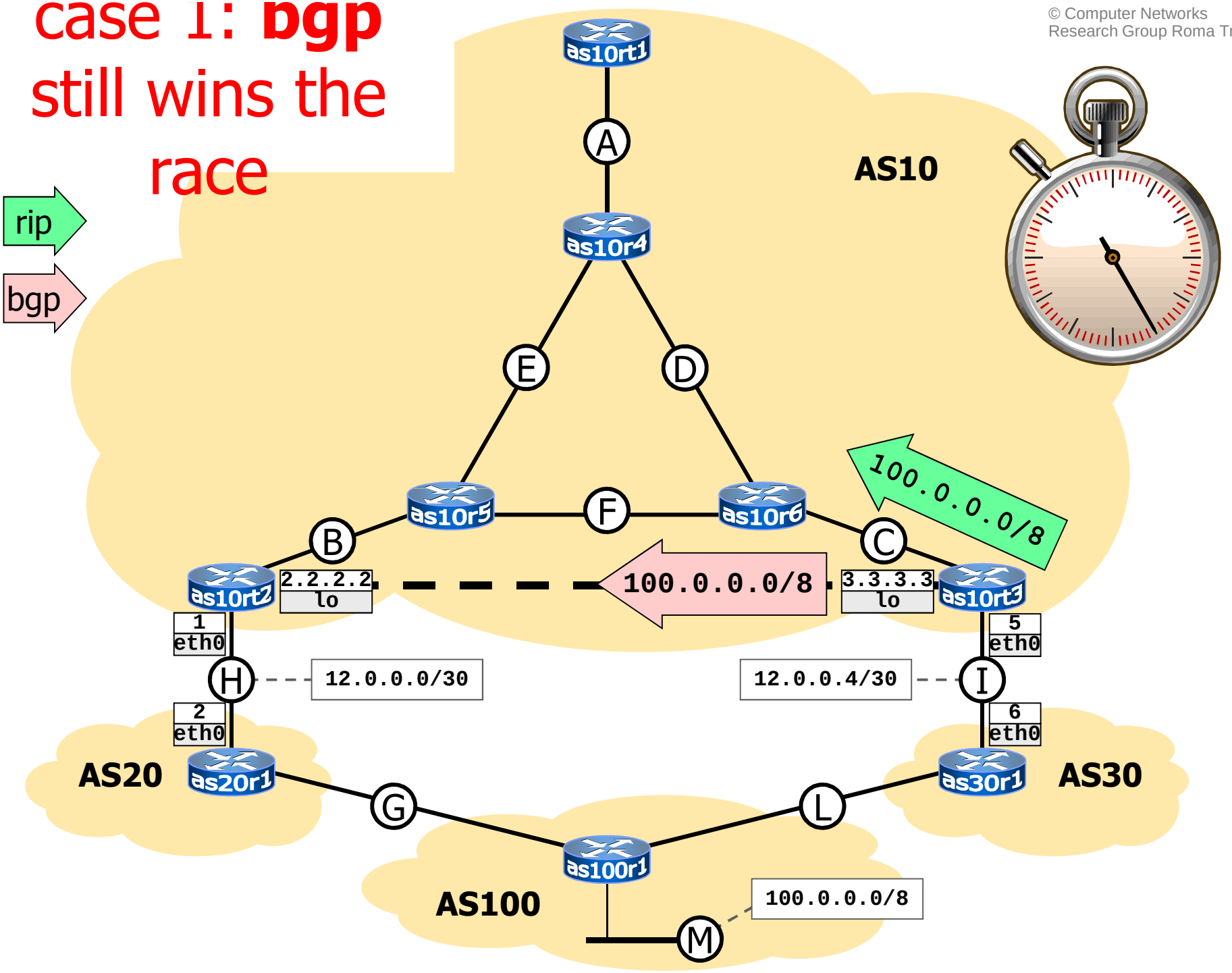






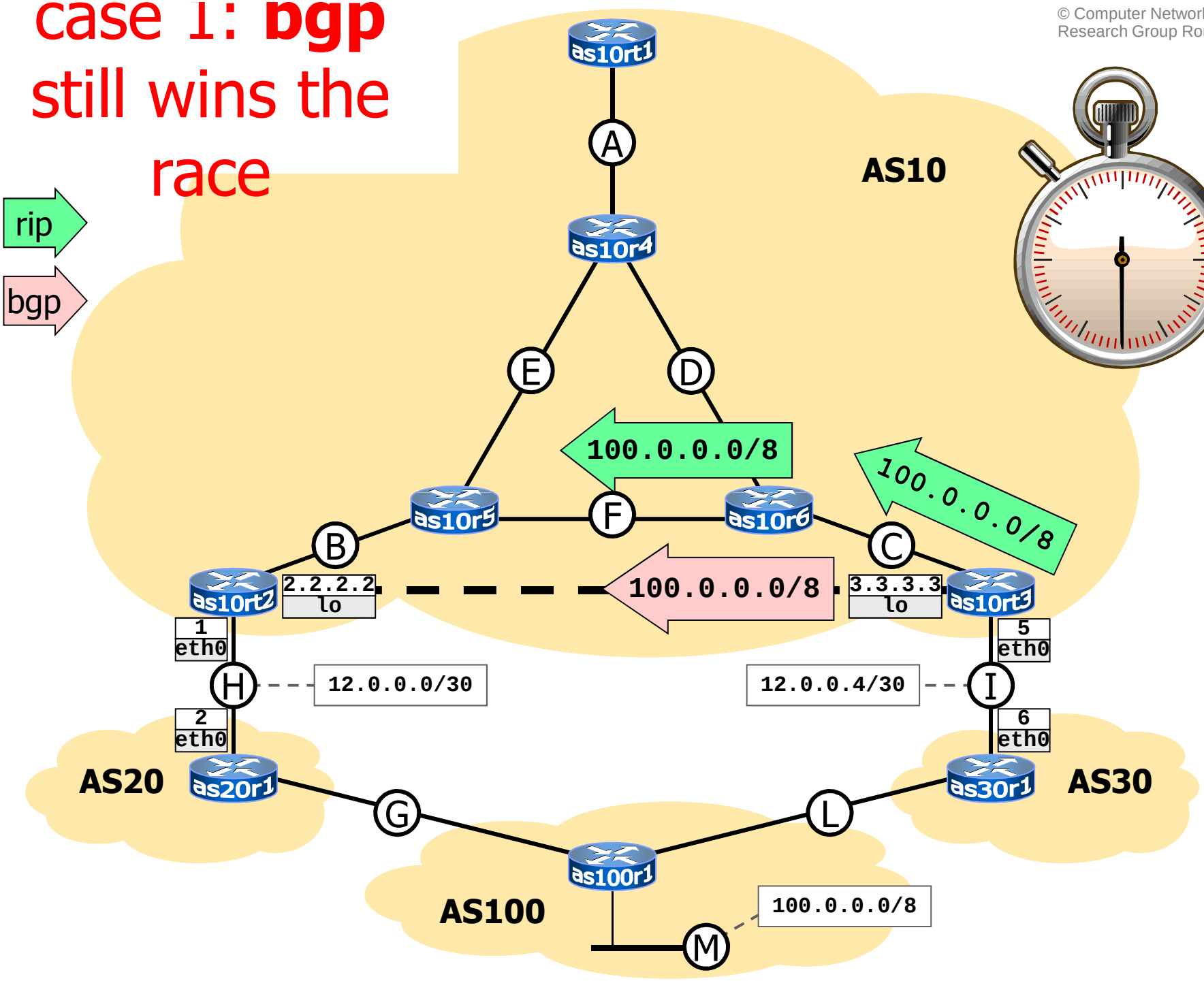
case 1: **bgp** still wins the race

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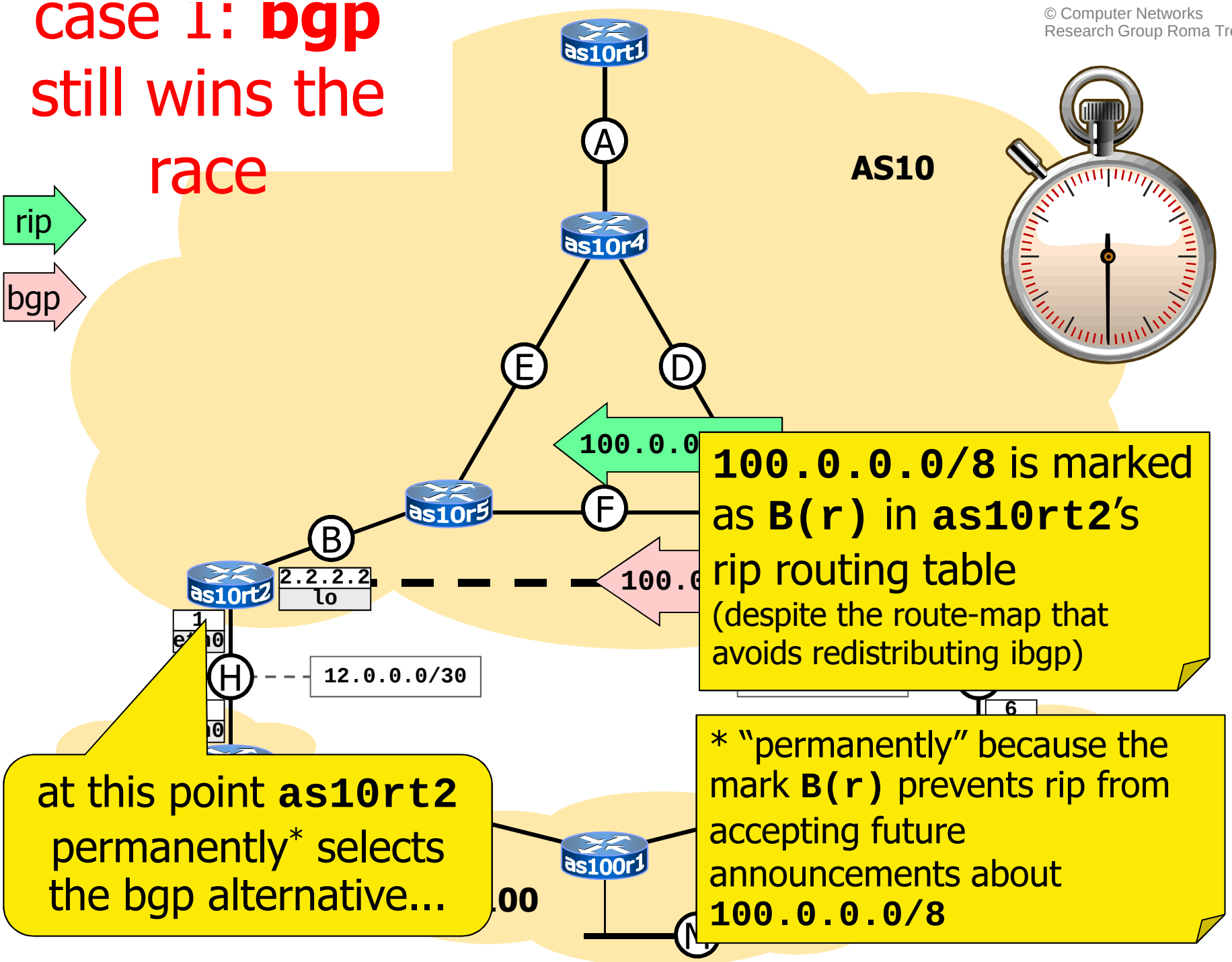
case 1: bgp still wins the race

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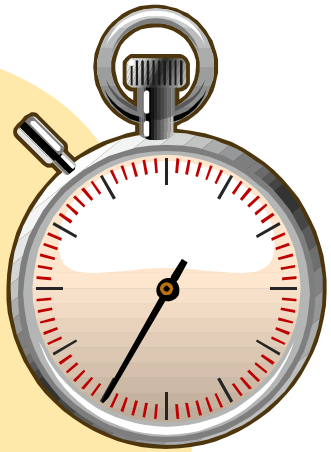
case 1: bgp still wins the race

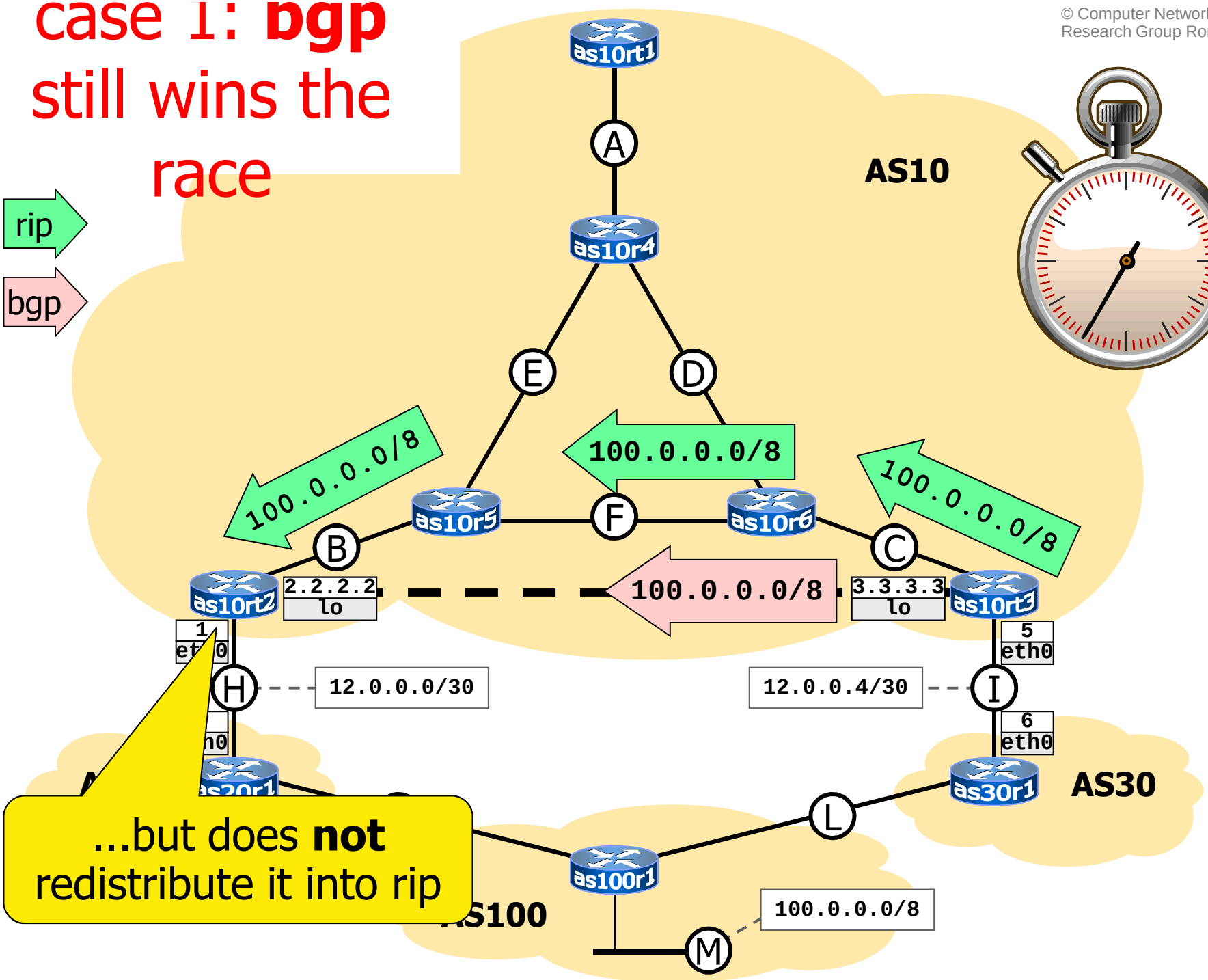
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case 1: **bgp** still wins the race

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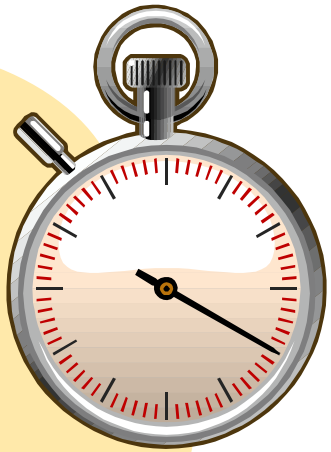


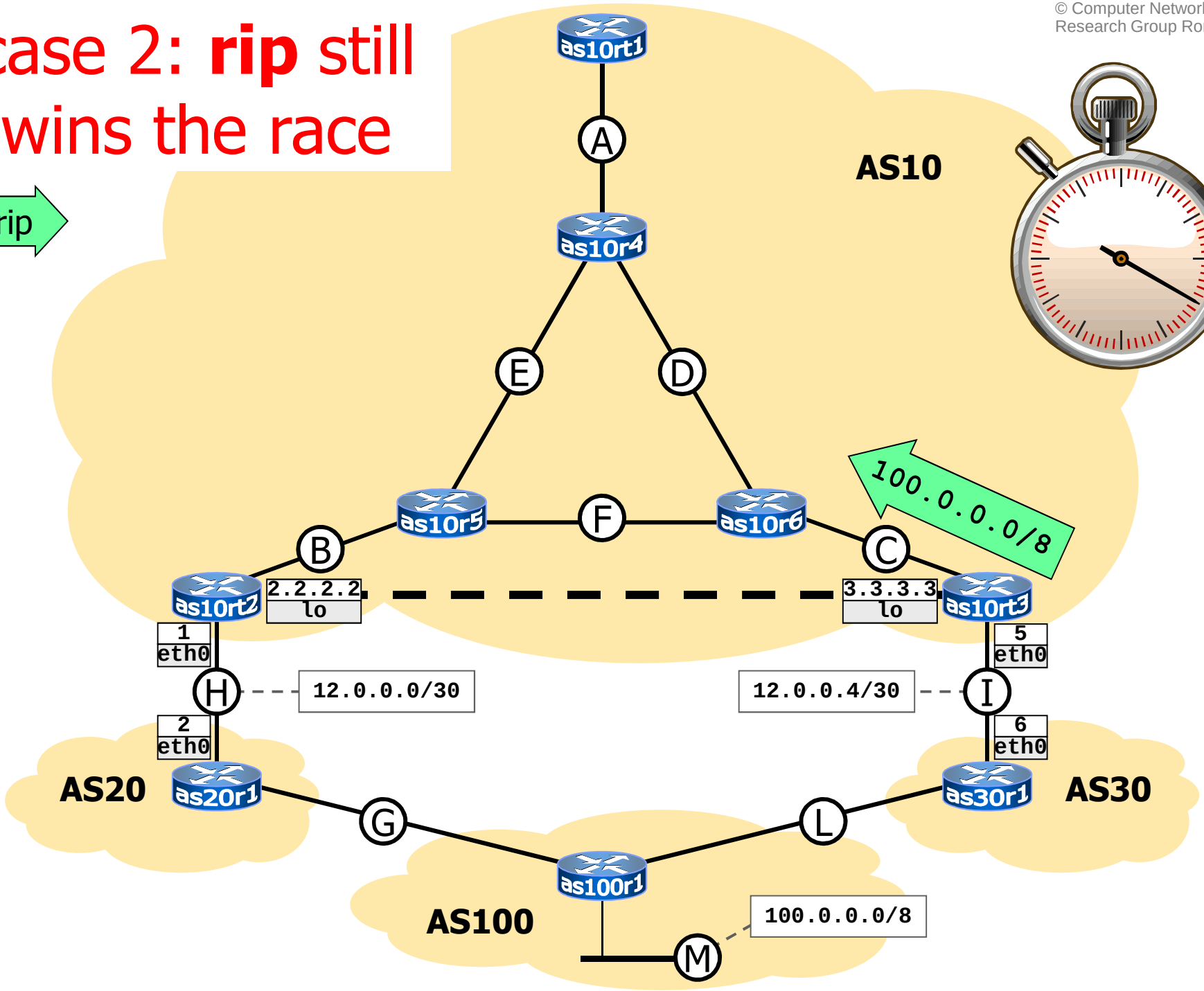


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case 2: **rip** still wins the race

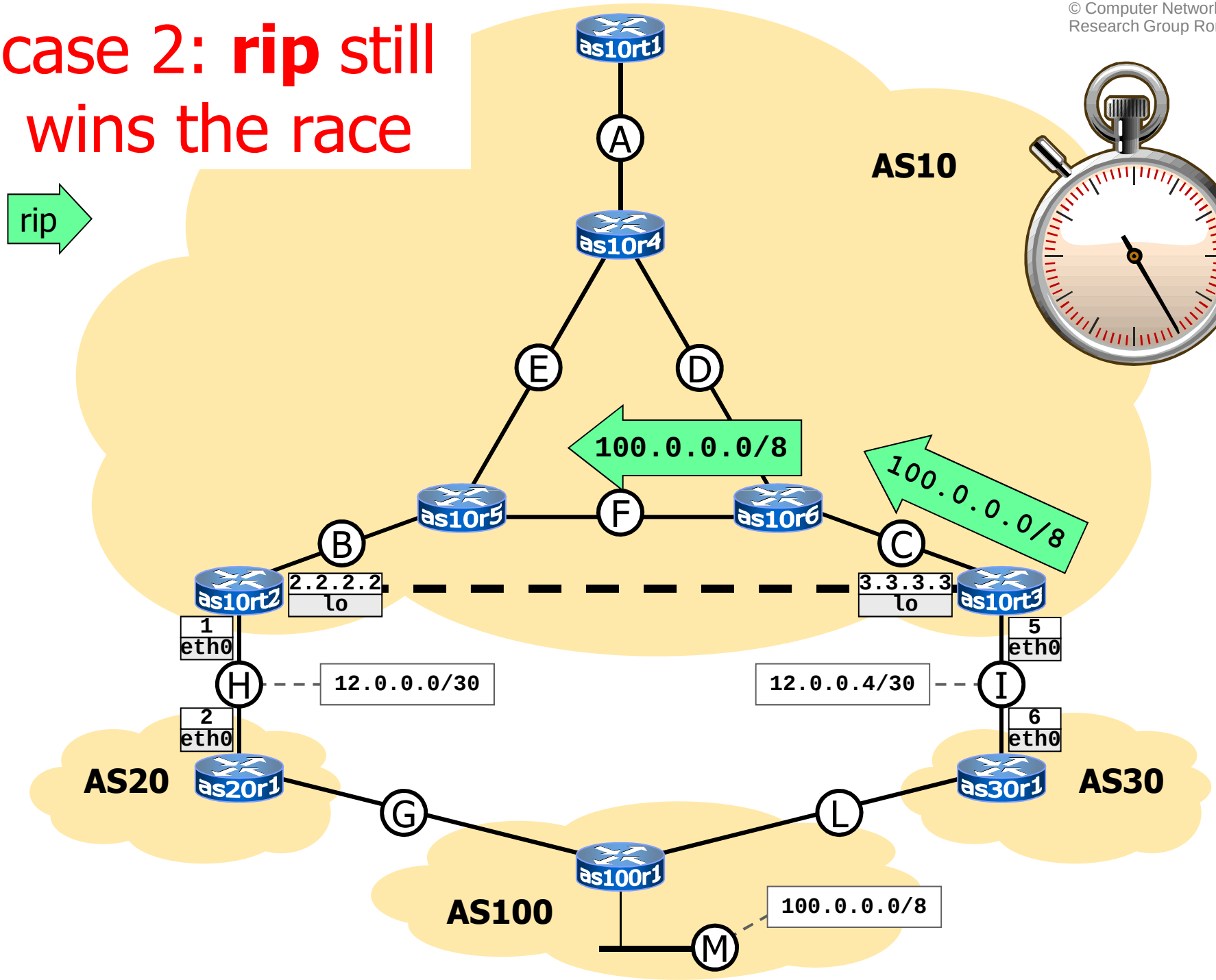






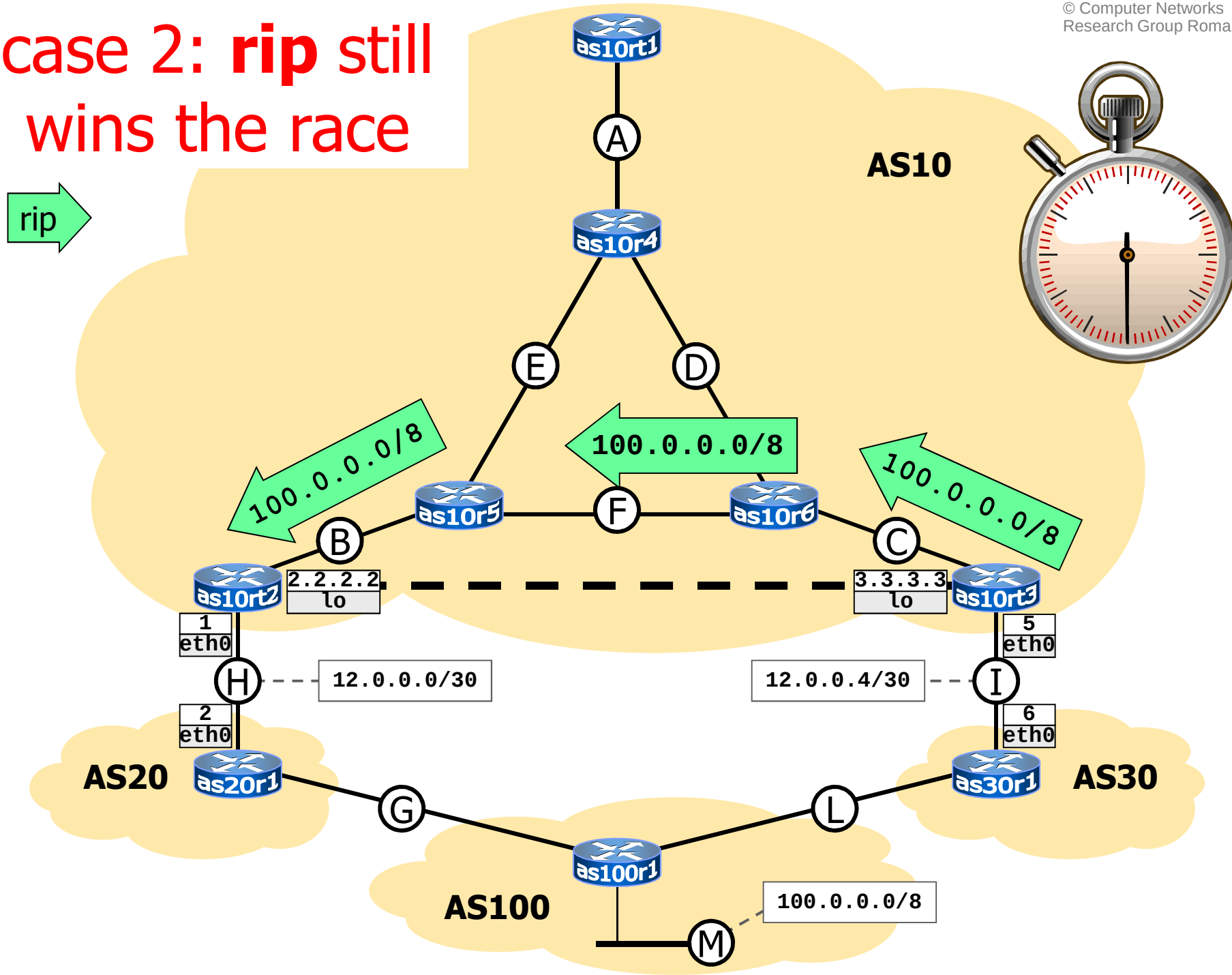
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case 2: **rip** still wins the race



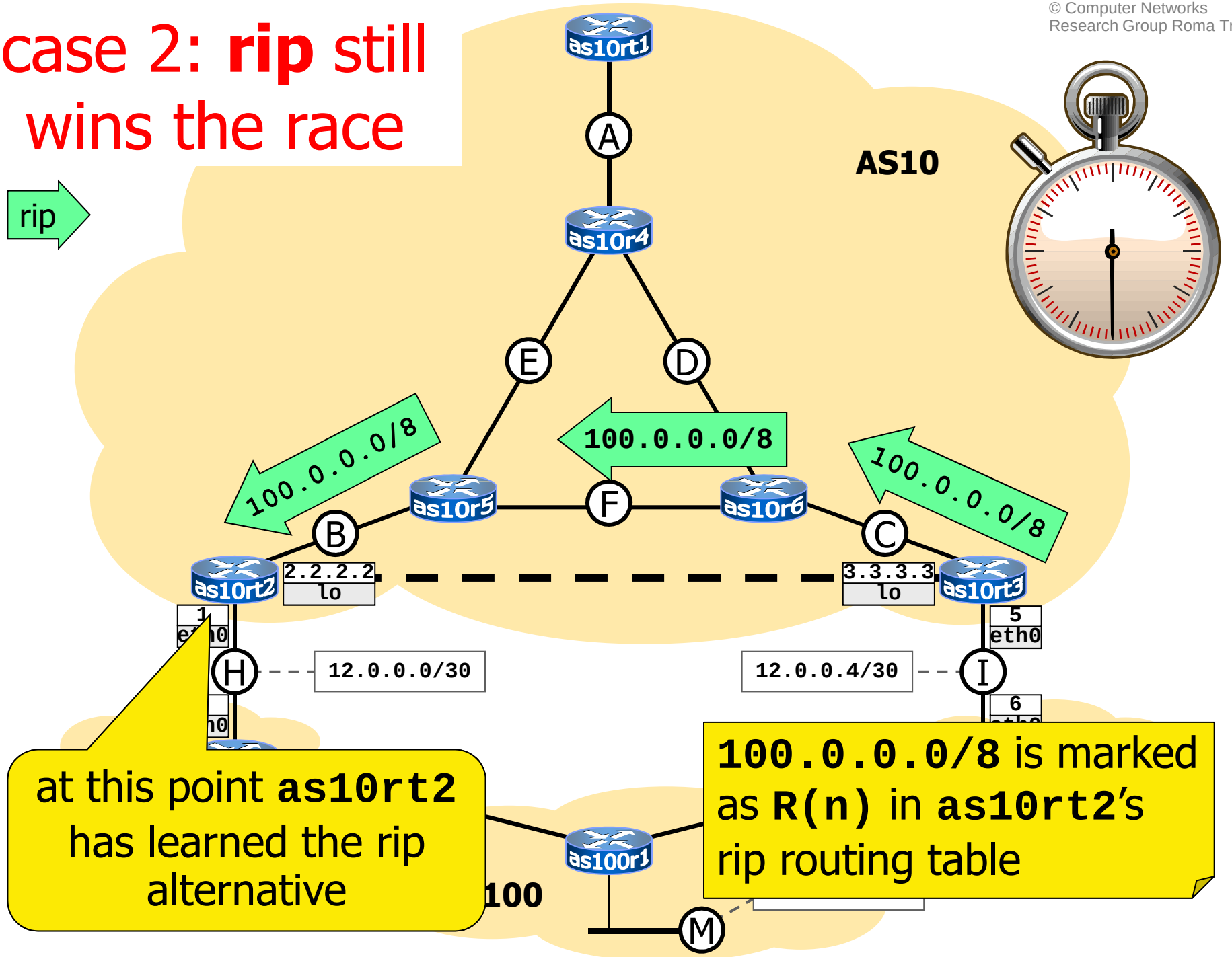
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case 2: **rip** still wins the race



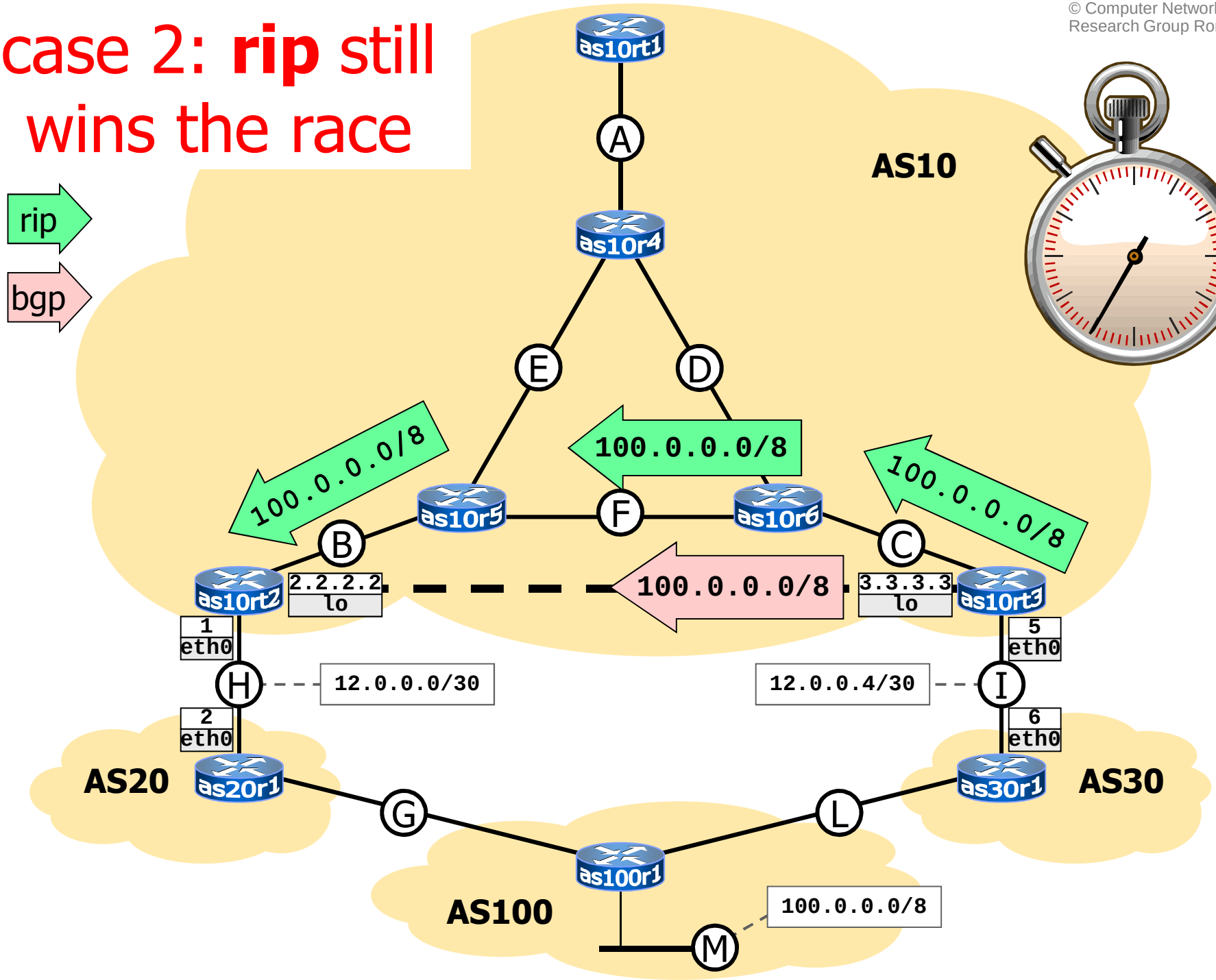
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case 2: **rip** still wins the race



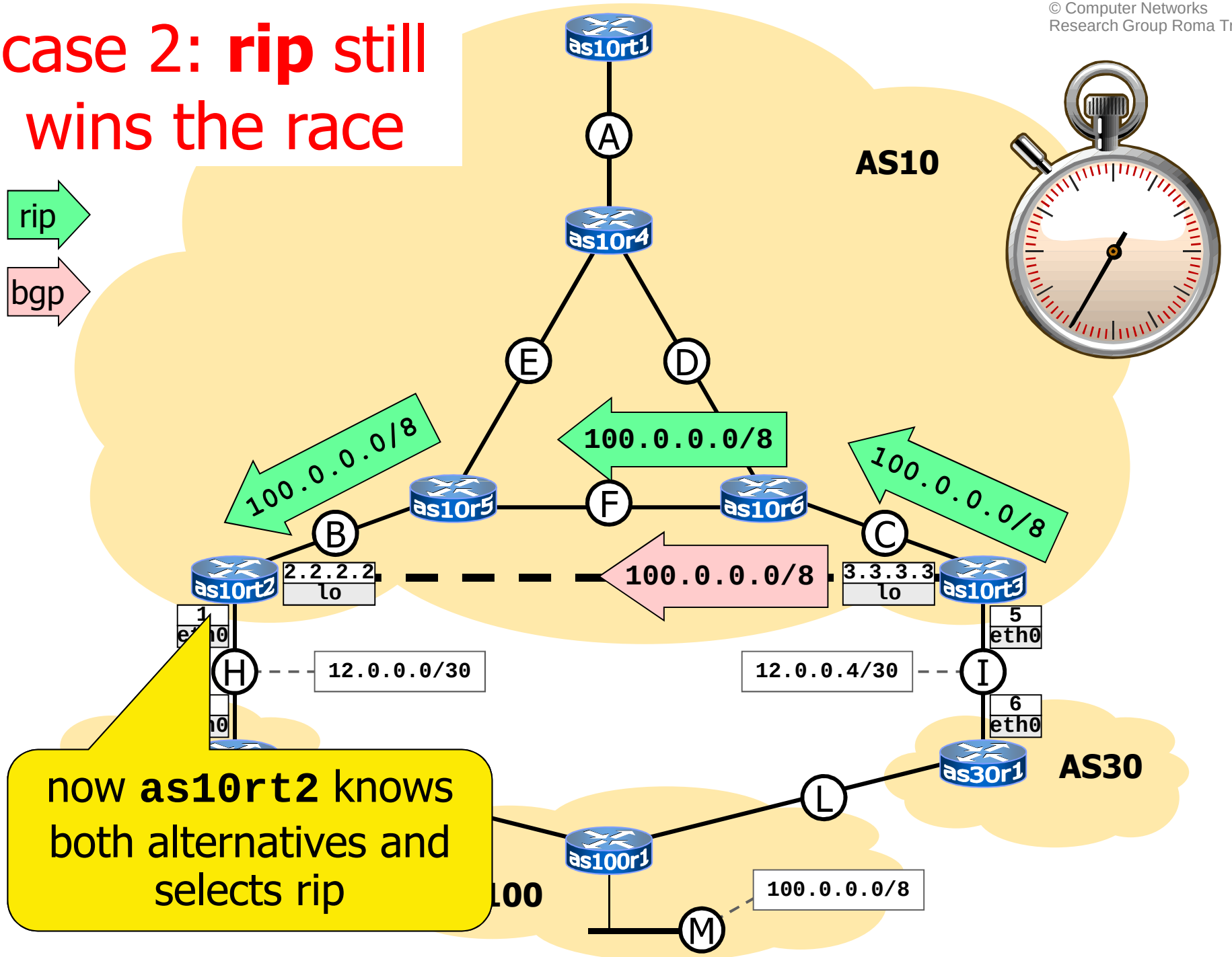
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case 2: **rip** still wins the race



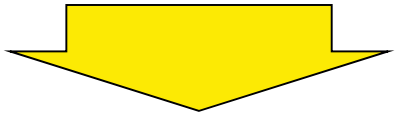
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case 2: **rip** still wins the race



timings may be treacherous

- there is nothing bad in choosing bgp rather than rip... or not?
- remember: if a bgp next-hop has been learned via bgp, the corresponding route is not selected in the routing table



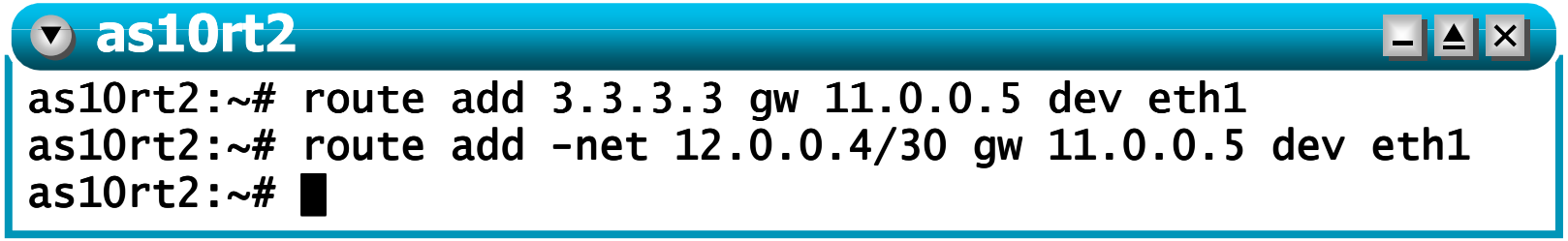
- if a router picks from bgp some route containing a bgp next-hop to a network, that network becomes unreachable!

timings may be treacherous

- example
  - **`as10rt2`** uses the bgp next-hop **`12.0.0.6`** to reach **`100.0.0.0/8`**
  - **`as10rt2`** *might* learn **`12.0.0.4/30`** via ibgp before learning it via rip
    - in this case, the rip alternative is never taken into account
  - let's see what happens in this case...

timings may be treacherous

- steps to reproduce the bad timing
  - add some kernel routes to let bgp peerings and routes stay there



timings may be treacherous

- steps to reproduce the bad timing
  - add some kernel routes to let bgp peerings and routes stay there
  - stop rip



timings may be treacherous

- steps to reproduce the bad timing
  - add some kernel routes to let bgp peerings and routes stay there
  - stop rip
  - wait for bgp routes to settle down (the routing table should look like the one in the following slide...)

timings may be treacherous

as10rt2

```
as10rt2:~# telnet localhost zebra
Trying 127.0.0.1...
Connected to as10rt2.
Escape character is '^]'.
...
Router> show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP, O - OSPF,
       B - BGP, > - selected route, * - FIB route

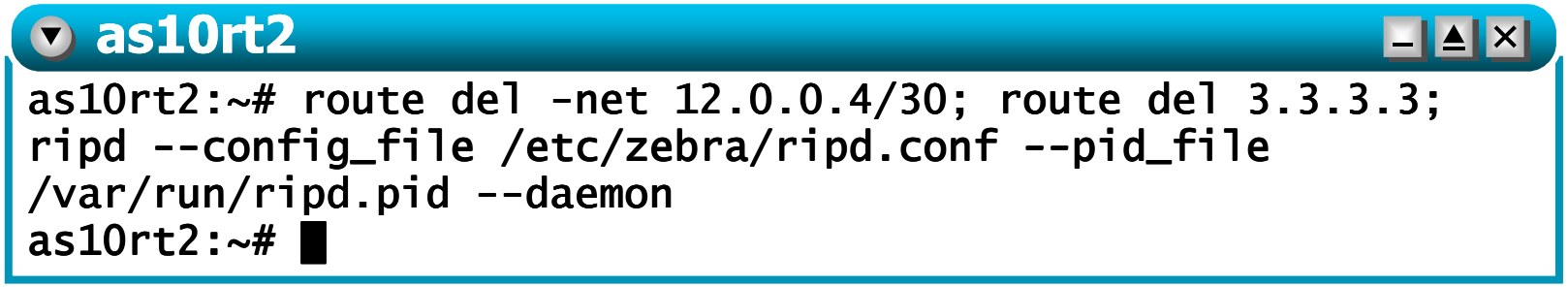
C>* 2.2.2.2/32 is directly connected, lo
K>* 3.3.3.3/32 via 11.0.0.5, eth1
C>* 11.0.0.4/30 is directly connected, eth1
C>* 12.0.0.0/30 is directly connected, eth0
B   12.0.0.4/30 [200/0] via 3.3.3.3, 00:00:03
K>* 12.0.0.4/30 via 11.0.0.5, eth1
B>* 12.0.0.8/30 [20/0] via 12.0.0.2, eth0, 00:26:57
B>* 12.0.0.12/30 [200/0] via 12.0.0.6, recursive via 11.0.0.5, eth1, 00:00:03
B>* 20.0.0.0/8 [20/0] via 12.0.0.2, eth0, 00:26:57
B>* 30.0.0.0/8 [200/0] via 12.0.0.6, recursive via 11.0.0.5, eth1, 00:00:03
B>* 100.0.0.0/8 [200/0] via 12.0.0.6, recursive via 11.0.0.5, eth1, 00:00:03
C>* 127.0.0.0/8 is directly connected, lo
Router>
```

routes are now known via bgp

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timings may be treacherous

- steps to reproduce the bad timing
  - add some kernel routes to let bgp peerings and routes stay there
  - stop rip
  - wait for bgp routes to settle down (the routing table should look like the one in the following slide...)
  - remove kernel routes and immediately start rip before bgp can even realize those routes have been lost



timings may be treacherous

as10rt2

```
Router> show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP, O - OSPF,
       B - BGP, > - selected route, * - FIB route

R>* 1.1.1.1/32 [120/4] via 11.0.0.5, eth1, 00:00:41
C>* 2.2.2.2/32 is directly connected, lo
R>* 3.3.3.3/32 [120/4] via 11.0.0.5, eth1, 00:00:41
R>* 10.0.0.0/30 [120/2] via 11.0.0.5, eth1, 00:00:41
R>* 10.0.0.4/30 [120/2] via 11.0.0.5, eth1, 00:00:41
R>* 10.0.0.8/30 [120/3] via 11.0.0.5, eth1, 00:00:41
R>* 11.0.0.0/30 [120/3] via 11.0.0.5, eth1, 00:00:41
C>* 11.0.0.4/30 is directly connected, eth1
R>* 11.0.0.8/30 [120/3] via 11.0.0.5, eth1, 00:00:41
C>* 12.0.0.0/30 is directly connected, eth0
B>* 12.0.0.4/30 [200/0] via 3.3.3.3, recursive via 11.0.0.5, eth1, 00:00:57
B>* 12.0.0.8/30 [20/0] via 12.0.0.2, eth0, 00:11:51
B>* 12.0.0.12/30 [200/0] via 12.0.0.6, recursive via 11.0.0.5, eth1, 00:00:57
B>* 20.0.0.0/8 [20/0] via 12.0.0.2, eth0, 00:11:51
B>* 30.0.0.0/8 [200/0] via 12.0.0.6, recursive via 11.0.0.5, eth1, 00:00:57
B>* 100.0.0.0/8 [200/0] via 12.0.0.6, recursive via 11.0.0.5, eth1, 00:00:57
C>* 127.0.0.0/8 is
Router>
```

at first, routes are still known via bgp

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timings may be treacherous

as10rt2

```
Router> show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP, O - OSPF,
       B - BGP, > - selected route, * - FIB route

R>* 1.1.1.1/32 [120/4] via 11.0.0.5, eth1, 00:00:41
C>* 2.2.2.2/32 is directly connected, lo
R>* 3.3.3.3/32 [120/4] via 1
R>* 10.0.0.0/30 [120/2] via
R>* 10.0.0.4/30 [120/2] via
R>* 10.0.0.8/30 [120/3] via
R>* 11.0.0.0/30 [120/3] via 11.0.0.5, eth1, 00:00:41
C>* 11.0.0.4/30 is directly connected, eth1
R>* 11.0.0.8/30 [120/3] via 11.0.0.5, eth1, 00:00:41
C>* 12.0.0.0/30 is directly connected, eth0
B>* 12.0.0.4/30 [200/0] via 3.3.3.3, recursive via 11.0.0.5, eth1, 00:00:57
B>* 12.0.0.8/30 [20/0] via 12.0.0.2, eth0, 00:11:51
B>* 12.0.0.12/30 [200/0] via 12.0.0.6, recursive via 11.0.0.5, eth1, 00:00:57
B>* 20.0.0.0/8 [20/0] via 12.0.0.2, eth0, 00:11:51
B>* 30.0.0.0/8 [200/0] via 12.0.0.6, recursive via 11.0.0.5, eth1, 00:00:57
B>* 100.0.0.0/8 [200/0] via 12.0.0.6, recursive via 11.0.0.5, eth1, 00:00:57
C>* 127.0.0.0/8 is directly connected, lo
Router>
```

in particular, the bgp next-hop **12.0.0.6** is known via bgp

timings may be treacherous

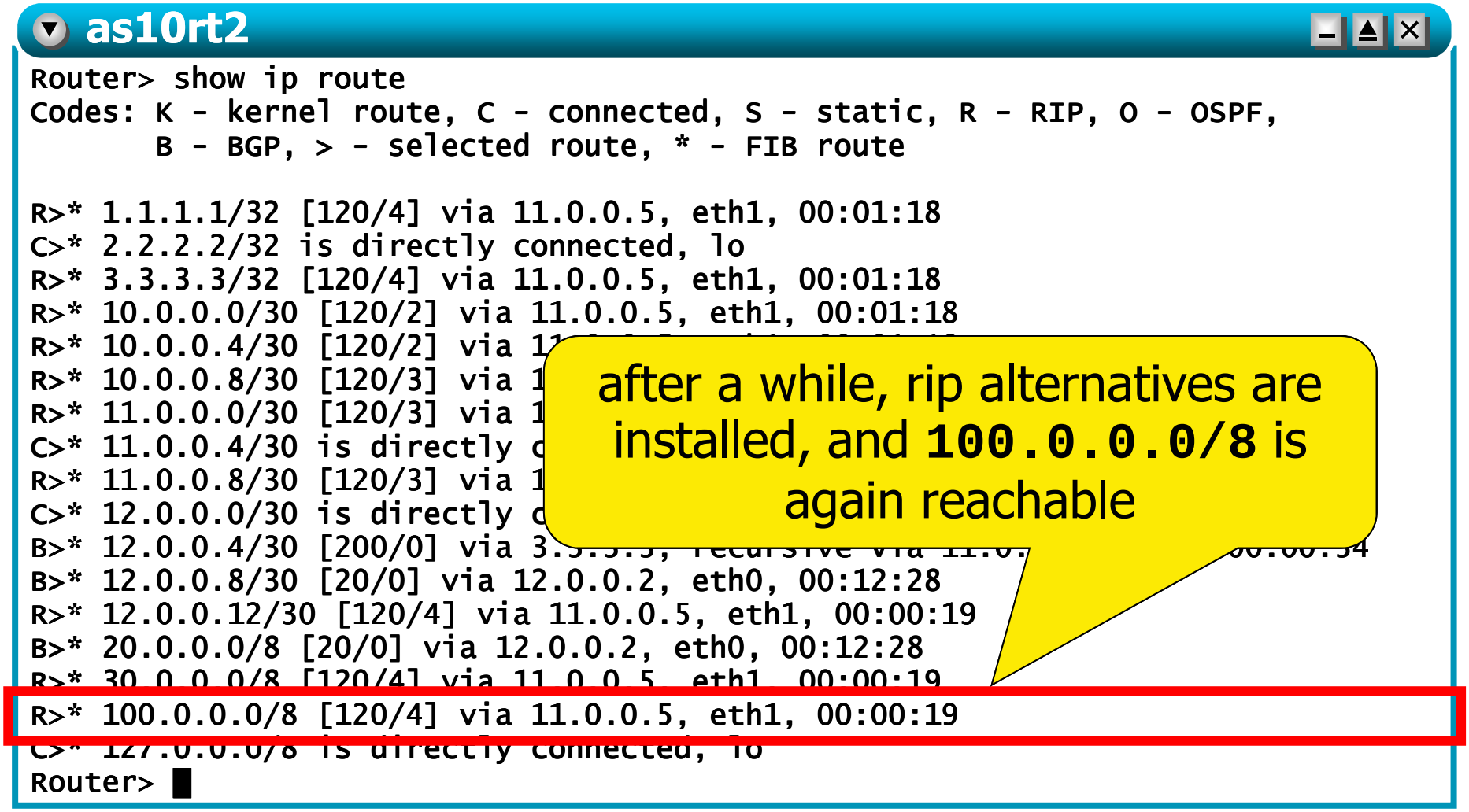
as10rt2

```
Router> show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP, O - OSPF,
       B - BGP, > - selected route, * - FIB route

R>* 1.1.1.1/32 [120/4] via 11.0.0.5, eth1, 00:00:46
C>* 2.2.2.2/32 is directly connected, lo
R>* 3.3.3.3/32 [120/4] via 11.0.0.5, eth1, 00:00:46
R>* 10.0.0.0/30 [120/2] via 11.0.0.5, eth1, 00:00:46
R>* 10.0.0.4/30 [120/2] via 11.0.0.5, eth1, 00:00:46
R>* 10.0.0.8/30 [120/3] via 11.0.0.5, eth1, 00:00:46
R>* 11.0.0.0/30 [120/3] via 11.0.0.5, eth1, 00:00:46
C>* 11.0.0.4/30 is directly connected, eth1
R>* 11.0.0.8/30 [120/3] via 11.0.0.5, eth1, 00:00:46
C>* 12.0.0.0/30 is directly connected, eth0
B>* 12.0.0.4/30 [200/0] via 3.3.3.3, recursive
B>* 12.0.0.8/30 [20/0] via 12.0.0.2, eth0, 00:1
B>* 20.0.0.0/8 [20/0] via 12.0.0.2, eth0, 00:11
C>* 127.0.0.0/8 is directly connected, lo
Router>
```

after some time bgp realizes that the recursive lookup fails for routes using a bgp learned next-hop

timings may be treacherous



timings may be treacherous

as10rt2

```
as10rt2:~# telnet localhost bgpd
Trying 127.0.0.1...
Connected to as10rt2.
Escape character is '^]'.
...
as10rt2-bgpd> show ip bgp
BGP table version is 0, local router ID is 12.0.0.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
Origin codes: i - IGP, e - EGP, ? - inc

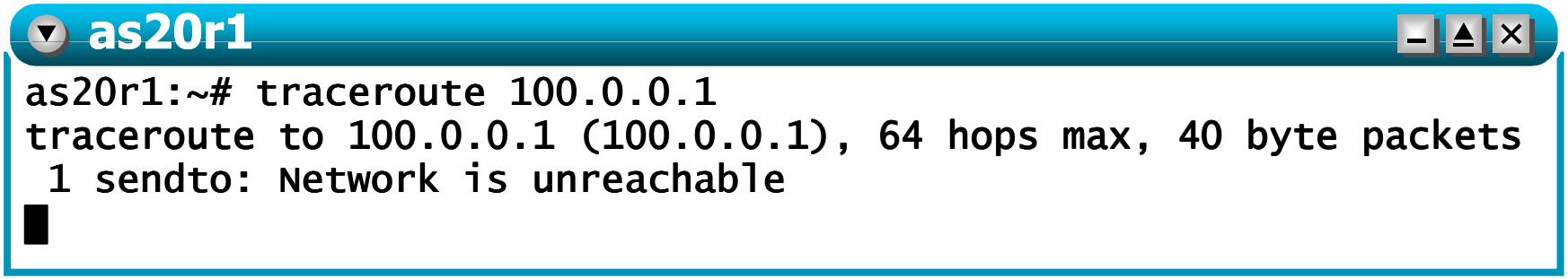
   Network          Next Hop
* i10.0.0.0         3.3.3.3
* i                 1.1.1.1
*>                  0.0.0.0
*> 12.0.0.0/30      0.0.0.0
*>i12.0.0.4/30      3.3.3.3
*> 12.0.0.8/30      12.0.0.2
* i12.0.0.12/30     12.0.0.6
*> 20.0.0.0         12.0.0.2
* i30.0.0.0         12.0.0.6
* i100.0.0.0        12.0.0.6

Total number of prefixes 8
```

however, **100.0.0.0/8** is no more available for interdomain routing

timings may be treacherous

- the outcome of this adverse timing is really undesirable



timings may be treacherous

- solutions?
  - filtering
    - do not announce the demarcation zone in ibgp
    - do not accept incoming bgp announcements for the demarcation zone
    - ...or simply do not announce the demarcation zone in bgp at all
  - avoid redistributing bgp
    - knowing something via bgp masks any rip alternative when **`redistribute bgp`** is used

timings may be treacherous



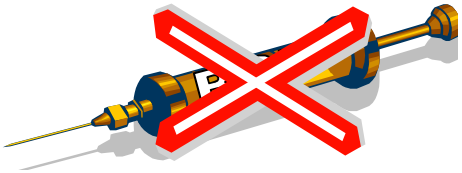
is it really *that* bad?



actually not

- ibgp peers reach each other by means of igp-learned routes
- such routes travel in the igp together with subnets containing ebgp next hops (because of `redistribute connected`)
- before ibgp peerings are established, ebgp next hops have always already been learned via igp (via distance vectors or link state advertisements)

conclusions



- never (ever!) inject ibgp into rip
  - routing table overgrowth
  - update churn
  - very bad interactions with unfortunate timings
  - routing policies may break routing consistency
    - forwarding loop

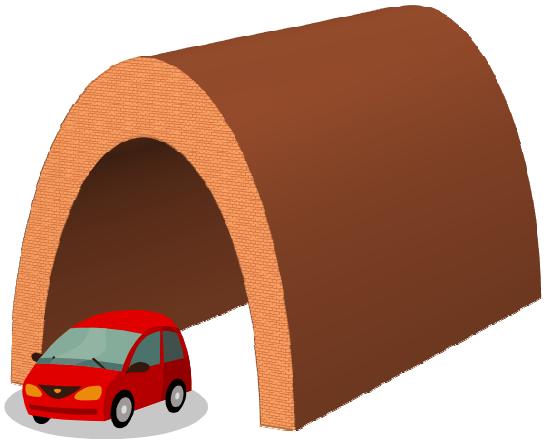
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Dipartimento di Informatica e Automazione
Computer Networks Research Group

choice 2

overlay

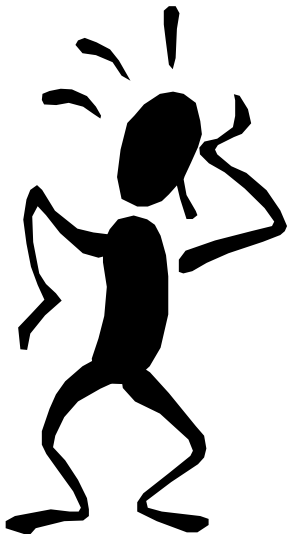
overlay

- ebgp is not redistributed into the igp
  - smaller routing tables
  - less igp churn
- ebgp next hops are reached via a direct link (tunnel)



(don’t forget) internal customers!

- border routers inject a statically configured default route in the igp
  - ensure reachability of the external world



transit as

- stop the current lab

host machine

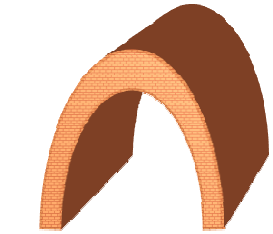
```
user@localhost:~$ cd netkit-lab_bgp-transit-as-forwarding-loop
user@localhost:~/netkit-lab_bgp-transit-as-forwarding-loop$ lcrash
```

- start the tunnel lab

host machine

```
user@localhost:~$ cd netkit-lab_bgp-transit-as-tunnel-ipip
user@localhost:~/netkit-lab_bgp-transit-as-tunnel-ipip$ lstart
```

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13.0.0.z

AS10

as10rt1

as10r4

10.0.0.w

as10r5

as10r6

as10rt2

as10rt3

11.0.0.x

AS20

as20r1

as30r1

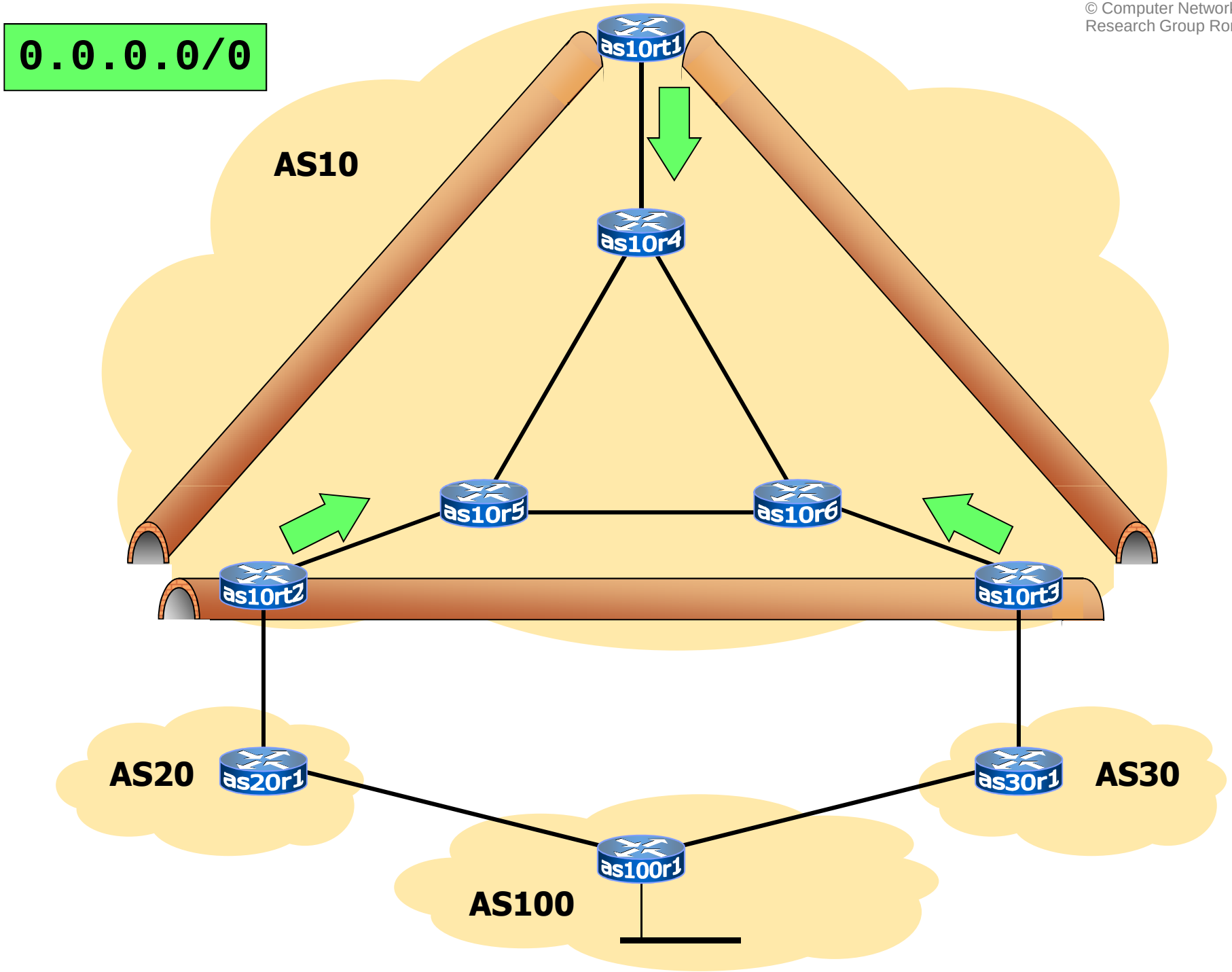
AS30

12.0.0.y

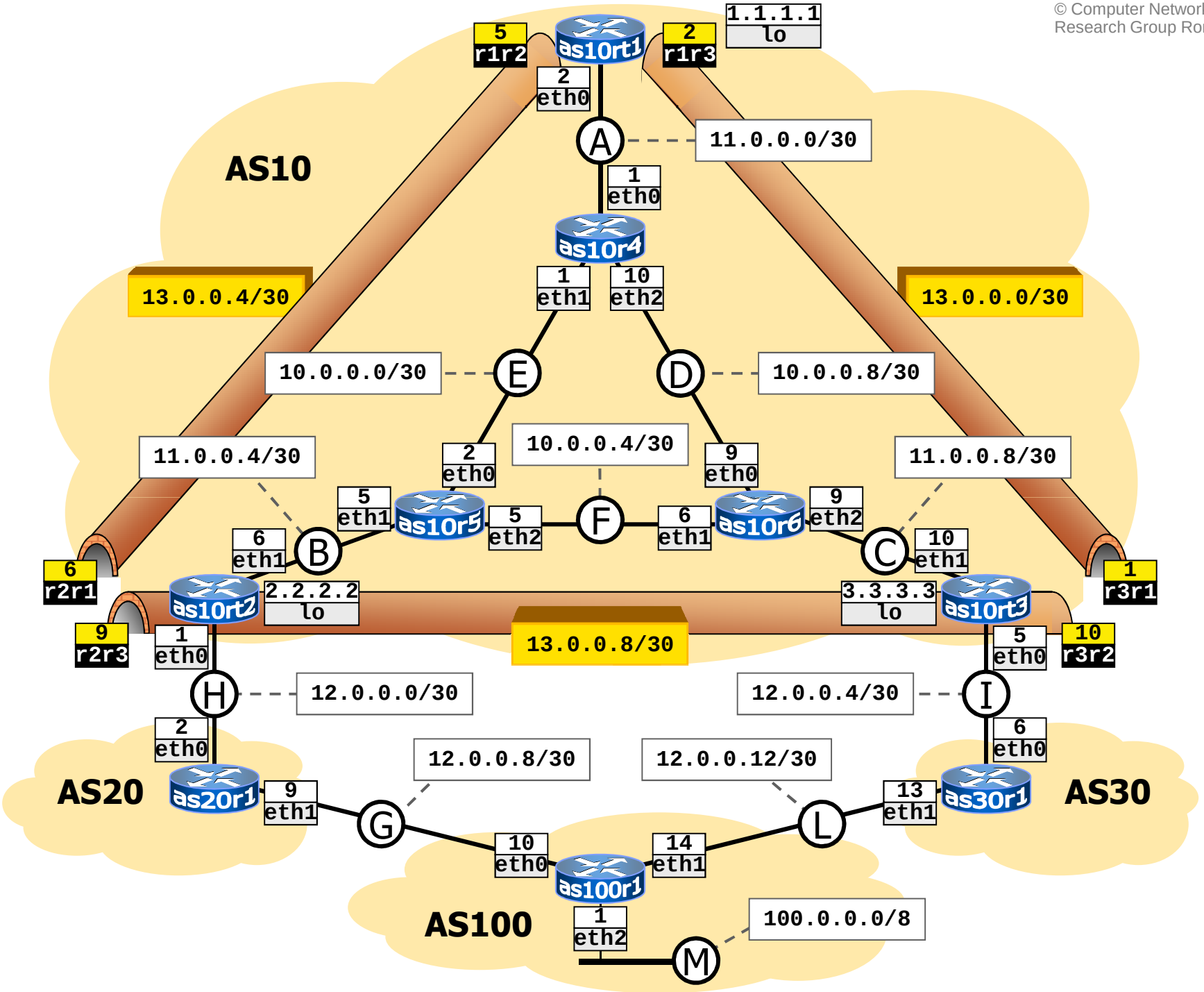
as100r1

AS100

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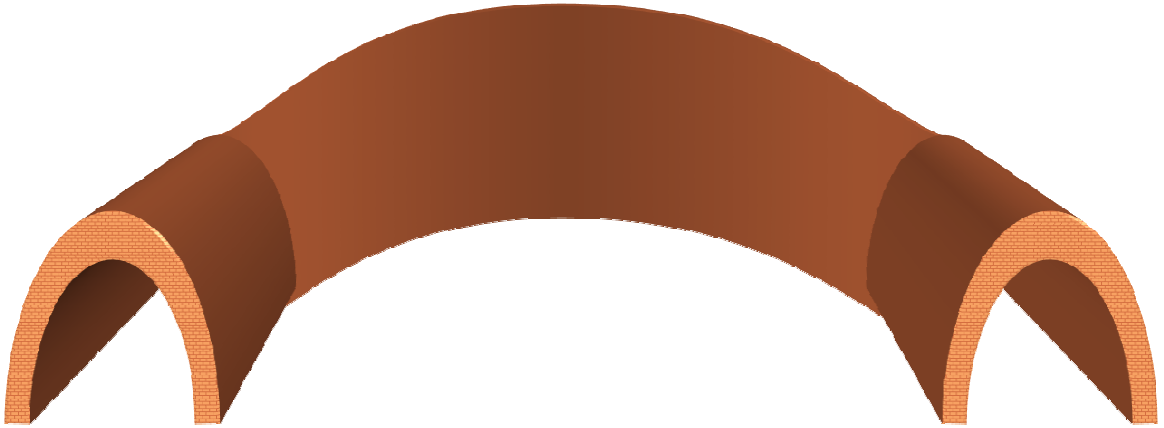


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configuring a tunnel

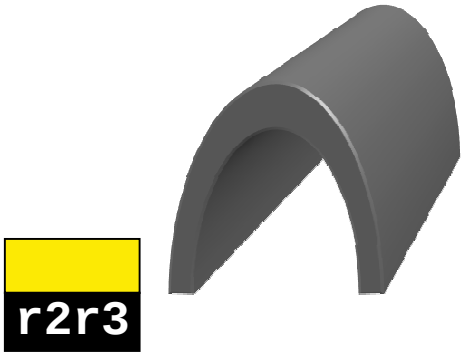


as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
ip tunnel add r2r1 mode ipip remote 1.1.1.1 local 2.2.2.2 ttl 10
ip link set r2r1 multicast on
ip addr add dev r2r1 13.0.0.6 peer 13.0.0.5
ifconfig r2r1 up
```

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configuring a tunnel



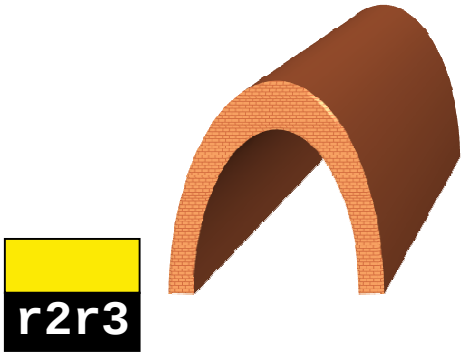
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r1 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig
ip tunnel                                  1.1.1.1 local 2.2.2.2 ttl 10
ip link s
ip addr a                                  3.0.0.5
ifconfig r2r1 up
```

endpoint name (appears as a virtual interface on the router

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configuring a tunnel



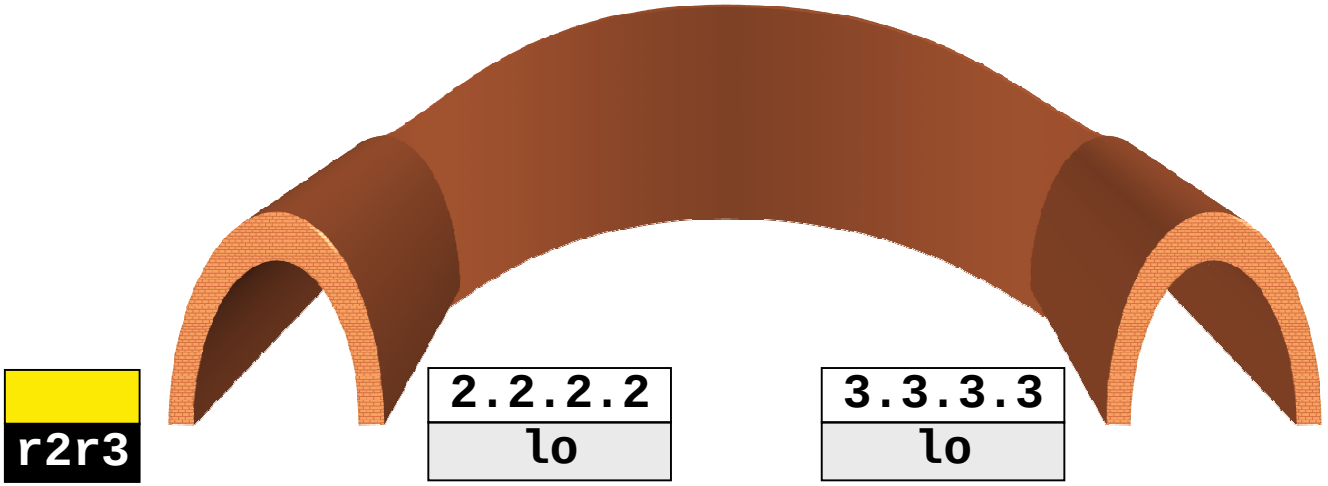
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.5 peer 13.0.0.6 ttl 10
ifconfig r2r3 up
ip tunnel add r2r1 mode ipip remote 1.1.1.1 local 2.2.2.2 ttl 10
ip link set r2r1 multicast on
ip addr add dev r2r1 13.0.0.6 peer 13.0.0.5
ifconfig r2r1 up
```

encapsulation type
(IP in IP)

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configuring a tunnel



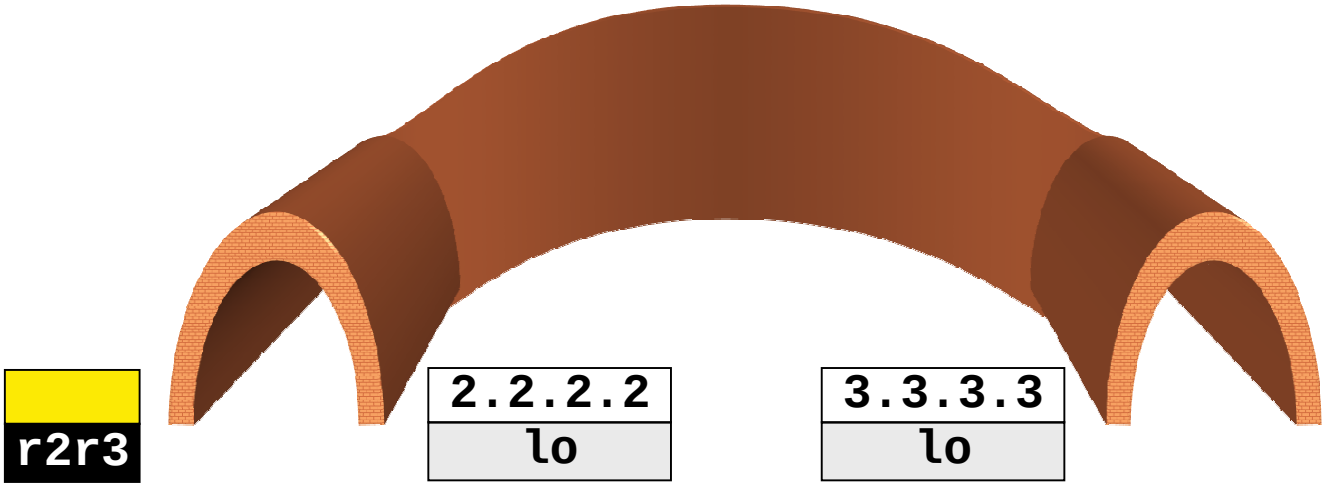
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9                              10
ifconfig r2r3 up
ip tunnel add r2r1 mode ipip r                             local 2.2.2.2 ttl 10
ip link set r2r1 multicast on
ip addr add dev r2r1 13.0.0.6 peer 13.0.0.5
ifconfig r2r1 up
```

tunnel endpoints

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configuring a tunnel



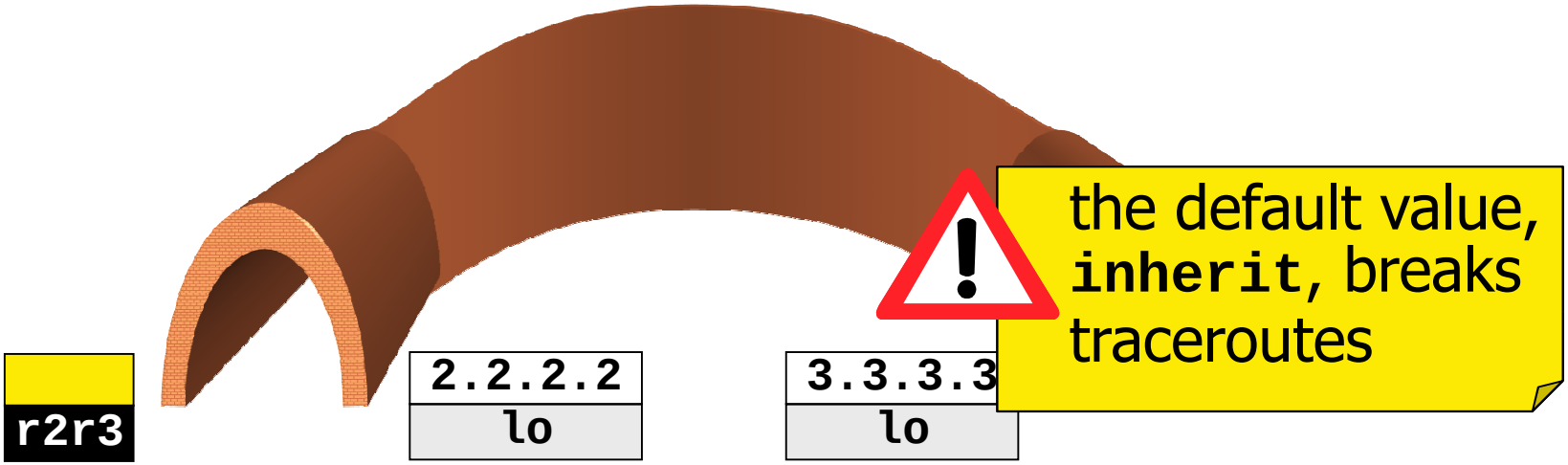
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
ip tunnel add r2r1 mode i
ip link set r2r1 multicas
ip addr add dev r2r1 13.0
ifconfig r2r1 up
```

ttl of the external envelope when a packet is encapsulated
(will expectedly travel through at most 4 hops, so any value >=4 should be fine)

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configuring a tunnel



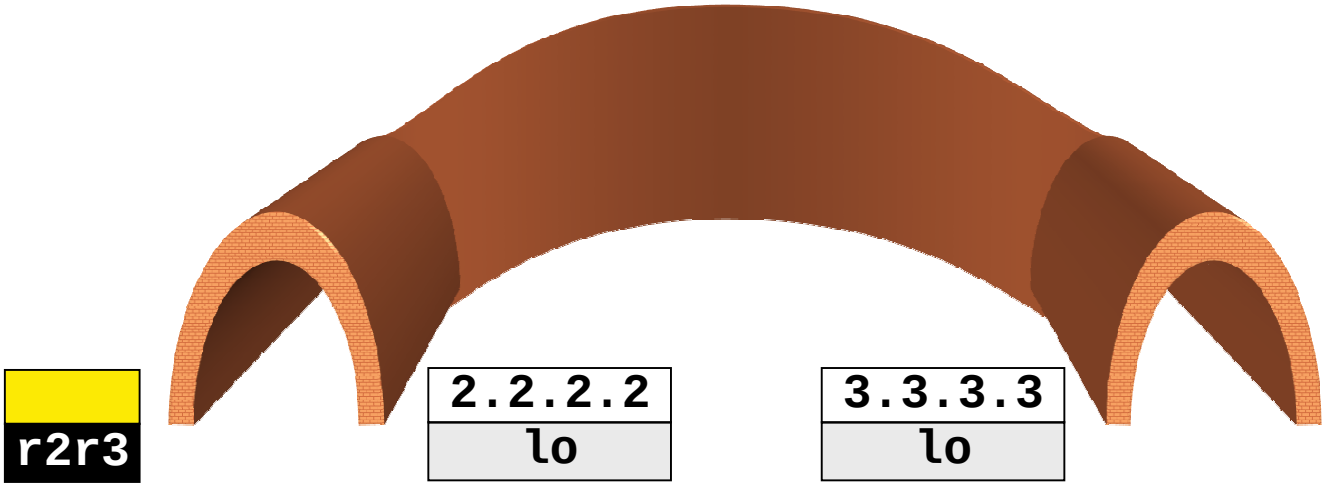
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
ip tunnel add r2r1 mode i
ip link set r2r1 multicas
ip addr add dev r2r1 13.0
ifconfig r2r1 up
```

ttl of the external envelope when a packet is encapsulated
(will expectedly travel through at most 4 hops, so any value >=4 should be fine)

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configuring a tunnel



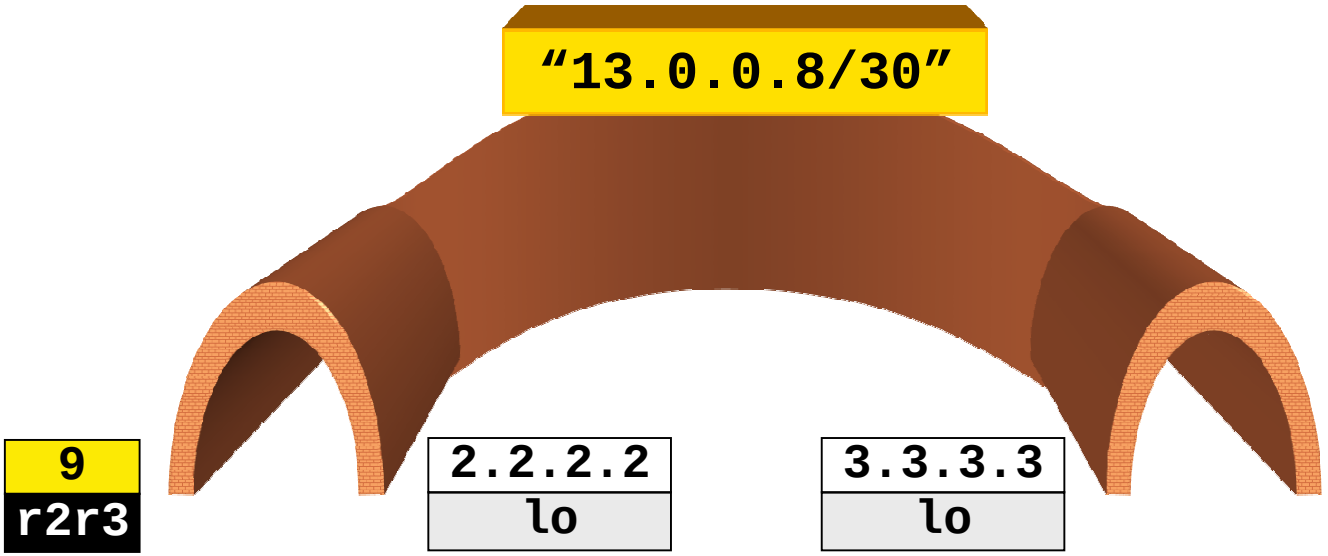
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 pee
ifconfig r2r3 up
ip tunnel add r2r1 mode ipip remot                    2.2 ttl 10
ip link set r2r1 multicast on
ip addr add dev r2r1 13.0.0.6 peer 13.0.0.5
ifconfig r2r1 up
```

rip uses multicast packets

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configuring a tunnel



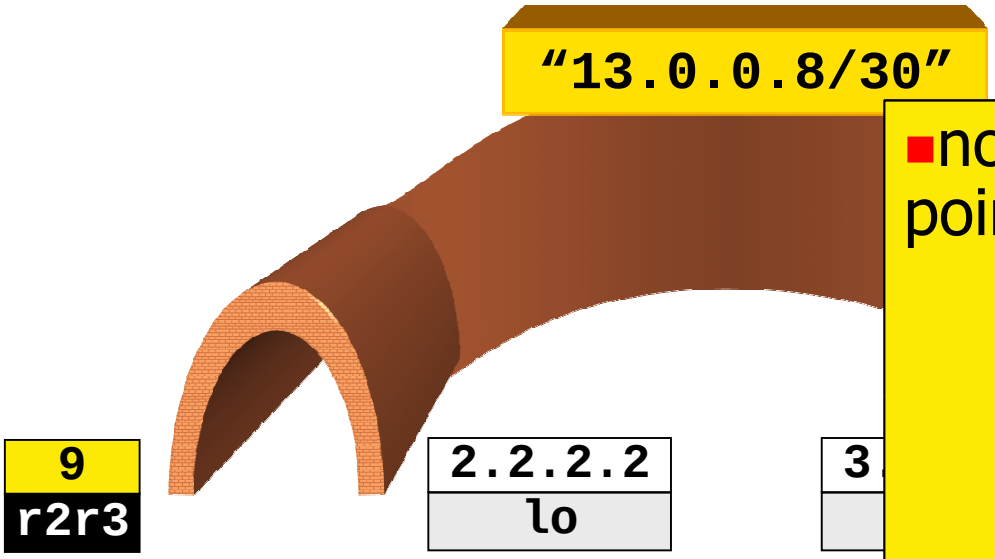
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
```

assign an ip address to the tunnel interface

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configuring a tunnel



- note: this is a pointopoint interface
  - speaking of “network” is senseless
  - we do it nevertheless to simplify the graphical layout

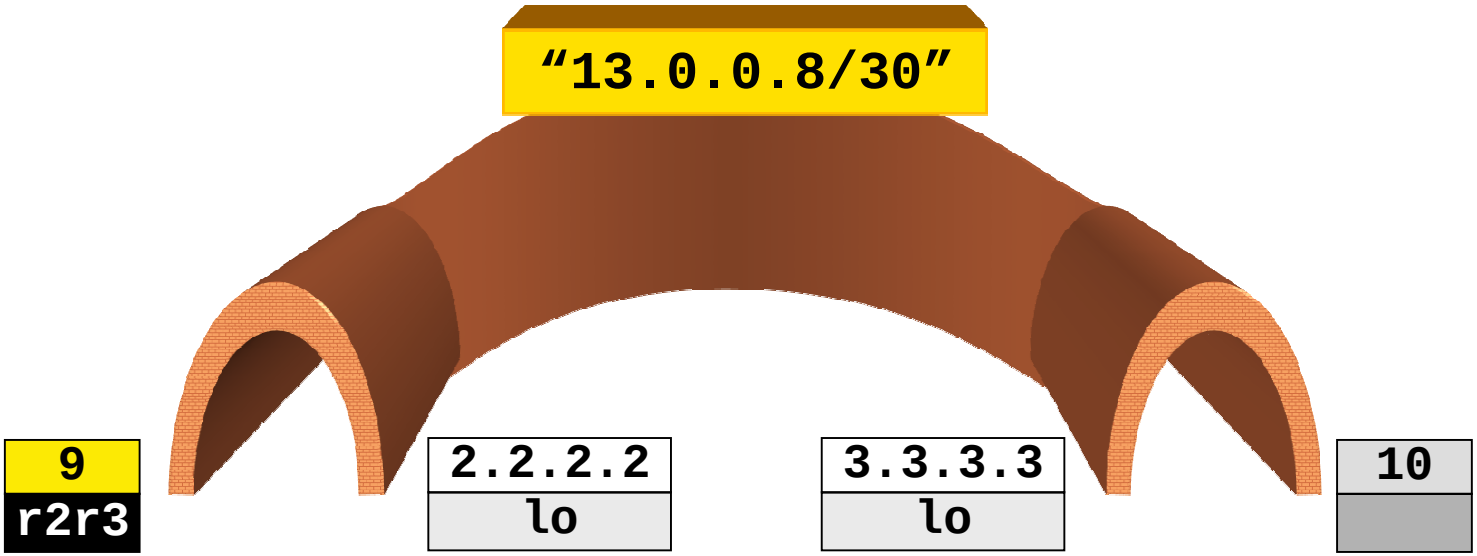
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
ip tunnel add r2r1 mod          remote 1.1.1.1 local 2.2.2.2 ttl 10
ip lin
ip add                                r 13.0.0.5
ifconf
```

assign an ip address to the tunnel interface

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configuring a tunnel



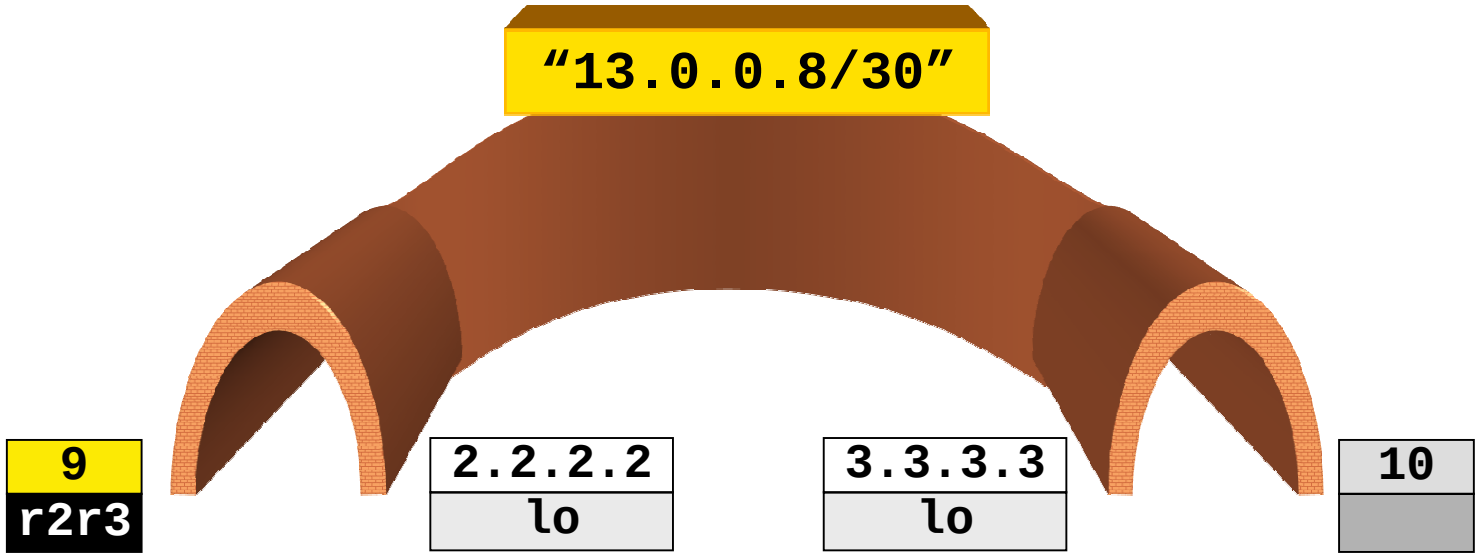
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
ip tun                .1.1.1 local 2.2.2.2 ttl 10
ip lin
ip add                .0.0.5
ifconf
```

for a pointopoint interface we can set the address of the other endpoint

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configuring a tunnel



as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
```

for a pointopoint interface we can set the address of the other endpoint

automatically inserts an entry in the routing table for **13.0.0.10**

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configuring a tunnel



note: failure to set the peer's address prevents rip from recognizing packets coming from the tunnel (possibly because it cannot match the sender's address with any of the local interface's subnets)

```
2007/10/30 11:27:25 RIP: RECV packet from
   13.0.0.10 port 520 on unknown
2007/10/30 11:27:25 RIP: packet comes from
   unknown interface
```

9
r2r3

interface name expected here

as10rt2 configuration

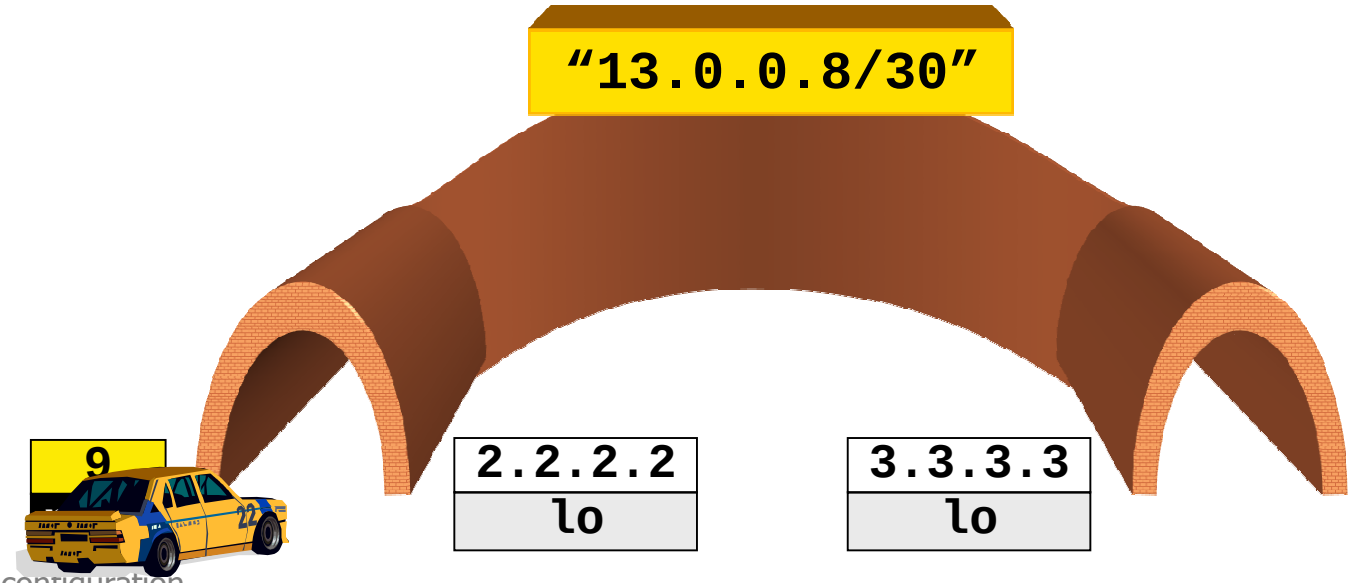
```
ip tunnel add r2r3 mode ipip remote 3.3.3.
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
```

for a pointopoint interface we can set the address of the other endpoint

automatically inserts an entry in the routing table for **13.0.0.10**

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configuring a tunnel



as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
ip tunnel add r2r1 mode ipip remote 1.1.1.1 local 2.2.2.2 ttl 10
ip link set r2r1 multi
ip addr add dev r2r1 13
ifconfig r2r1 up
```

switch the tunnel interface on

tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-list externalNetworks out r2r1
 distribute-list externalNetworks out r2r3
 distribute-list internalNetworks out eth1
 route 0.0.0.0/0
!
access-list externalNetworks permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```

tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-list externalNetworks out r2r1
 distribute-list externalNetworks out r2r3
 distribute-list internalNetworks out eth1
 route 0.0.0.0/0
!
access-list externalNetworks permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```



tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-list externalNe
 distribute-list externalNe
 distribute-list internalNet
 route 0.0.0.0/0
!
access-list externalNetworks permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```

distribute also **lo:1**'s address (but not **lo**'s address because it is within the reserved **127.0.0.0/8**)

tunnels and routing

as10rt2 ripd configuration

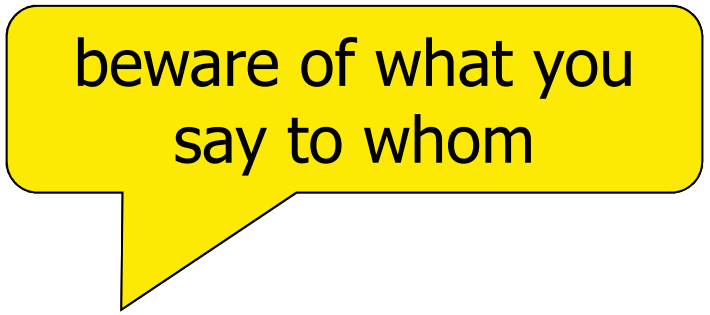
```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-list externalNe
 distribute-list externalNe
 distribute-list interna
 route 0.0.0.0/0
!
access-list externalNetworks permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```

propagate a statically configured default route *inside* the transit as (offers Internet connectivity to internal routers, if required)

tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-list externalNetworks out r2r1
 distribute-list externalNetworks out r2r3
 distribute-list internalNetworks out eth1
 route 0.0.0.0/0
!
access-list externalNetworks permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```

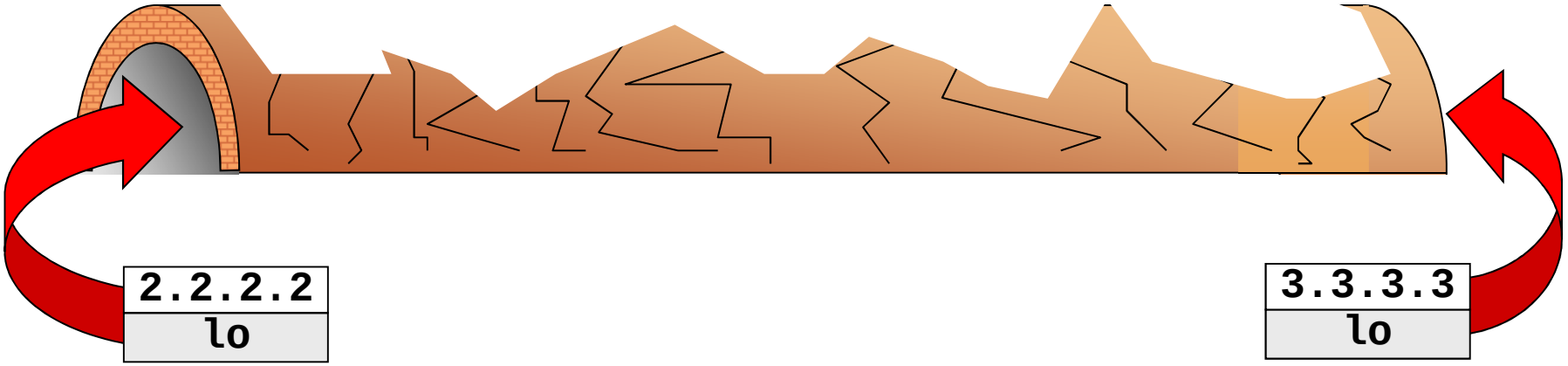


tunnels and routing



- we cannot announce the tunnel's endpoints inside the tunnel

tunnels and routing



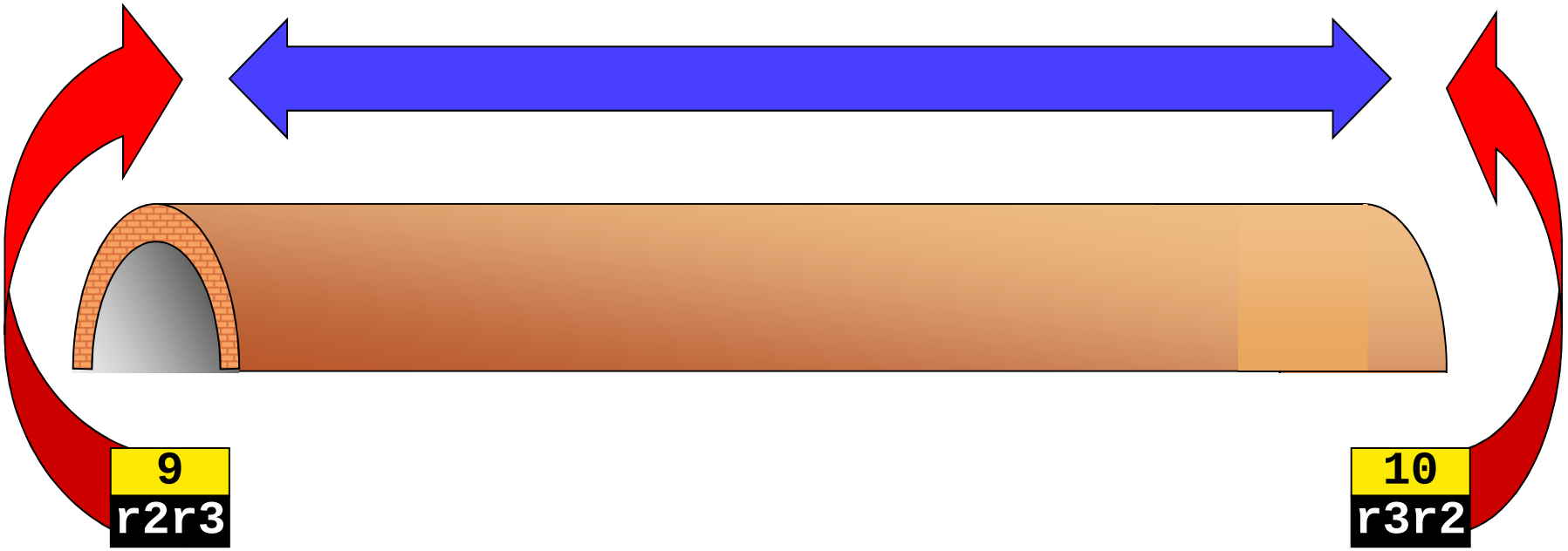
- we cannot announce the tunnel's endpoints inside the tunnel
  - that would tear down the tunnel!

tunnels and routing



- we shouldn’t announce the tunnel’s interfaces outside the tunnel

tunnels and routing



- we shouldn’t announce the tunnel’s interfaces outside the tunnel
  - traffic might flow outside the tunnel

tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-l                                  ut r2r1
 distribute-l                                  ut r2r3
 distribute-l                                  ut eth1
 route 0.0.0.0/0
!
access-list externalNetworks permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```

access-lists instruct rip about what to propagate

tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-l                                    1
 distribute-l                                    3
 distribute-l                                    1
 route 0.0.0.0/0
!
access-list externalNetworks permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```

ebgp next hops (in this case **as20r1**) are announced inside the tunnel

tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-list externalNetworks out r2r1
 distribute-l                                  3
 distribute-l                                  1
 route 0.0.0.
!
access-list externa          works permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```

ebgp next hops (in this case **as20r1**) are *not* announced outside the tunnel

tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-list externalNetworks out r2r1
 distribute-list externalNetworks out r2r3
 distribute-list internalNetworks out eth1
 route 0.0.0.0/0
!
access-list externalNetworks per
access-list externalNetworks den
access-list internalNetworks den
access-list internalNetworks den
access-list internalNetworks per
```

note: the same routing behavior could be obtained using static routes

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tunnels and routing

- check the zebra routing table on **as10rt3**

as10rt3

```
Router> show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP, O - OSPF,
       B - BGP, > - selected route, * - FIB route

R>* 1.1.1.1/32 [120/4] via 11.0.0.9, eth1, 03:2
R>* 2.2.2.2/32 [120/4] via 11.0.0.9, eth1, 03:2
C>* 3.3.3.3/32 is directly connected, lo
R>* 10.0.0.0/30 [120/3] via 11.0.0.9, eth1, 03:
R>* 10.0.0.4/30 [120/2] via 11.0.0.9, eth1, 03:27:03
R>* 10.0.0.8/30 [120/2] via 11.0.0.9, eth1, 03:27:0
R>* 11.0.0.0/30 [120/3] via 11.0.0.9, eth1, 03:27:0
R>* 11.0.0.4/30 [120/3] via 11.0.0.9, eth1, 03:27:
C>* 11.0.0.8/30 is directly connected, eth1
R>* 12.0.0.0/30 [120/2] via 13.0.0.9, r3r2, 03:26:44
B   12.0.0.0/30 [200/0] via 2.2.2.2, recursive via 11.0.0.9, eth1, 03:26:58
C>* 12.0.0.4/30 is directly connected, eth0
B>* 12.0.0.8/30 [200/0] via 12.0.0.2, recursive via 13.0.0.9, r3r2, 03:26:38
B>* 12.0.0.12/30 [20/0] via 12.0.0.6, eth0, 03:26:24
C>* 13.0.0.2/32 is directly connected, r3r1
C>* 13.0.0.9/32 is directly connected, r3r2
B>* 20.0.0.0/8 [200/0] via 12.0.0.2, recursive via 13.0.0.9, r3r2, 03:26:38
B>* 30.0.0.0/8 [20/0] via 12.0.0.6, eth0, 03:26:24
B>* 100.0.0.0/8 [20/0] via 12.0.0.6, eth0, 03:26:24
C>* 127.0.0.0/8 is directly connected, lo
```

destinations routed through the tunnel

tunnels and routing

- **as10rt2** prefers the egress point **as10rt3**

as10rt2

```
as10rt2:~# traceroute -s 12.0.0.1 100.0.0.1
traceroute to 100.0.0.1 (100.0.0.1) from 12.0.0.1, 64 hops max, 40
byte packets
 1  13.0.0.10 (13.0.0.10)  3 ms  3 ms  2 ms
 2  12.0.0.6 (12.0.0.6)  2 ms  4 ms  5 ms
 3  100.0.0.1 (100.0.0.1)  2 ms  2 ms  2 ms
```

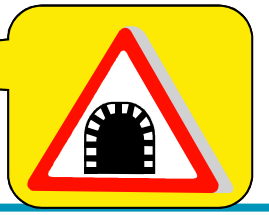
tunnels and routing

- **as10rt2** prefers the egress point **as10rt3**

did we already mention you should use a source address that is reachable from outside the transit as?

as10rt2

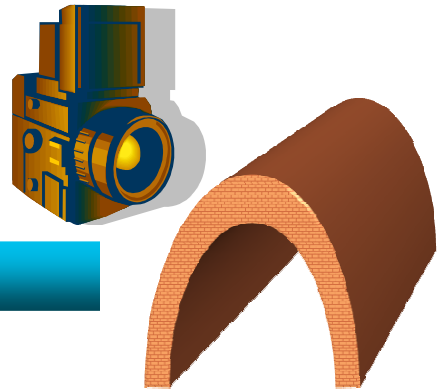
```
as10rt2:~# traceroute -s 12.0.0.1 100.0.0.1
traceroute to 100.0.0.1 (100.0.0.1) from 12.0.0.1, 64 hops max, 40
byte packets
 1  13.0.0.10 (13.0.0.10)  3 ms  3 ms  2 ms
 2  12.0.0.6 (12.0.0.6)  2 ms  4 ms  5 ms
 3  100.0.0.1 (100.0.0.1)  2 ms  2 ms  2 ms
```



- now **as10rt3** is directly reached via the tunnel

tunnels and routing

- a look outside the tunnel



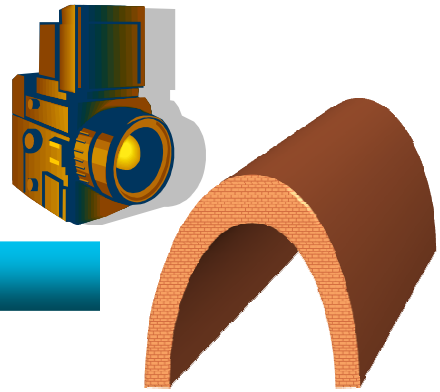
as10rt2

```
as10rt2:~# ip tunnel show r2r3
r2r3: ip/ip  remote 3.3.3.3  local 2.2.2.2  ttl 10
as10rt2:~# ip link show eth0
1: eth0: <BROADCAST,MULTICAST,UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether fe:fd:0c:00:00:01 brd ff:ff:ff:ff:ff:ff
as10rt2:~# ip link show r2r3
7: r2r3@NONE: <POINTOPOINT,MULTICAST,NOARP,UP> mtu 1480 qdisc noqueue
    link/ipip 2.2.2.2 peer 3.3.3.3
as10rt2:~#
```

the tunnel is active

tunnels and routing

- a look outside the tunnel

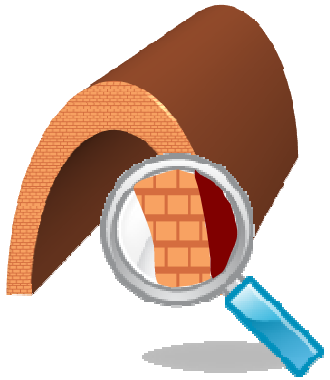


as10rt2

```
as10rt2:~# ip tunnel show r2r3
r2r3: ip/ip  remote 3.3.3.3  local 2.2.2.2  ttl 10
as10rt2:~# ip link show eth0
1: eth0: <BROADCAST,MULTICAST,UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether fe:fd:0c:00:00:01 brd ff:ff:ff:ff:ff:ff
as10rt2:~# ip link show r2r3
7: r2r3@NONE: <POINTOPOINT,MULTICAST,NOARP,UP> mtu 1480 qdisc noqueue
    link/ipip 2.2.2.2 peer 3.3.3.3
as10rt2:~#
```

the tunnel's **mtu** is 20 bytes smaller because of the additional ip header

tunnels and routing



- a look inside the tunnel

as10rt2

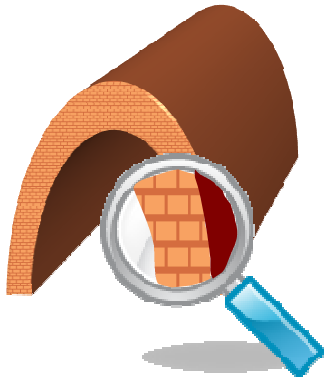
```
as10rt2:~# ping -I 12.0.0.1 100.0.0.1
PING 100.0.0.1 (100.0.0.1) from 12.0.0.1 : 56(84) bytes of data.
64 bytes from 100.0.0.1: icmp_seq=1 ttl=63 time=1.43 ms
64 bytes from 100.0.0.1: icmp_seq=3 ttl=63 time=1.41 ms
```

as10r5

```
as10r5:~# tcpdump -i eth1
tcpdump: verbose output suppressed, use -v or -vv for full protocol
decode
listening on eth1, link-type EN10MB (Ethernet), capture size 96 bytes
14:58:30.602023 IP 2.2.2.2 > 3.3.3.3: IP 12.0.0.1 > 100.0.0.1: icmp
64: echo request seq 27 (ipip-proto-4)
14:58:31.612680 IP 2.2.2.2 > 3.3.3.3: IP 12.0.0.1 >
64: echo request seq 28 (ipip-proto-4)
```

packets are encapsulated

tunnels and routing



- a look inside the tunnel

as10rt2

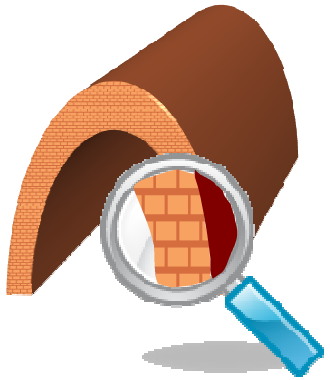
```
as10rt2:~# ping -I 12.0.0.1 100.0.0.1
PING 100.0.0.1 (100.0.0.1) from 12.0.0.1 : 56(84) bytes of data.
64 bytes from 100.0.0.1: icmp_seq=1 ttl=63 time=1.43 ms
64 bytes from 100.0.0.1: icmp_seq=3 ttl=63 time=1.41 ms
```

as10r5

```
as10r5:~# tcpdump -i eth1
tcpdump: verbose output suppressed, use -v or -vv for full protocol
decode
listening on eth1, link-type EN10MB (Ethernet), capture size 96 bytes
14:58:30.602023 IP 2.2.2.2 > 3.3.3.3: IP 12.0.0.1 > 100.0.0.1: icmp
64: echo request seq 27 (ipip-proto-4)
14:58:31.612680 IP 2.2.2.2 > 3.3.3.3: IP 12.0.
64: echo request seq 28 (ipip-proto-4)
```

outer ip addresses correspond to the tunnel endpoints

tunnels and routing



- a look inside the tunnel

as10rt2

```
as10rt2:~# ping -I 12.0.0.1 100.0.0.1
PING 100.0.0.1 (100.0.0.1) from 12.0.0.1 : 56(84) bytes of data.
64 bytes from 100.0.0.1: icmp_seq=1 ttl=63 time=1.43 ms
64 bytes from 100.0.0.1: icmp_seq=3 ttl=63 time=1.41 ms
```

as10r5

```
as10r5:~# tcpdump -i eth1
tcpdump: verbose output suppressed, use -v or -vv for full protocol
decode
listening on eth1, link-type EN10MB (Ethernet), capture size 96 bytes
14:[illegible]2 > 3.3.3.3: IP 12.0.0.1 > 100.0.0.1: icmp
64:[illegible]ipip-proto-4)
14:[illegible]2 > 3.3.3.3: IP 12.0.0.1 > 100.0.0.1: icmp
64:[illegible]ipip-proto-4)
```

inner ip addresses correspond to the real source and destination

transit as: routing tables

as10r6

```
as10r6-ripd> show ip rip
Codes: R - RIP, C - connected, O - OSPF, B - BGP
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

     Network            Next Hop         Metric From            Time
R(n) 0.0.0.0/0          11.0.0.10             2 11.0.0.10       02:48
R(n) 1.1.1.1/32         10.0.0.10             3 10.0.0.10       02:59
R(n) 2.2.2.2/32         10.0.0.5              3 10.0.0.5        02:49
R(n) 3.3.3.3/32         11.0.0.10             2 11.0.0.10       02:48
R(n) 10.0.0.0/30        10.0.0.5              2 10.0.0.5        02:49
C(i) 10.0.0.4/30        0.0.0.0               1 self
C(i) 10.0.0.8/30        0.0.0.0               1 self
R(n) 11.0.0.0/30        10.0.0.10             2 10.0.0.10       02:59
R(n) 11.0.0.4/30        10.0.0.5              2 10.0.0.5        02:49
C(i) 11.0.0.8/30        0.0.0.0               1 self
as10r6-ripd>
```

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transit as: routing tables

as10r6

```
as10r6-ripd> show ip rip
Codes: R - RIP, C - connected, O - OSPF, B - BGP
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

     Network            Next Hop         Metric From              Time
R(n) 0.0.0.0/0          11.0.0.10                                 48
R(n) 1.1.1.1/32         10.0.0.10                                 59
R(n) 2.2.2.2/32         10.0.0.5                                  49
R(n) 3.3.3.3/32         11.0.0.10                                 48
R(n) 10.0.0.0/30        10.0.0.5                                  49
C(i) 10.0.0.4/30        0.0.0.0
C(i) 10.0.0.8/30        0.0.0.0
R(n) 11.0.0.0/30        10.0.0.10                                 59
R(n) 11.0.0.4/30        10.0.0.5                                  49
C(i) 11.0.0.8/30        0.0.0.0
as10r6-ripd>
```

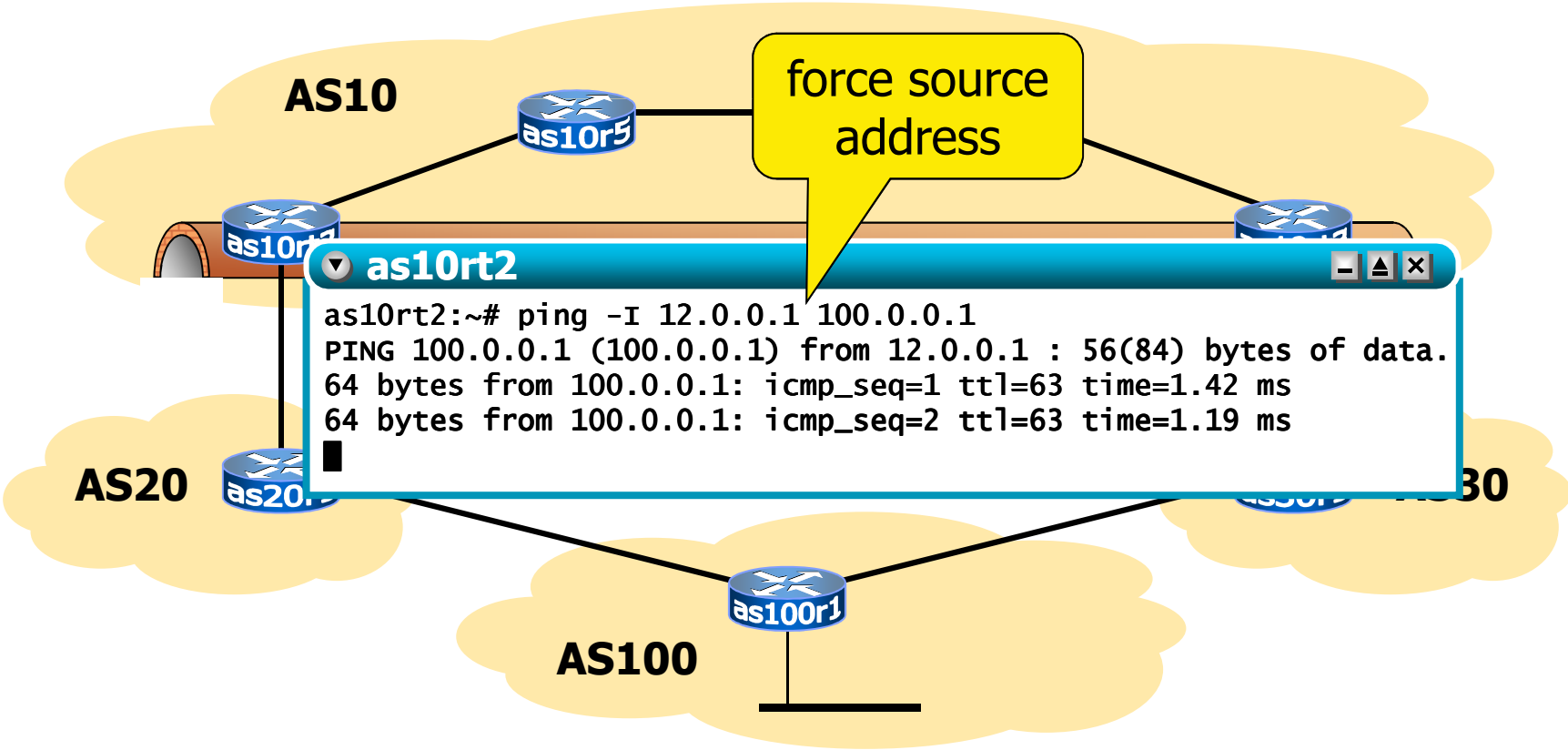
internal routers only know about internal destinations

(the default route is only there to offer Internet access, if required)

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transit as: asymmetric routing

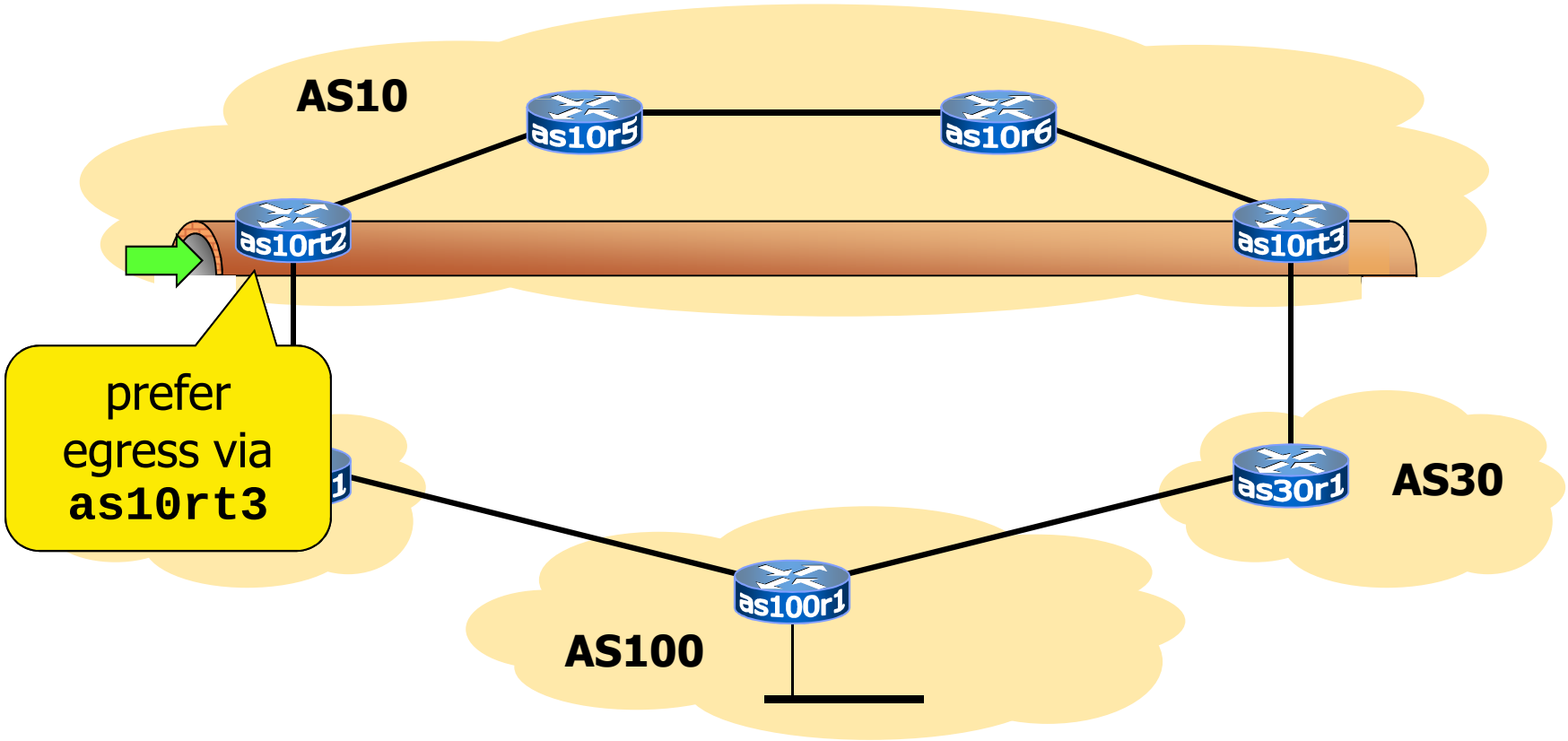
- bgp routing policies make routing asymmetric



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transit as: asymmetric routing

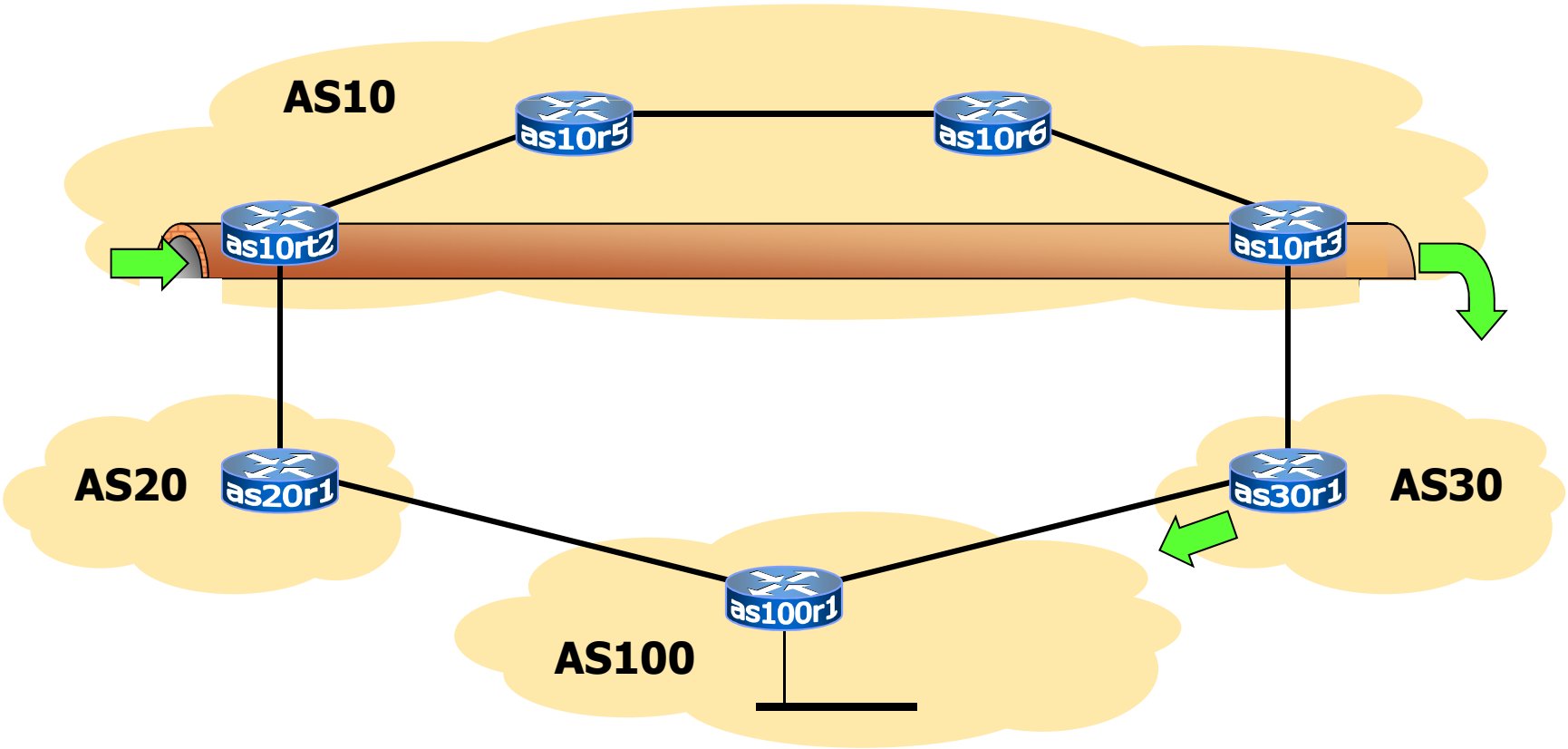
- bgp routing policies make routing asymmetric



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transit as: asymmetric routing

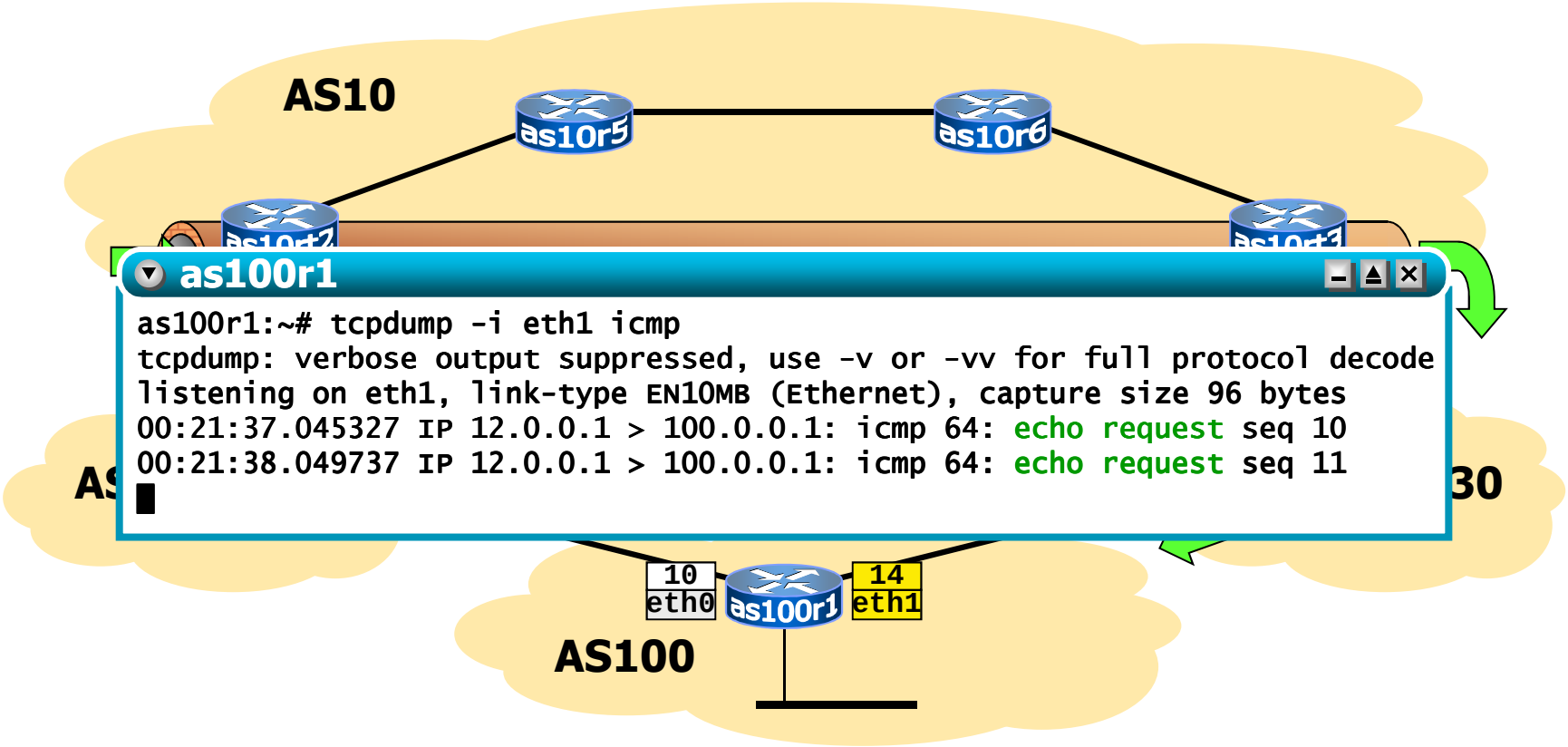
- bgp routing policies make routing asymmetric



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transit as: asymmetric routing

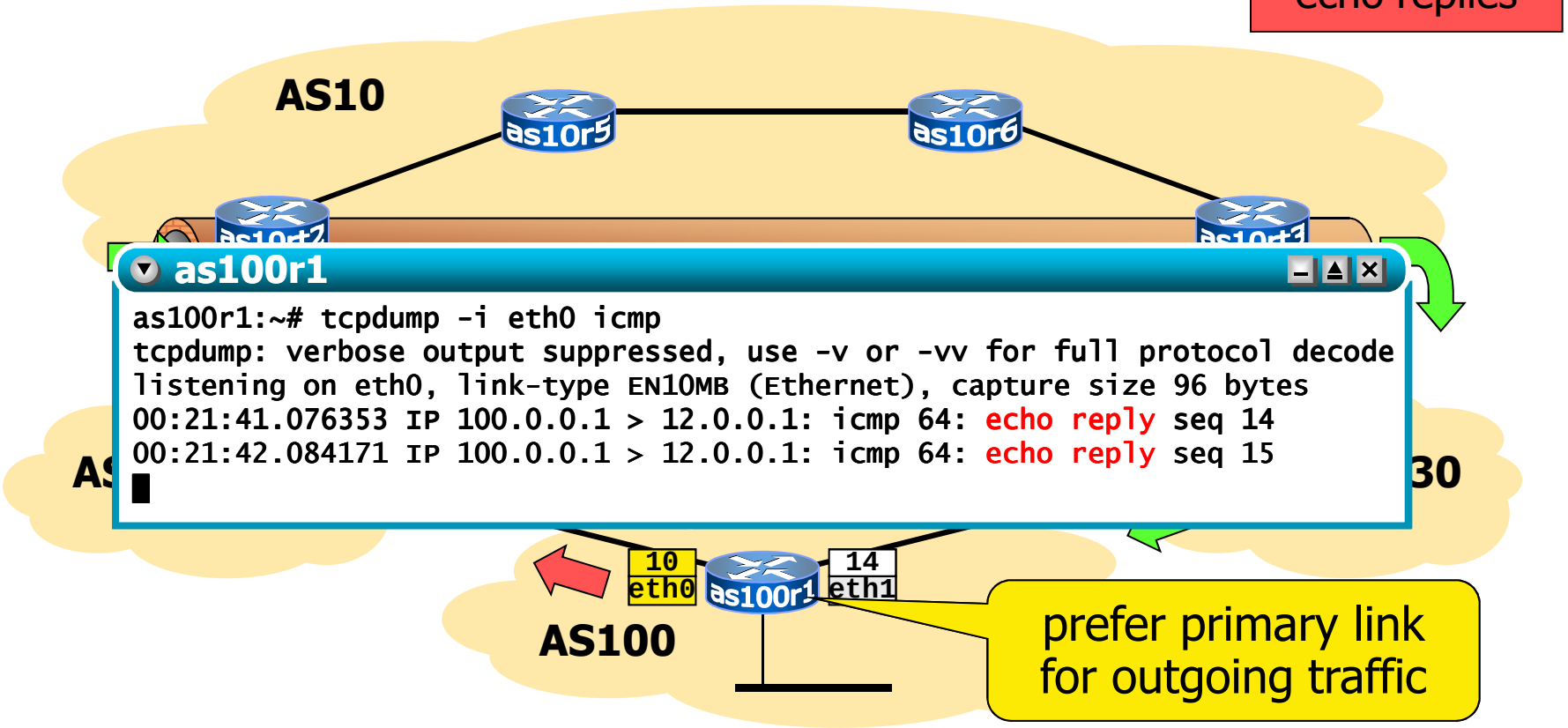
- bgp routing policies make routing asymmetric



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transit as: asymmetric routing

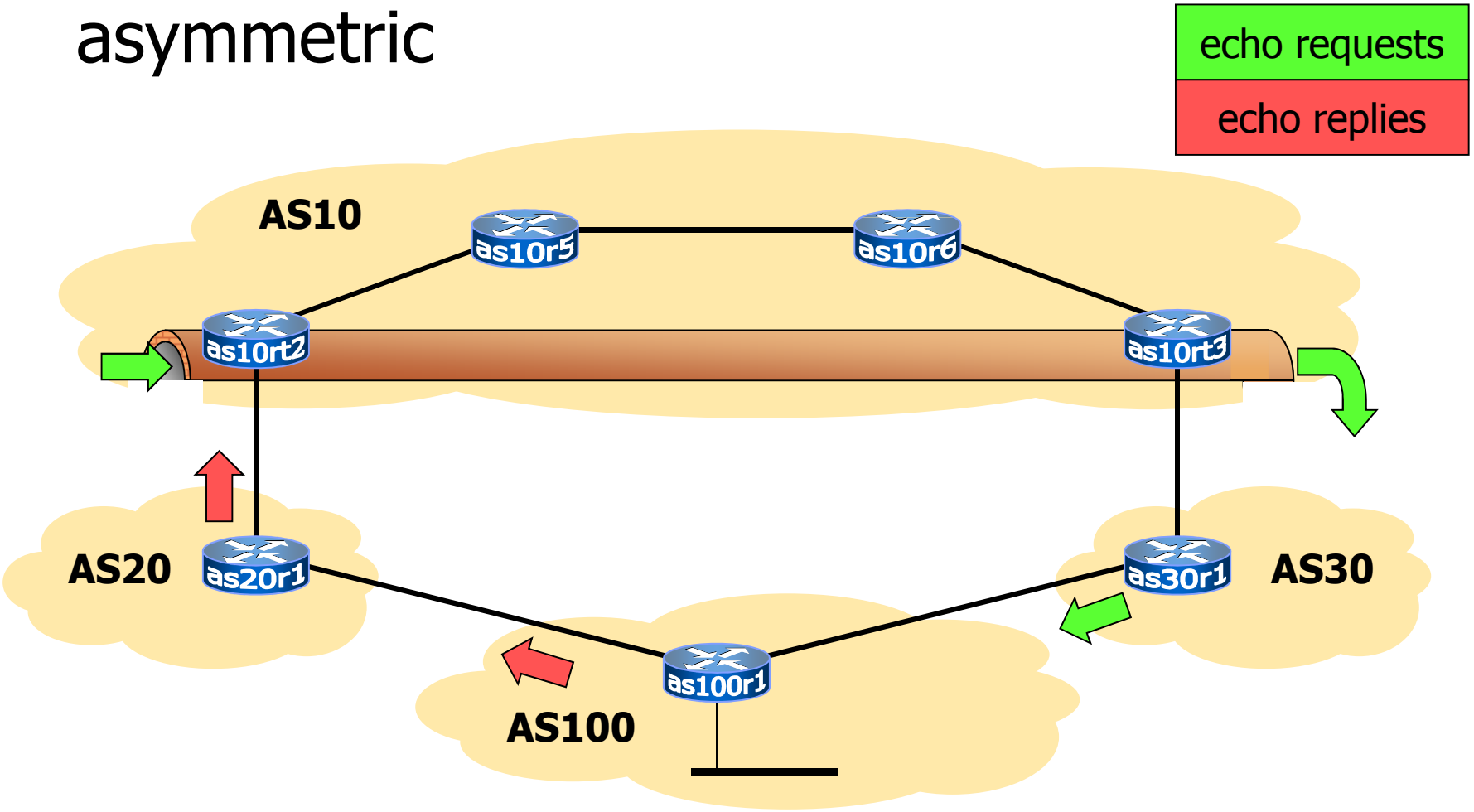
- bgp routing policies make routing asymmetric



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transit as: asymmetric routing

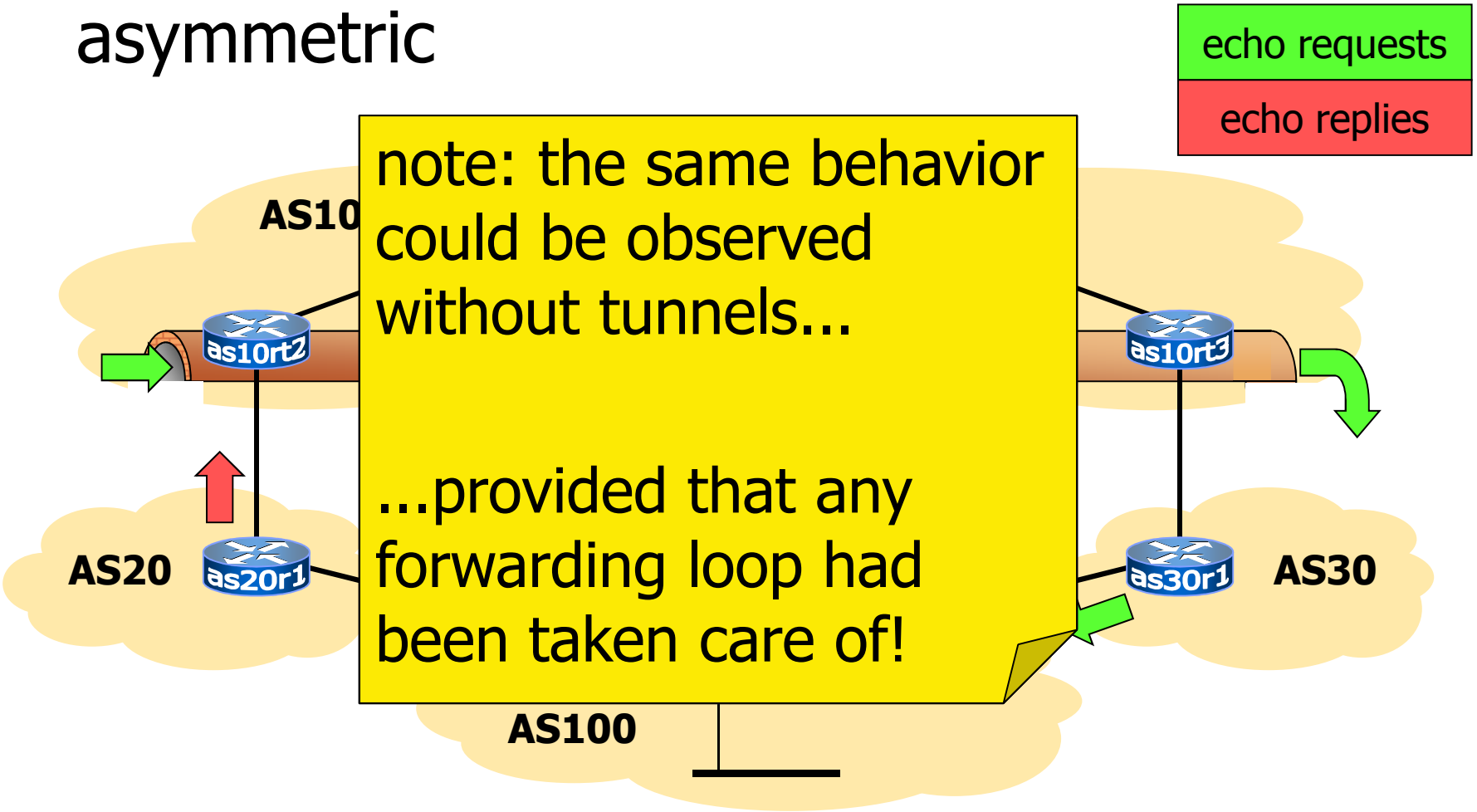
- bgp routing policies make routing asymmetric



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transit as: asymmetric routing

- bgp routing policies make routing asymmetric



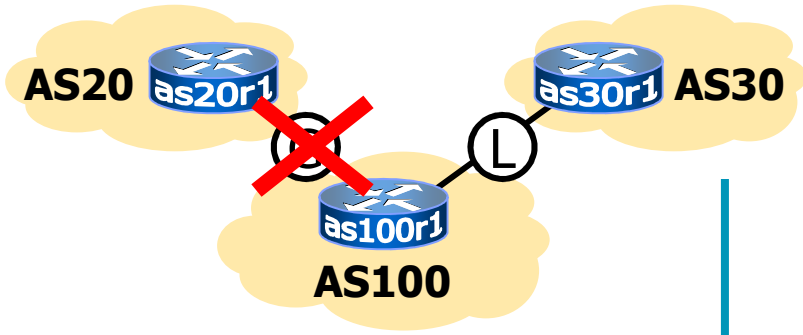
transit as: playing with the backup

- let's bring **as100**'s primary link down (again!)
  - expected result: traffic from **as20r1** to **as100r1** should traverse the transit as

as20r1

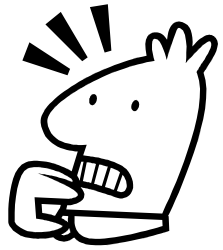
```
as20r1:~# telnet localhost bgpd
Trying 127.0.0.1...
Connected to as20r1.
Escape character is '^]'.

Hello, this is zebra (version 0.94).
...
as20r1-bgpd> enable
as20r1-bgpd# configure terminal
as20r1-bgpd(config)# router bgp 20
as20r1-bgpd(config-router)# neighbor 12.0.0.10 shutdown
```



transit as: playing with the backup

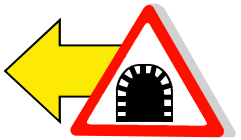
- wait for the routing to converge
  - fingers crossed...



- check the reachability of **100.0.0.0/8**

as20r1

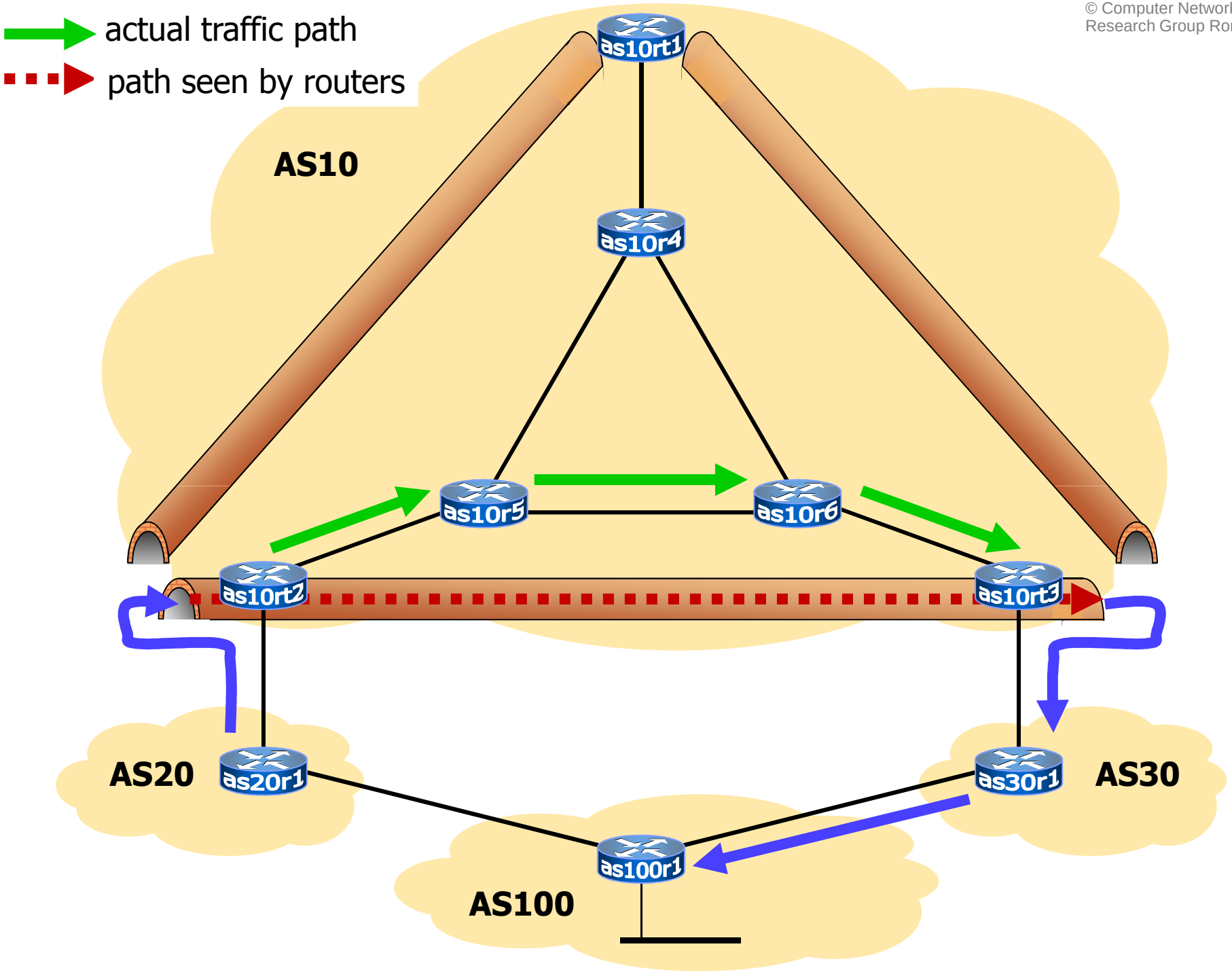
```
as20r1:~# traceroute 100.0.0.1
traceroute to 100.0.0.1 (100.0.0.1), 64 hops max, 40 byte packets
 1  12.0.0.1 (12.0.0.1)  2 ms  2 ms  1 ms
 2  13.0.0.10 (13.0.0.10)  2 ms  3 ms  2 ms
 3  12.0.0.6 (12.0.0.6)  3 ms  3 ms  3 ms
 4  100.0.0.1 (100.0.0.1)  3 ms  3 ms  3 ms
as20r1:~#
```





  - traffic is now traversing the transit as!

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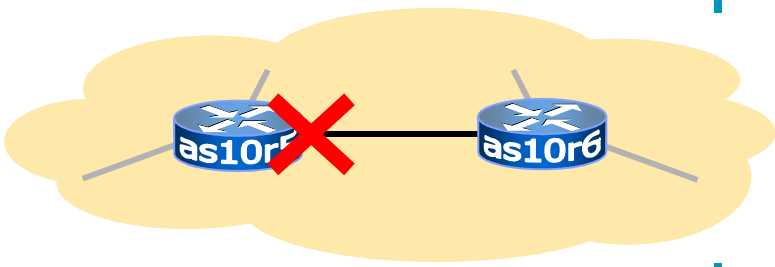
transit as: “rubbery” tunnels

- breaking an internal link does not tear the tunnels down
  - (as long as the transit as is not partitioned)

as10r5

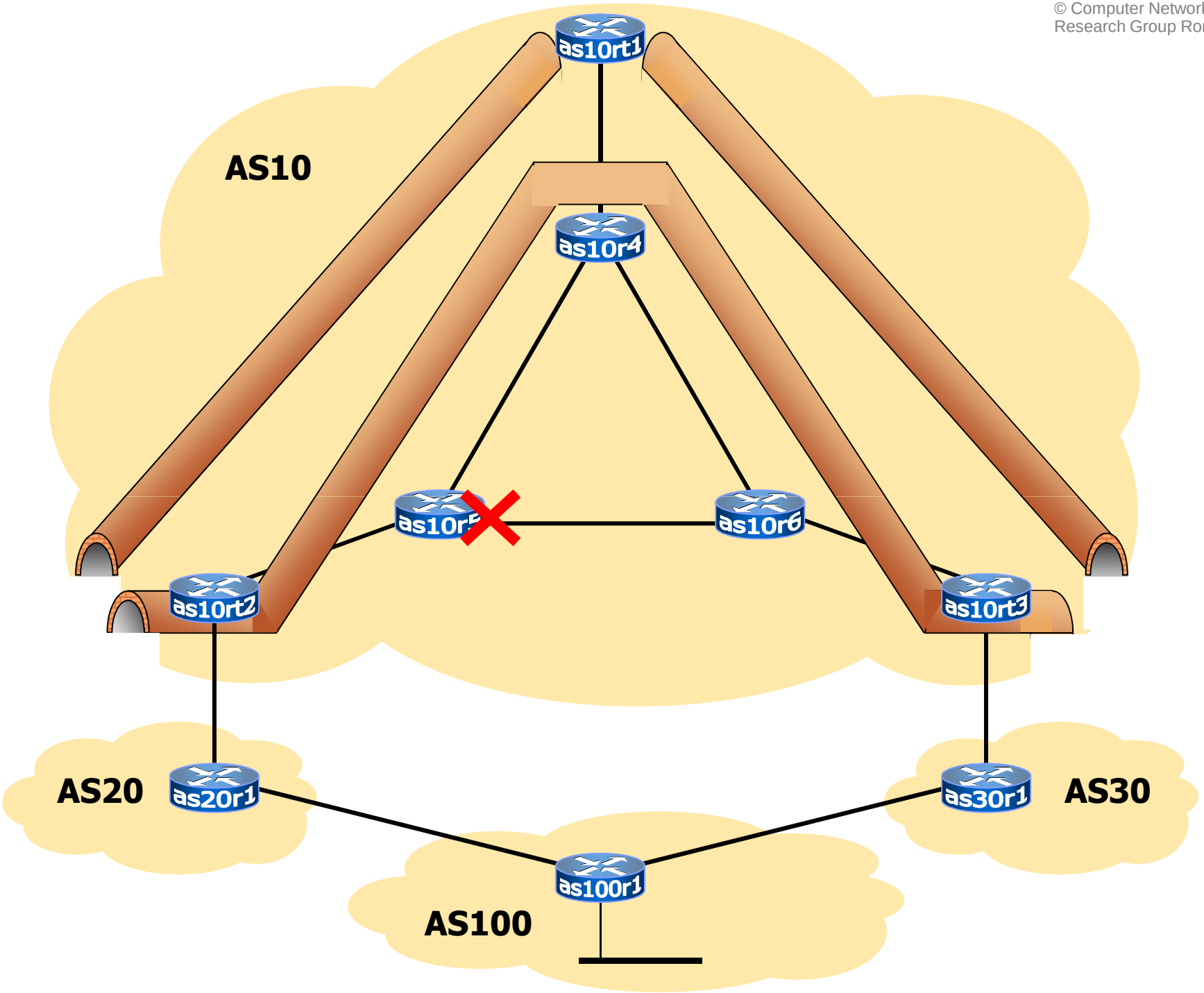
```
as10r5:~# ifconfig eth2 down
```






```
as10r5:~# traceroute 10.0.0.6
traceroute to 10.0.0.6 (10.0.0.6), 64 hops max, 40 byte packets
 1  10.0.0.1 (10.0.0.1)  1 ms  1 ms  1 ms
 2  10.0.0.6 (10.0.0.6)  1 ms  2 ms  2 ms
as10r5:~#
```

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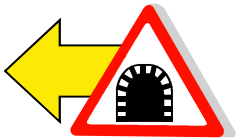


transit as: “rubbery” tunnels

- wait for the routing to converge
  - be **really** patient
- check the reachability of **`100.0.0.0/8`**

as20r1

```
as20r1:~# traceroute 100.0.0.1
traceroute to 100.0.0.1 (100.0.0.1), 64 hops max, 40 byte packets
 1  12.0.0.1 (12.0.0.1)  1 ms  1 ms  1 ms
 2  13.0.0.10 (13.0.0.10)  2 ms  4 ms  2 ms
 3  12.0.0.6 (12.0.0.6)  3 ms  3 ms  3 ms
 4  100.0.0.1 (100.0.0.1)  6 ms  3 ms  4 ms
as20r1:~#
```



  - traffic is still able to traverse the transit as

conclusions

- an overlay network is better
  - smaller routing tables on internal routers
  - less churn
  - predictable interplay between igp and egp
- sample implementation: tunnels
  - directed to the egress points
- support internal customers
  - static default route injected into rip
- observations
  - bgp peerings could be established on the tunnel interfaces
  - tunnels are as robust as the underlying igp